

Packers

Halliburton production packers meet the needs of the oil and gas industry with innovation, performance, and value. Whether it is a standard packer from our extensive product line or a unique specialized design for a hostile environment, Halliburton is the preferred supplier of production packers.

Halliburton has a packer for every type of completion: permanent packers rated up to 20,000 psi (138 MPa) at 475°F (246°C); retrievable packers suitable for 15,000 psi (103.4 MPa) at 400°F (204°C); dual packers for stacked or injection wells; thermal packers for steam service; as well as packers for multilateral wells, horizontal wells, deepwater completions, SmartWell® completion systems, and sand control.

Finite element analysis and full-scale testing in Halliburton's world-class test facility help ensure the packer performs as designed. The High Temperature Test Facility (HTTF) tests packers to ISO14310/API 11D1 requirements for all "V" class ratings. The Deep Well Simulator allows packers to be tested under actual downhole pressures and temperatures with applied tubing loads. Two test wells verify tool compatibility and packer operating sequences. Oilfield metallurgy and elastomer development experts keep Halliburton at the forefront of these technical fields.

Halliburton packers are manufactured to a quality program standard that meets the requirements of ISO 9001. All threads are gauged and inspected according to API or proprietary standards.

This chapter describes several types of packers used in subsurface completions. Descriptions of permanent production packers, retrievable production packers, and packer completion accessories are included.

Halliburton offers a complete line of packers and accessory equipment to help plan your well completions. Besides packers for production installations, Halliburton provides a wide choice of isolation, utility, and special application packers, offering a packer solution for every well completion type. To determine which packer is best suited for your well, contact your local Halliburton representative.



Carrollton Test Facility featuring two test wells for full scale packer testing



Using the Deep Well Simulator, Halliburton can test new packer designs for use in high-temperature, high-pressure wells.

Packer Performance Envelopes

Performance envelopes provide a graphical means by which packer capabilities can be effectively communicated. Accurate packer performance is determined based on the combined effects of loading which includes pressure, temperature, and axial forces.

In order to provide a graphical reference for the loading scenarios, Halliburton plots performance envelopes unique to packer designs.

Quadrants labeled in the envelope example below represent the following:

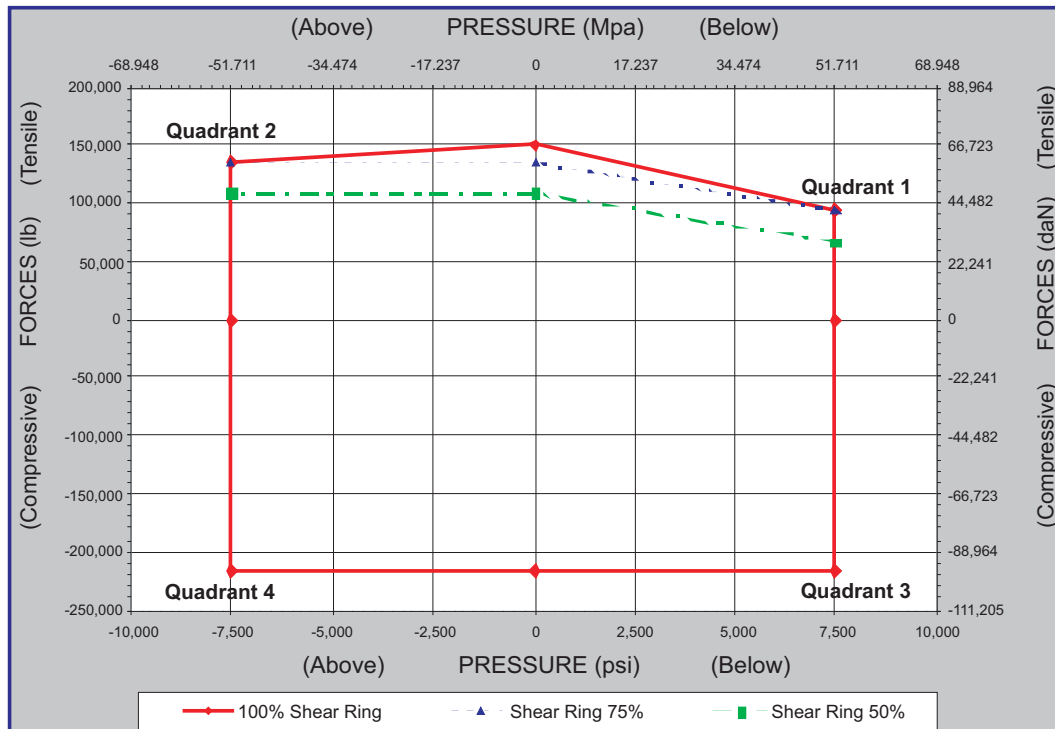
- Quadrant 1 – Pressure from *Below* with *Tension*
- Quadrant 2 – Pressure from *Above* with *Tension*
- Quadrant 3 – Pressure from *Below* with *Compression*
- Quadrant 4 – Pressure from *Above* with *Compression*

ISO 14310 / API 11D1 Design Validation Grades

Validation levels should be used to describe development test requirements for new packers. ISO 14310/API 11D1 design validation grades are defined below.

- Grade V6 = Minimum grade (supplier specified)
- Grade V5 = Liquid test
- Grade V4 = Liquid test with axial loads
- Grade V3 = Liquid test with axial loads and temperature cycling
- Grade V2 = Gas test with axial loads
- Grade V1 = Gas test with axial loads and temperature cycling
- Grade V0 = V1 with special acceptance criteria (zero bubble)

Halliburton typically supplies V3 and V0 validation grades. Selection of a specific validation grade is best determined with the consultation of a Halliburton field representative.



Operating envelopes are used for combined packer performance ratings.

Packer Application/Selectivity Table

Packer Description	Packer Type	Catalog Page	Single-String	Setting Method				Primary Applications									
				Wireline-Set	Hydraulic-Set	Tubing / Mechanical-Set	Hydrostatic-Set	Releasing Method	Production	Tubing-Conveyed Perforating	High-Pressure, High-Temperature	Single-Trip Completions	Injection	Thermal	Bridge Plug Application	Coiled Tubing Completions	Open Hole
Permanent Sealbore Type Packers																	
Wireline-Set Perma-Series®	AWB (1) (2), TWS	6-4	x	x	3			4	x	x	x		x	x	x		
Hydraulic-Set Perma-Series	MHR (1)	6-13	x		x			4	x	x	x	x	x	x			
Hydrostatic-Set Perma-Series	HHR (1)	6-15	x		x		x	4	x	x	x	x					
Retrievable Sealbore Type Packers																	
Wireline-Set Versa-Trieve®	VBA, VGP, VTA	6-28	x	x	3			5	x	x	x		x	x	x		
Hydraulic-Set HP/HT Retrievable	HPH/HPS	6-22	x		x			5	x	x	x	x	x				
Hydro-Set, Hydro-Release	HP1	6-24	x		x			13	x	8	x	x	x				
Retrievable Hydraulic-Set Packers																	
Hydraulic-Set Perma-Lach®	PHL	6-44	x		x			6	x	8	x	x	x				
Hydraulic-Set Single Production	RH	6-46	x		x			6	x	8		x	x				
	G-77	6-48	x		x			6	x	8		x	x				
	ZoneGuard®	6-52	x		x			6									x
	AHC	6-17	x		x			7	x	8		x	x				
Hydraulic-Set Variable Bore	MFT	6-26			x			7		x	x	x	x				
Hydraulic-Set Dual Production	RDH	6-50			x			6	x	8		x	x				
Hydrostatic-Set Single Production	HHC	6-18	x				x	7	x	8		x	x				
Hydraulic-Set Coiled Tubing	HGO	6-63	x	x				6	x							x	
Retrievable Wireline-Set Packers																	
Wireline-Set Retrievable	Versa-Set®	6-36	x	x				9	x	8	x		x		x		
Mechanical-Set Retrievable																	
Mechanical-Set Double Grip	Versa-Set	6-36	x			x		9, 11	x	8	x	x	x		x		
	PLS	6-39	x			x		9, 11	x	8	x	x	x		x		
	G-6	6-42	x			x		9, 11	x	8		x	x	x	x		
Mechanical-Set Single Grip	GO™	6-63	x			x		6				x				x	
	R4™	6-64	x			x		6, 10, 11	x			x	x				

Application Notes

- Various optional receiving head and bottom sub connections are available.
- Optional twin-flow conversion units and chemical injection heads for dual completions.
- May also be set using a hydraulic setting tool.
- Packer is retrieved by milling. This packer features short milling distance and is non-rotating.
- Retrieve using retrieving tool.
- Retrieve packer by straight pull on tubing.
- Some models may be retrieved by cutting the packer mandrel to release.
- Limited application - contact Halliburton Completion Specialist.
- Packer releases with 1/4 to 1/3 right-hand tubing turn at the packer.
- Retrieve packer by slacking off on tubing then straight pull when set in tension.
- Emergency straight shear release
- Hydrostatic assist
- Punch-to-release

Permanent Production Packers

Wireline-Set Perma-Series® Packers

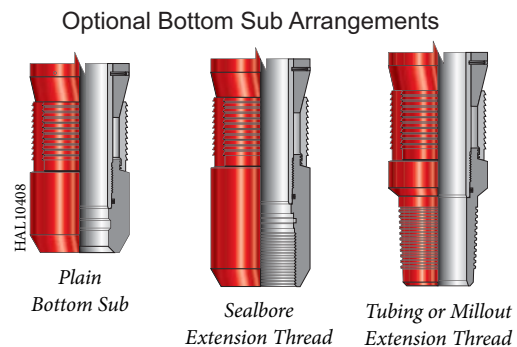
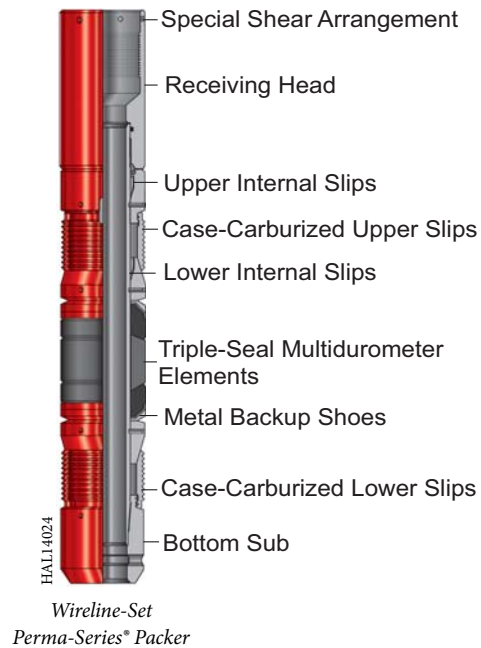
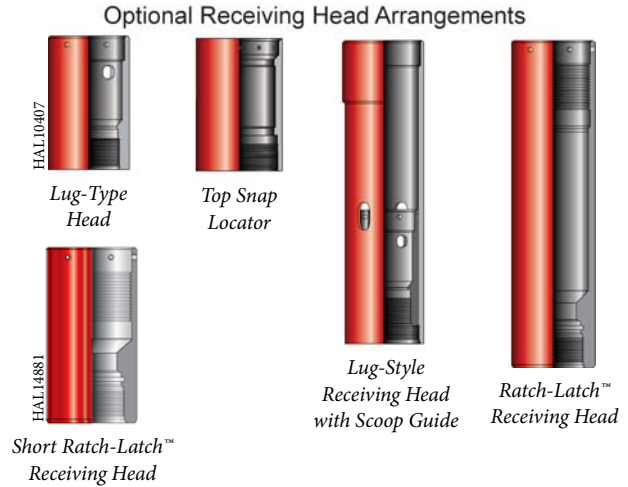
Halliburton wireline-set Perma-Series® packers are effective differential production packers for single or multizone completions. These permanent packers may be slickline, electric line, or hydraulic set on the workstring. They are also designed to leave the packer bore free of all setting devices and maintain a large fluid-bypass area through the packer. Operating envelopes are a graphical representation of the safe combination of differential pressure and applied tubing loads and are available for most Perma-Series packers.

Applications

- High formation, unloading differential, or stimulation pressure areas
- Pulling tubing without unseating packers
- High bottomhole temperatures
- Downhole conditions demanding special running clearance
- Twin-flow chemical injections systems

Features

- Upper internal slip design retains the setting force throughout the packer assembly; lower internal slips prevent upward movement of the upper wedge while running
- Internal locking-slip ring aids in fast removal, saving rig time if milling becomes necessary
- Triple-seal multidurometer seal element
- Metal backup shoes
- Case-carburized lower slips
- Bottom sub
- Contoured c-slip design provides maximum casing contact with uniform slip engagement
- Case-carburized slips provide controlled hardness depth and maximum hardness across tooth section
- Core of slip retains original metallic structure for easy milling



Benefits

- Maintains a large fluid-bypass area through the packer
- Saves rig time
- Helps ensure easy milling because the slip core retains its original metallic structure
- Provides flexibility with multiple receiving head and bottom sub options
- Can be adapted for high-pressure/high-temperature applications

Options

- Optional receiving head arrangements include short Ratch-Latch™ assembly or lug-style receiving head for vertical wells, short Ratch-Latch assembly or lug-style receiving head with scoop guide for deviated wells, and Ratch-Latch receiving head for high tensile strength or maximum flow capability
- Optional bottom sub arrangements include plain bottom sub for sump packer applications, sealbore extension thread for additional seal surface, and tubing or millout extension thread for tailpipe accessories
- Seal element package is available in standard up to 325°F (163°C) and special up to 550°F (288°C) packages

Schematic 1

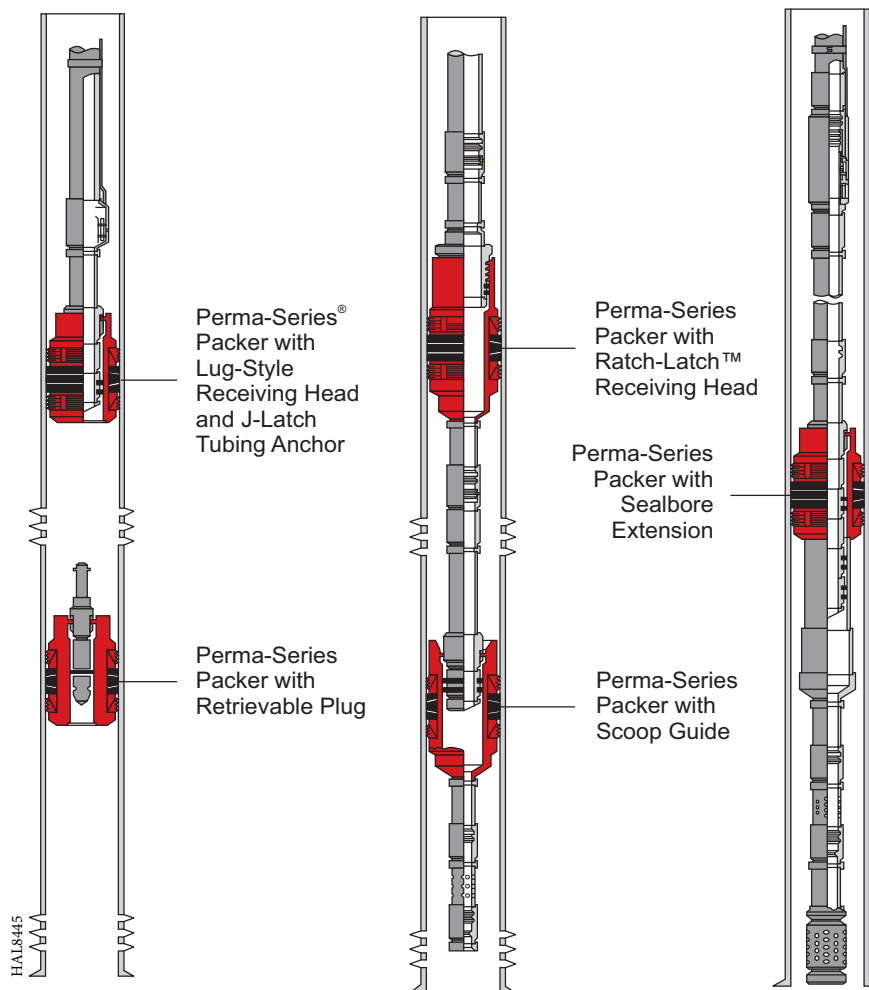
Serving as temporary bridge plugs, tubing anchors, and test tools

Schematic 2

Isolating multizone completions

Schematic 3

Permitting tubing movement and deploying tubing-conveyed perforating systems



Ordering Information

Specify: casing size and weight; packer bore, receiving head arrangement (short Ratch-Latch™ or lugs with a scoop guide, or Ratch-Latch), bottom sub arrangement (blank, threaded for tubing, sealbore or extensions, or millout extensions), bottom thread, if applicable, tubing size; weight; grade/material; and thread; service environment (standard, %H₂S, %CO₂, amines/other chemicals, chloride content, pressures, temperatures, etc.); maximum differential pressure requirement.

Part Number Prefix: 212AWB, 212BWB, 212AWD, 212BWD, 212TWB, 212TWD, 212TWR, 212TWS

Wireline-Set Perma-Series® Production Packers

Casing Size		Casing Weight		Packer OD		Sealbore ID		Seal Unit ID*					
in.	mm	lb/ft	kg/m	in.	mm	in.	mm	in.	mm	in.	mm	in.	mm
4 1/2	114.30	9.5-12.6	14.14-18.75	3.79	96.27	2.55	64.90	1.68	42.60	1.81	45.97	1.920	48.77
		9.5-12.6**	14.14-18.75	3.79	96.27	2.75	69.85	1.91	48.51				
		9.5-13.5	14.14-20.09	3.72	94.49	2.37	60.33	1.53	38.86	1.74	44.07		
		13.5-15.1	20.09-22.47	3.64	92.46	2.55	64.90	1.68	42.60	1.81	45.97	1.920	48.77
		13.5-15.1**	20.09-22.47	3.64	92.46	2.75	69.85	1.91	48.51				
		13.5-16.8	20.09-25.00	3.60	91.44	2.37	60.33	1.53	38.86	1.74	44.07		
5	127.00	15-21	22.32-31.25	3.96	100.58	2.55	64.90	1.68	42.60	1.81	45.97	1.920	48.77
		15-21**	22.32-31.25	3.96	100.58	3.12	79.25	2.39	60.71				
		23.2-24.2	34.53-36.01	3.79	96.27	2.55	64.90	1.68	42.60	1.81	45.97	1.920	48.77
		23.2-24.2**	34.53-36.01	3.79	96.27	2.75	69.85	1.91	48.51				
5 1/2	139.70	13-20	19.35-29.76	4.54	115.32	2.75	69.85	1.83	46.48	1.93	48.95		
		13-20	19.35-29.76	4.54	115.32	3.00	76.20	1.93	48.95	2.24	56.90	2.330	59.18
		13-20**	19.35-29.76	4.54	115.32	3.50	88.90	2.51	63.75				
		20-26	29.76-38.69	4.36	110.74	2.75	69.85	1.83	46.48	1.93	48.95		
		20-26	29.76-38.69	4.36	110.74	3.00	76.20	1.93	48.95	2.24	56.90	2.330	59.18
		20-26**	29.76-38.69	4.36	110.74	3.50	88.90	2.51	63.75				
		23-28.4	34.23-42.26	4.26	108.20	3.00	76.20	1.93	48.95	2.24	56.90	2.330	59.18
		23-28.4**	34.23-42.26	4.26	108.20	3.12	79.25	2.39	60.71				
6 5/8	168.28	17-32	25.30-47.62	5.47	138.89	2.75	69.85	1.83	46.48				
		17-32	25.30-47.62	5.47	138.89	3.25	82.55	2.35	59.69				
		17-32**	25.30-47.62	5.47	138.89	4.25	107.95	3.25	82.55	1.93	48.95		
		20-24	29.76-35.72	5.69	144.45	4.00	101.60	2.97	75.44				
		20-24**	29.76-35.72	5.69	144.45	4.75	120.65	3.85	97.79				
7	177.80	17-20	25.30-29.76	6.25	158.75	4.00	101.60	2.97	75.44				
		17-23	25.30-34.23	6.18	156.97	3.25	82.55	2.35	59.69				
		17-23**	25.30-34.23	6.18	156.97	4.25	107.95	3.25	82.55				
		20-26	29.76-38.69	6.00	152.40	4.00	101.60	2.97	75.44				
		23-38	34.23-56.55	5.69	144.45	2.75	69.85	1.83	46.48				
		23-38	34.23-56.55	5.69	144.45	3.25	82.55	2.35	59.69				
		23-38**	34.23-56.55	5.69	144.45	4.25	107.95	3.25	82.55	1.93	48.95		
		23-32	34.23-47.62	5.88	149.23	4.00	101.60	2.97	75.44				
		23-32**	34.23-47.62	5.88	149.23	5.00	127.00	3.93	100.03				
		32-38	47.62-56.55	5.69	144.45	4.00	101.60	2.97	75.44				
		32-38**	47.62-56.55	5.69	144.45	4.75	120.65	3.85	97.79				
		32-45.4	47.62-67.56	5.47	138.89	2.75	69.85	1.83	46.48				
		32-45.4	47.62-67.56	5.47	138.89	3.25	82.55	2.35	59.69				
		32-45.4**	47.62-67.56	5.47	138.89	4.25	107.95	3.25	82.55				
7 5/8	193.68	26.4-39	39.29-58.04	6.38	161.93	2.75	69.85	1.83	46.48				
		26.4-39	39.29-58.04	6.38	161.93	3.25	82.55	2.35	59.69				
		26.4-39	39.29-58.04	6.38	161.93	4.00	101.60	2.97	75.44				
		33.7-39	50.15-58.04	6.38	161.93	4.50	114.30	2.97	75.44				
		33.7-39**	50.15-58.04	6.38	161.93	5.25	133.35	4.00	101.60	1.92	48.77		
		33.7-45.3	50.15-67.41	6.18	156.97	3.25	82.55	2.35	59.69				
		33.7-45.3**	50.15-67.41	6.18	156.97	4.25	107.95	3.25	82.55				
		33.7-45.3	50.15-67.41	6.25	158.75	4.00	101.60	2.97	75.44				
8 5/8	219.08	36-49	53.57-72.92	7.26	184.40	4.00	101.60	2.97	75.44				
9 5/8	244.48	36-47	53.57-69.94	8.42	213.87	6.00	152.40	4.86	123.44				
		36-59.4	53.57-88.40	8.12	206.25	3.25	82.55	2.35	59.69				
		36-59.4	53.57-88.40	8.12	206.25	4.00	101.60	2.97	75.44				
		36-59.4	53.57-88.40	8.12	206.25	5.00	127.00	3.85	97.79				
		36-59.4**	53.57-88.40	8.12	206.25	6.50	165.10	5.00	127.00				
		40-53.5	59.53-79.62	8.22	208.79	6.00	152.40	4.86	123.44				
		40-53.5**	59.53-79.62	8.22	208.79	7.00	177.80	6.13	155.58				
		47-58.4	69.94-86.90	8.16	223.11	6.00	152.40	4.86	123.44				

*Thread type and size control seal unit ID

**Upper Ratch-Latch™ head style only

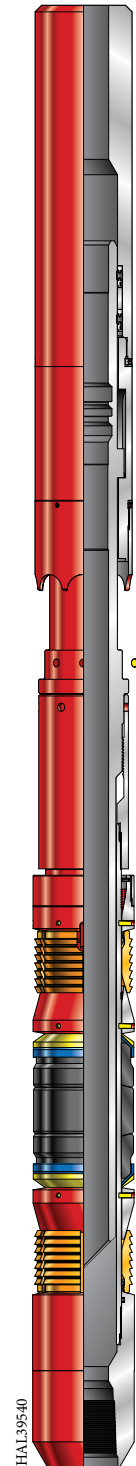
TWS Wireline-Set, Perma-Series® Production Packers

The Halliburton TWS Permanent Packer is a permanent millable packer set by wireline in the well casing to divert casing-to-tubing flow. A Ratch-Latch™ thread is provided within the head. The packer has an internal slip under the top wedge that guards against presetting the packer while running it downhole. Once the packer is set, it cannot be released. However, the packer can be milled out. The bottom sub contains a tubing connection for adding accessories below the packer.

If pressure differential reversal occurs after the packer is set, the internal slip inhibits movement of the mandrel. The lug on the top slip support and the internal slip keep the packer from rotating during milling operations.

Anchor Latch Kit

The Halliburton Seal Anchor is an economical threaded-type latch installed at the surface in a hydraulic-set packer with a Ratch-Latch™ head. It is used to anchor the tubing to the packer and tailpipe by locking into the Ratch-Latch head of the packer with left-hand rotation. Disengagement from the packer is accomplished by right-hand rotation of the tubing. Once the seal anchor is released from the packer, a Ratch-Latch locator sub is used to re-latch the packer.



*TWS Packer with
OSA Seal Anchor*

4 1/2-in. 9.5-12.6 lb and 5 -in. 23.2-24.2 lb 2.55 Bore TWS Packer and Accessories

Packer and Accessories	Maximum OD	Sealbore ID	Minimum ID	Short Ratch-Latch™ Thread	Top Thread Connection	Bottom Thread Connection	Length	Mandrel/Seal Material	Service/ Temperature Rating*	O-ring Connection
TWS Permanent Packer (80 ksi)	3.79	2.550	1.92	3.095-8	n/a	2 3/8 API-EU box	30.32	Alloy 80 ksi	H ₂ S/325°F	***
TWS Permanent Packer (125 ksi)	3.79	2.550	1.92	3.095-8	n/a	2 3/8 API-EU box	30.32	Alloy 125 ksi	STD/325°F	***
Setting Adapter Kit	3.780	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
Connector Blank (field conversion)	3.79 blank	n/a	1.92 blank	n/a	2 15/16 12 UN	Blank	7.25	Alloy 80 ksi	H ₂ S	91QV1234-H†
Ratch-Latch	3.094	n/a	1.92	3.095-8	2 3/8 API-EU pin	2 1/4 12 UN box	15.00	Alloy 80 ksi	H ₂ S	91QV1138-H†
Ratch-Latch (125 ksi)	3.094	n/a	1.92	3.095-8	2 3/8 API-EU pin	2 1/4 12 UN box	15.00	Alloy 125 ksi	STD	91QV1138-H/ 91QBP1138-B backup ring†
Ratch-Latch Premium**	3.094	n/a	1.92	3.095-8	2 3/8 4.6 New VAM pin	2 1/4 12 UN box	15.00	Alloy 80 ksi	H ₂ S	n/a
Ratch-Latch Premium** (125 ksi)	3.094	n/a	1.92	3.095-8	2 3/8 4.6 New VAM pin	2 1/4 12 UN box	15.00	Alloy 125 ksi	STD	91QV1138-H/ 91QBP1138-B backup ring†
MSN Seal Unit (80 ksi)	2.603 (seals)	n/a	1.92	n/a	2 1/4 12 UN pin	2 1/4 12 UN pin	18.13/ 13.90 (makeup)	Alloy 80 ksi/ Seal Nitrile	STD/325°F	n/a
MSN Seal Unit (125 ksi)	2.603 (seals)	n/a	1.92	n/a	2 1/4 12 UN pin	2 1/4 12 UN pin	18.13/ 13.90 (makeup)	Alloy 125 ksi/ Seal Nitrile	STD/325°F	n/a
RTR Seal Unit (80 ksi)	2.573 (seals)	n/a	1.92	n/a	2 1/4 12 UN pin	2 1/4 12 UN pin	18.13/ 13.90 (makeup)	Alloy 80 ksi/ Seal RTR	H ₂ S/350°F	n/a
RTR Seal Unit (125 ksi)	2.573 (seals)	n/a	1.92	n/a	2 1/4 12 UN pin	2 1/4 12 UN pin	18.13/ 13.90 (makeup)	Alloy 125 ksi/ Seal RTR	STD/350°F	n/a
AFLAS® V-Pkg (4 req field conversion)	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	H ₂ S/ 100-400°F	n/a
Muleshoe Guide	2.52	n/a	1.92	n/a	2 1/4 12 UN box	n/a	4.95 (nom.)	Alloy 80 ksi	H ₂ S	n/a
OSA (80 ksi)	3.766	1.875	1.875	n/a	Blanked	n/a	51.95	Low Alloy 80 ksi	H ₂ S 40-350°F	n/a
OSA (110 ksi)	3.766	1.875	1.875	n/a	Blanked	n/a	51.95	Super Chrome 110 ksi	STD/CO ₂ 40-350°F	n/a

*Contact your local Halliburton representative for specifics regarding temperature ratings.

**Ratch-Latch Premium Mandrel 101411679 (212N4546)

**Ratch-Latch Premium Mandrel (125 ksi) 101445086 (212N4559)

***Applicable o-rings and backup rings are part of the packer assembly.

†O-rings and backup rings are required to seal but are not included.

AFLAS is a registered trademark of Asahi Glass Co., Ltd.

4 1/2-in. 13.5-15.1 lb 2.55 Bore TWS Packer and Accessories

Packer and Accessories	Maximum OD	Sealbore ID	Minimum ID	Short Ratch-Latch™ Thread	Top Thread Connection	Bottom Thread Connection	Length	Mandrel/Seal Material	Service/ Temperature Rating*	O-ring Connection
TWS Permanent Packer (80 ksi)	3.64	2.550	1.92	2.950-8	N/A	2 3/8 API-EU Box	30.32	Alloy 80 ksi	H ₂ S/325°F	***
TWS Permanent Packer (125 ksi)	3.64	2.550	1.92	2.950-8	N/A	2 3/8 API-EU Box	30.32	Alloy 125 ksi	STD/325°F	***
Setting Adapter Kit	3.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Connector Blank (field conversion)	3.64 Blank	N/A	1.92 Blank	N/A	2 15/16 12 UN	Blank	8.25 (nom.)	Alloy 80 ksi	H ₂ S	91QV1234-H†
Ratch-Latch	2.875	N/A	1.92	2.950-8	2 3/8 API-EU Pin	2 1/4 12 UN box	15.00	Alloy 80 ksi	H ₂ S	91QV1138-H†
Ratch-Latch (125 ksi)	2.875	N/A	1.92	2.950-8	2 3/8 API-EU Pin	2 1/4 12 UN box	15.00	Alloy 125 ksi	STD	91QV1138-H/ 91QBP1138-B backup ring†
Ratch-Latch Premium**	2.875	N/A	1.92	2.950-8	3 3/8 4.6 New VAM Pin	2 1/4 12 UN Box	15.00	Alloy 80 ksi	H ₂ S	91QV1138-H†
Ratch-Latch Premium** (125 ksi)	2.875	N/A	1.92	2.950-8	3 3/8 4.6 New VAM Pin	2 1/4 12 UN Box	15.00	Alloy 25 ksi	STD	91QV1138-H/ 91QBP1138-B backup ring†
MSN Seal Unit (80 ksi)	2.603 (Seals)	N/A	1.92	N/A	2 1/4 12 UN Pin	2 1/4 12 UN Pin	18.13/ 13.90 (Makeup)	Alloy 80 ksi/ Seal Nitrile	STD/325°F	N/A
MSN Seal Unit (125 ksi)	2.603 (Seals)	N/A	1.92	N/A	2 1/4 12 UN Pin	2 1/4 12 UN Pin	18.13/ 13.90 (Makeup)	Alloy 125 ksi/ Seal Nitrile	STD/325°F	N/A
RTR Seal Unit (80 ksi)	2.753 (Seals)	N/A	1.92	N/A	2 1/4 12 UN Pin	2 1/4 12 UN Pin	18.13/ 13.90 (Makeup)	Alloy 80 ksi/ Seal RTR	H ₂ S/350°F	N/A
RTR Seal Unit (125 ksi)	2.753 (Seals)	N/A	1.92	N/A	2 1/4 12 UN Pin	2 1/4 12 UN Pin	18.13/ 13.90 (Makeup)	Alloy 125 ksi/ Seal RTR	STD/350°F	N/A
AFLAS® V-Pkg (4 req field conversion)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	H ₂ S/ 100-400°F	N/A
Muleshoe Guide	2.52	N/A	1.92	N/A	2 1/4 12 UN Box	N/A	4.95 (Nom.)	Alloy 80 ksi	H ₂ S	N/A
OSA	3.7	1.875	1.875	N/A	Blanked	N/A	33.13	Low Alloy 80 and 125 ksi	STD 40-250°F	N/A

*Contact your local Halliburton representative for specifics regarding temperature ratings.

**Ratch-Latch Premium Mandrel 101415554 (212N4551)

**Ratch-Latch Premium Mandrel (125 ksi) 101455424 (212N4563)

***Applicable o-rings and backup rings are part of the packer assembly.

†O-rings and backup rings are required to seal but are not included.

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5 1/2-in. 20-26 lb 2.75 Bore TWS Packer and Accessories

Packer and Accessories	Maximum OD	Sealbore ID	Minimum ID	Short Ratch-Latch™ Thread	Top Thread Connection	Bottom Thread Connection	Length	Mandrel/Seal Material	Service/ Temperature Rating*	O-ring Connection
TWS Permanent Packer (80 ksi)	4.36	2.750	1.92	3.690-8	N/A	2 3/8 API-EU box	32.59	Alloy 80 ksi	H ₂ S/325°F	***
TWS Permanent Packer (125 ksi)	4.36	2.750	1.92	3.690-8	N/A	2 3/8 API-EU box	32.59	Alloy 125 ksi	STD/325°F	***
Setting Adapter Kit	4.34	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Connector Blank (field conversion)	4.36 Blank	N/A	1.92 Blank	N/A	3 3/8-12 UN	Blank	8.25 (Nom.)	Alloy 80 ksi	H ₂ S	91QV1236-H†
Ratch-Latch	3.69	N/A	1.92	3.690-8	2 3/8 API-EU pin	2 3/8 API-NU box	16.00	Alloy 80 ksi	H ₂ S	N/A
Ratch-Latch (125 ksi)	3.69	N/A	1.92	3.690-8	2 3/8 API-EU pin	2 3/8 API-NU box	16.00	Alloy 125 ksi	STD	N/A
Ratch-Latch Premium**	3.69	N/A	1.92	3.690-8	2 3/8 4.6 New VAM pin	2 3/8 API-NU box	16.00	Alloy 80 ksi	H ₂ S	N/A
Ratch-Latch Premium** (125 ksi)	3.69	N/A	1.92	3.690-8	2 3/8 4.6 New VAM pin	2 3/8 API-NU box	16.00	Alloy 125 ksi	STD	N/A
MSN Seal Unit (80 ksi)	2.795 (Seals)	N/A	1.92	N/A	2 3/8 API-NU pin	2 3/8 API-NU pin	18.13/ 13.90 (makeup)	Alloy 80 ksi/ Seal Nitrile	STD/325°F	N/A
MSN Seal Unit (125 ksi)	2.795 (Seals)	N/A	1.92	N/A	2 3/8 API-NU pin	2 3/8 API-NU pin	18.13/ 13.90 (Makeup)	Alloy 125 ksi/ Seal Nitrile	STD/325°F	N/A
RTR Seal Unit (80 ksi)	2.768 (Seals)	N/A	1.92	N/A	2 3/8 API-NU pin	2 3/8 API-NU pin	18.13/ 16.50 (Makeup)	Alloy 80 ksi/ Seal RTR	H ₂ S/350°F	N/A
RTR Seal Unit (125 ksi)	2.768 (Seals)	N/A	1.92	N/A	2 3/8 API-NU pin	2 3/8 API-NU pin	18.13/ 16.50 (Makeup)	Alloy 125 ksi/ Seal RTR	STD/350°F	N/A
AFLAS® V-Pkg (4 req field conversion)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	H ₂ S/ 100-400°F	N/A
Muleshoe Guide	2.71	N/A	1.92	N/A	2 3/8 API-NU box	N/A	4.95 (Nom.)	Alloy 80 ksi	H ₂ S	N/A
OSA (80 ksi)	4.390	1.875	1.875	N/A	Blanked	N/A	51.95	Low Alloy 80 ksi	H ₂ S/ 40-350°F	N/A
OSA (110 ksi)	4.390	1.875	1.875	N/A	Blanked	N/A	33.13	Super Chrome 110 ksi	STD/CO ₂ 40-350°F	N/A

*Contact your local Halliburton representative for specifics regarding temperature ratings.

**Ratch-Latch Premium Mandrel 101419818 (212N5534)

**Ratch-Latch Premium Mandrel (125 ksi) 101444322 (212N5556)

***Applicable o-rings and backup rings are part of the packer assembly.

†O-rings and backup rings are required to seal but are not included.

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7-in. 23-38 lb 2.750 Bore TWS Packer and Accessories

Packer and Accessories	Maximum OD	Sealbore ID	Minimum ID	Short Ratch-Latch™ Thread	Top Thread Connection	Bottom Thread Connection	Length	Mandrel/Seal Material	Service/ Temperature Rating*	O-ring Connection
TWS Permanent Packer (80 ksi)	5.687	2.750	1.92	3.690-8	N/A	2 3/8 API-EU Box	38.69	Alloy 80 ksi	H ₂ S/325°F	***
TWS Permanent Packer (125 ksi)	5.687	2.750	1.92	3.690-8	N/A	2 3/8 API-EU Box	38.69	Alloy 125 ksi	STD/325°F	***
Setting Adapter Kit	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Connector Blank (field conversion)	5.687 Blank	N/A	1.92 Blank	n/a	3 3/4 10 UNS	Blank	8.25 (Nom.)	Alloy 80 ksi	H ₂ S	91QV1342-H†
Ratch-Latch	3.69	N/A	1.92	3.690-8	2 3/8 API-EU Pin	2 3/8 API-NU Box	16.00	Alloy 80 ksi	H ₂ S	N/A
Ratch-Latch (125 ksi)	3.69	N/A	1.92	3.690-8	2 3/8 API-EU Pin	2 3/8 API-NU Box	16.00	Alloy 125 ksi	STD	N/A
Ratch-Latch Premium**	3.69	N/A	1.92	3.690-8	2 3/8 4.6 New VAM Pin	2 3/8 API-NU Box	16.00	Alloy 80 ksi	H ₂ S	N/A
Ratch-Latch Premium** (125 ksi)	3.69	n/a	1.92	3.690-8	2 3/8 4.6 New VAM Pin	2 3/8 API-NU Box	16.00	Alloy 125 ksi	STD	N/A
MSN Seal Unit (80 ksi)	2.795 (Seals)	n/a	1.92	n/a	2 3/8 API-NU Pin	2 3/8 API-NU Pin	18.13/ 13.90 (Makeup)	Alloy 80 ksi/ Seal Nitrile	STD/325°F	N/A
MSN Seal Unit (125 ksi)	2.795 (Seals)	N/A	1.92	N/A	2 3/8 API-NU Pin	2 3/8 API-NU Pin	18.13/ 13.90 (Makeup)	Alloy 125 ksi/ Seal Nitrile	STD/325°F	N/A
RTR Seal Unit (80 ksi)	2.768 (Seals)	N/A	1.92	N/A	2 3/8 API-NU Pin	2 3/8 API-NU Pin	18.13/ 16.50 (Makeup)	Alloy 80 ksi/ Seal RTR	H ₂ S/350°F	N/A
RTR Seal Unit (125 ksi)	2.768 (Seals)	N/A	1.92	N/A	2 3/8 API-NU Pin	2 3/8 API-NU Pin	18.13/ 16.50 (Makeup)	Alloy 125 ksi/ Seal RTR	STD/350°F	N/A
AFLAS® V-Pkg (4 req field conversion)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	H ₂ S/ 100-400°F	N/A
Muleshoe Guide	2.71	N/A	1.92	N/A	2 3/8 API-NU Box	N/A	4.95 (Nom.)	Alloy 80 ksi	H ₂ S	N/A
OSA	5.515	1.875	1.875	N/A	Blanked	N/A	54.46	Low Alloy 80 ksi	H ₂ S/ 40-350°F	N/A

*Contact your local Halliburton representative for specifics regarding temperature ratings.

**Ratch-Latch Premium Mandrel 101419818 (212N5534)

**Ratch-Latch Premium Mandrel (125 ksi) 101444322 (212N5556)

***Applicable o-rings and backup rings are part of the packer assembly.

†O-rings and backup rings are required to seal but are not included.

AFLAS is a registered trademark of Asahi Glass Co., Ltd.

Setting Adapter Kits and Hydraulic Setting Tools

Setting Adapter Kits

Halliburton adapter kits are designed to be attached to the packer and either a conventional explosive-type wireline setting tool or a hydraulic setting tool. When the setting tool is activated, the setting adapter kit sets the packer. When prescribed setting force is applied to the packer, setting pins in the wireline adapter kit separate to release setting equipment from the packer so it can be retrieved.

Hydraulic Setting Tools

Hydraulic setting tools allow packers normally run and set on electric wireline to be run and set on a workstring instead.

Various configurations of setting adapter kits and setting tools are possible. Figure 1 features a standard electric wireline adapter kit where the packer and adapter kit are connected directly to the hydraulic setting tool.

In Figure 2, the packer is connected directly to the hydraulic setting tool so no adapter kit is required. This setting tool incorporates a seal in the packer bore that enables an annulus test against the packer before expending the ball and seat.

Figure 3 shows a hydraulic setting tool designed to be used only on certain Perma-Series® packers. It provides an additional anti-preset mechanism by limiting downward movement of the tube guide and top slip carrier until the hydraulic setting mechanism is activated. A stabilizer bar with pack-off is provided to test the annulus against the packer before shearing off the packer and expending the ball seat. No tubing rotation is required for disengagement from the packer.

In all three cases, the setting ball and seat are set to shear at a predetermined pressure, which exposes a port to allow tubing-to-casing communication. The operator can then pull a dry workstring.



Figure 1

Figure 2

Figure 3

Hydraulic Setting Tools

Hydraulic-Set Perma-Series® Packers

The Halliburton hydraulic-set Perma-Series® packer is a single-trip permanent millable production version of the proven Perma-Series packer. It is ideal for highly deviated and/or single-trip production and injection applications.

This packer includes a one-piece mandrel and sealbore, which eliminates a potential leak path. It has a low profile for greater running clearance to help reduce problems that may occur when running in highly deviated and horizontal wells. Premium, metal-to-metal (MTM) thread connections add to the dependability of the packer.

Ease of use has been proven in flanged-up, single-trip completions common in offshore operations. This packer sets with low pressure, major components are rotationally locked, and materials are suitable for milling. No tubing manipulation is required to set the packer, and the hydraulic-set Perma-Series packer is compatible with standard Halliburton accessory equipment.

An upper polished bore receptacle (PBR) or overshot tubing seal divider can be connected to the top of the packer with either a seal anchor or Ratch-Latch™ assembly. The upper PBR assembly can be retrieved with either connection.

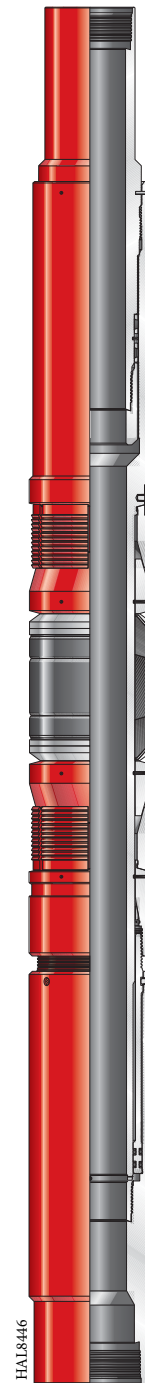
The internal seal assembly is run in place and shear-pinned to allow space out and fluid displacement.

Applications

- Highly deviated production and/or injection areas
- Single-trip production and/or injection wells
- Offshore completions that require a single-trip packer

Features

- One-piece mandrel and sealbore
- Case-carburized slips
- Metal-to-metal backup shoes with 360° casing wall contact
- Ratch-Latch™ sealbore with seal anchor
- Triple-seal multidurometer element package
- Single-piece bottom sub and piston housing
- MTM premium thread connections standard on the packer, seal anchor, and Ratch-Latch assembly
- Minimal number of pressure-containing o-ring seals
- Large bore through packer and seal assembly
- Low profile
- Self-contained hydraulic-setting system
- No mandrel movement



Hydraulic-Set
Perma-Series® Packer

Benefits

- Potential leak paths minimized
- No tubing manipulation required during setting
- Greater running clearance because of low profile of external components
- Compatible with standard accessory equipment
- Easy to use in flanged-up completions common to offshore operations

Options

- Lower sealbore (short) Ratch-Latch™ head
- Upper sealbore Ratch-Latch head (largest bore)
- Metal-to-metal backoff sub-head
- Compression-when-set capability
- No mandrel movement

Hydraulic-Set Perma-Series® Production Packers

Casing Size		Casing Weight		Packer OD		Packer Sealing Bore							
						Upper Sealbore		Packer and Seal Assembly Minimum ID*		Lower Sealbore		Packer and Seal Assembly Minimum ID*	
in.	mm	lb/ft	kg/m	in.	mm	in.	mm	in.	mm	in.	mm	in.	mm
5	127.00	15-21	22.32-31.25	3.96	100.58	3.12	79.25	1.91	48.51	—	—	—	—
5 1/2	139.70	13-20	19.35-29.76	4.54	115.32	3.50	88.90	2.36	59.94	2.38	60.33	1.74	44.07
		20-23	29.76-34.22	4.45	113.03	3.50	88.90	2.36	59.94	2.38	60.33	1.74	44.07
6 5/8	168.28	17-32	25.30-47.62	5.47	138.89	4.25	107.95	3.17	80.52	3.25	82.55	2.35	59.69
		17-20	25.30-29.76	5.88	149.23	5.00	127.00	3.88	98.43	4.00	101.60	2.88	73.03
7	177.80	17-23	25.30-34.22	6.18	156.97	4.25	107.95	3.17	80.52	3.25	82.55	2.35	59.69
		17-20	25.30-29.76	6.25	158.75	5.00	127.00	3.88	98.43	4.00	101.60	2.88	73.03
		23-32	34.22-47.62	5.88	149.23	5.00	127.00	3.88	98.43	4.00	101.60	2.88	73.03
		23-38	34.22-56.55	5.69	144.45	4.25	107.95	3.17	80.52	3.25	82.55	2.35	59.69
		32-44	47.62-65.48	5.47	138.89	4.25	107.95	3.17	80.52	3.25	82.55	2.35	59.69
7 5/8	193.68	26.4-39	39.29-58.04	6.38	161.93	5.00	127.00	3.88	98.43	4.00	101.60	2.88	73.03
		33.7-45.3	50.15-67.41	6.25	158.75	5.00	127.00	3.88	98.43	4.00	101.60	2.88	73.03
		33.7-47.3	50.15-70.39	6.18	156.97	4.25	107.95	3.17	80.52	3.25	82.55	2.35	59.69
9 5/8	244.48	36-59.4	53.57-88.40	8.13	206.38	6.50	165.10	4.85	123.19	5.00	127.00	3.88	98.43
		40-47	59.53-69.94	8.42	213.87	7.00	177.80	6.00	152.40	6.00	152.40	4.85	123.19
		53.50	79.62	8.31	211.07	7.25	184.15	6.02	152.98	6.00	152.40	4.85	123.19
10 3/4	273.05	55.5-60.7	82.59-90.33	9.28	235.71	7.00	177.80	6.00	152.40	6.00	152.40	4.85	123.19
		60.7-65.7	90.33-97.77	9.28	235.71	7.00	177.80	6.00	152.40	6.00	152.40	4.85	123.19

*Minimum IDs may vary depending on tubing thread connection. Other sizes available on request.

Ordering Information

Specify: casing size and weight, packer bore, bottom thread, minimum ID, locator type (Ratch-Latch™ or seal anchor), service environment (%H₂S, %CO₂, temperature, pressures, inhibitors, possibility of acid treatments, etc.), differential pressure rating requirements, special material requirements.

Part Number Prefix: 212MHR, 212MHB, 212THT, 212MNR, 212MNB, 212TNT

Hydrostatic-Set Permanent Production Packers

The Halliburton hydrostatic-set permanent packer is a single-string packer ideal for highly deviated well applications or where conventional packer plug setting techniques are not possible. Based on the most advanced permanent packers available in the industry, the Perma-Series® packer, this technology saves both time and money, and reduces intervention risk.

Applications

- Highly deviated wells or where conventional packer plug setting techniques are not possible (non-perforated wells)

Features

- Holds pressure above and below
- Advanced rupture disk technology (available in 500 psi increments with temperature compensation curves)
- Dual rupture disks for reliability
- Built-in anti-preset features (hydrostatic module biased in the closed position)
- Independent contingency setting mechanism
- Can be set in heavy completion/drilling fluids

Benefits

- Reduces packer setting costs
- Reduces rig time
- Uses available hydrostatic pressure
- Minimizes wellbore intervention to set packer setting plug

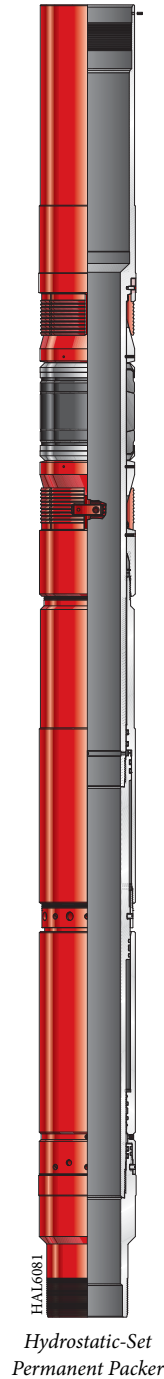
Operation

- Run on tubing string
- Set using available hydrostatic wellbore pressure and applied surface pressure without plugs
- Removed with conventional milling tools

Hydrostatic-Set Permanent Packer

Casing Size		Weight Range		Maximum OD		Minimum ID	
in.	mm	lb/ft	kg/m	in.	mm	in.	mm
5	127.00	24.1	35.86	3.84	97.54	2.19	55.62
7	177.80	29-32	43.15-47.62	5.90	149.86	3.87	98.29
		32-35	47.62-52.08	5.80	147.32	3.83	97.28
7 5/8	193.68	26.4-39	39.29-58.04	6.37	161.93	3.87	98.29
		33.7-39	50.15-58.04				
9 5/8	244.50	43.5-53.5	64.73-79.62	8.31	211.07	4.85	123.18
		43.5-47	64.73-69.94	8.42	213.87	6.00	152.39
		53.5	79.62	8.33	211.58	6.00	152.39
9 7/8	250.83	62.7-68	93.30-101.18	8.31	211.07	4.46	113.28

Part Number Prefixes: 212HHR, 212HHB, 212HHT, 212HNR, 212HNB, 212HNT



Hydrostatic-Set Permanent Packer

Hydraulic-Set Retrievable Production Packers

This section describes Halliburton hydraulic-set AHR and AHC production packers as well as HHC and XHHC hydrostatic-set packers.

Hydraulic-Set AHR Packer

The AHR packer is a retrievable, hydraulic-set production packer set in casing to divert casing-to-tubing flow. It is set by pressuring up the tubing string against a plugging device below the packer.

Bi-directional, case-carburized slips hold the packer against well pressures from above and below. Spring-loaded internal slips maintain the packer in the set position.

The AHR packer is a straight shear-to-release packer that is convertible to cut to release. The shear release value is adjustable.

Applications

- Multizone production applications
- Single trip applications
- Multi-string completions with control line feed through
- Electric submersible pump completions

Features

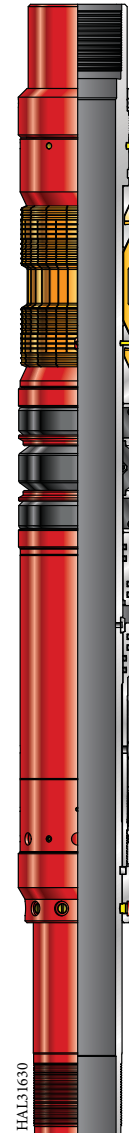
- API 11D1-V3 validated
- Single-trip capability
- Barrel slip design
- Single-piece mandrel construction
- Triple seal multidurometer element
- Straight pull or cut to release

Benefits

- Multiple packers can be set close together
- Easily converted between straight pull to release or cut to release
- Meets API 11D1 specifications
- Various casing sizes available

Hydraulic Set AHR Packer

Casing Size		Casing Weight		Packer OD		Packer ID	
in.	mm	lb/ft	kg/m	in.	mm	in.	mm
5 1/2	139.7	20 - 23	29.76 - 34.22	4.50	114.3	2.35	59.69
7	177.8	20 - 26	29.76 - 38.69	6.1	154.93	3.875	98.42
		23 - 29	34.22 - 43.15	6	152.39	3.875	98.42
		26 - 32	38.69 - 47.62	5.93	150.62	3.875	98.42
9 5/8	244.48	40 - 47	59.53 - 69.94	8.47	215.13	4.73	120.14
		47 - 53.5	69.94 - 79.62	8.31	211.07	4.73	120.14



Hydraulic-Set AHR Packer

Hydraulic-Set AHC Packer

The AHC packer is a hydraulic-set, cut-to-release production packer. The unique design allows either the packer ID to be maximized or to have multiple bores or control line feed-throughs. The large bore and no mandrel movement when setting make this packer ideally suited for multizone stacked packer applications.

The AHC packer can be configured as either a straight-shear release retrievable packer, or, where high pressures are expected, it can be set up as cut to release. In the cut-to-release design, the packer mandrel is cut, and the packer with tailpipe is retrieved without leaving any packer components in the well.

Applications

- Multizone production applications
- Single-trip completions
- Multi-string completions with control line feed-through
- Electric submersible pump completions
- High-pressure and TCP applications with cut-to-release option

Features

- Large ID for maximum tubing size or multiple bores
- Single-trip capability
- Triple-seal multidurometer element package
- Bi-directional, case-carburized external slips
- Single-piece mandrel construction
- Straight pull (in some cases) or cut to release

Benefits

- Does not require tubing manipulation to set
- Multiple packers can be set close together
- Can be easily redressed with a minimum of parts
- Works well in all types of casing
- Maximum protection from leaks

Options

- Premium metallurgy and elastomers
- Upper tubing connection or Ratch-Latch™ head
- Optional element packages for high-temperature or completion fluid compatibility
- Straight shear or cut to release

Hydraulic-Set AHC Packer

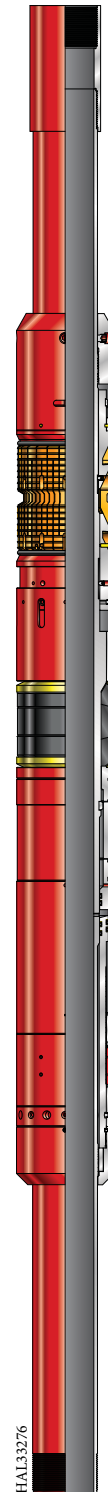
Casing Size		Casing Weight		Packer OD		Packer ID	
in.	mm	lb/ft	kg/m	in.	mm	in.	mm
4 1/2*	114.30	9.5-11.6	14.14-17.26	3.83	97.28	1.89	47.88
5 1/2*	139.70	13-17	19.34-25.30	4.72	119.88	2.88	73.15
		20-23	29.76-34.22	4.48	113.92	2.88	73.15
7	177.80	26-29	38.69-43.16	5.96	151.38	2.95	74.93
		32-35	47.62-52.09	5.78	146.81	3.47	88.13
		32-35	47.62-52.09	5.82	147.82	3.91	99.31
7 5/8*	193.68	29-39	43.16-58.03	6.44	163.57	3.93	99.82
9 5/8	244.48	40-47	59.53-69.94	8.47	215.13	4.79	121.66
		43.5-53	64.74-78.86	8.30	210.82	4.79	121.66
13 3/8	339.73	68-72	101.20-107.14	12.13	308.10	8.52	216.41

*These sizes do not feature the barrel slip.

Ordering Information

Specify: casing size and weight; tubing size; weight; grade; and thread; service environment (standard, %H₂S, %CO₂, amines/other chemicals, chloride content, pressures, temperatures, etc.); maximum differential pressure requirement.

Part Number Prefix: 12AHC



Hydraulic-Set AHC Packer

Hydrostatic-Set Retrievable Production Packers

The Halliburton HHC and XHHC hydrostatic-set retrievable packers are single-trip packers ideal for large bore, deviated well applications or where conventional packer plug setting techniques are not possible. Based on the highly reliable AHC retrievable packer platform, the HHC packer saves both time and money, and reduces intervention risk.

Applications

- Highly deviated wells or where conventional packer plug setting techniques are not possible (non-perforated wells)
- Where a large bore retrievable packer design is required

Features

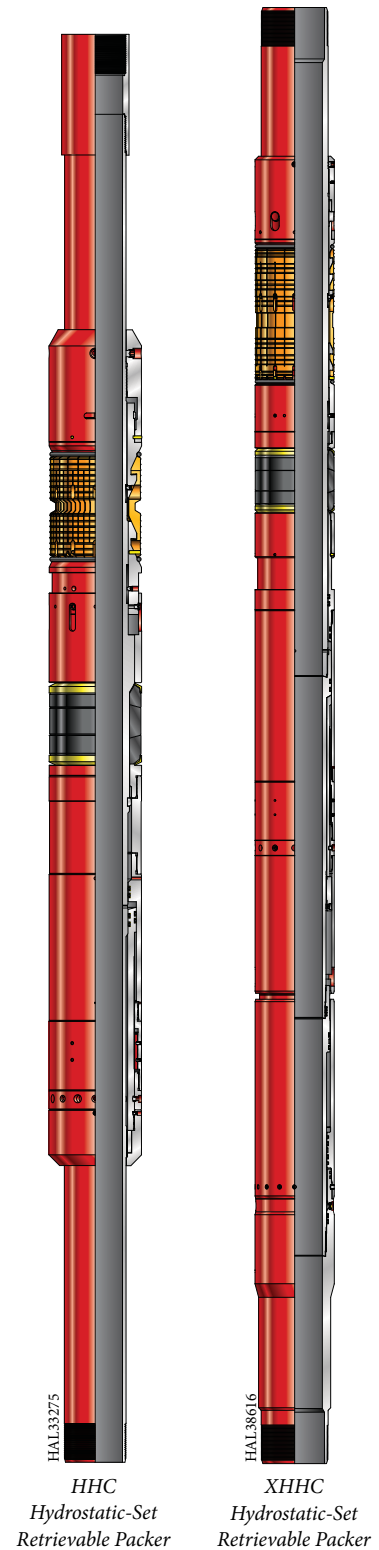
- Holds pressure above and below
- Patented barrel slips for effective load distribution
- Advanced rupture disk technology (available in 500 psi increments with temperature compensation curves)
- Dual rupture disks for reliability
- Built-in anti-preset features (hydrostatic module is biased in the closed position)
- Independent contingency setting mechanism
- Removed from the wellbore using cut-to-release technology

Benefits

- Reduces packer setting costs
- Reduces rig time
- Uses available hydrostatic pressure to initiate setting process
- Eliminates wellbore intervention to set packer setting plug
- Reliable cut-to-release design eliminates lengthy milling operations during retrieval operations

Operation

- Run on tubing string
- Set using available hydrostatic wellbore pressure and applied surface pressure without plugging devices
- Removed by cutting mandrel in predetermined cut zone using chemical or mechanical cutting tools



HHC Hydrostatic-Set Retrievable Packer

Casing Size		Weight Range		Maximum OD		Minimum ID	
in.	mm	lb/ft	kg/m	in.	mm	in.	mm
7	177.80	29 - 32	43.85 - 47.62	5.93	150.62	3.87	98.29
		32 - 35	47.62 - 52.08	5.82	147.82	3.85	97.79
9 5/8	244.50	43.5 - 53.5	64.73 - 79.62	8.31	211.06	4.79	121.41
		43.5 - 47	64.73 - 69.94	8.44	214.37	6.00	152.36
		47 - 53.5	69.94 - 79.62	8.35	212.08	6.00	152.36
10 3/4	273.05	55.5 - 60.7	82.59 - 90.33	9.45	240.03	4.59	116.59
		60.7 - 65.7	90.33 - 97.77	9.38	238.25	4.73	120.14
13 3/8	339.73	68 - 72	101.20 - 107.14	12.13	308.10	6.02	152.91

Part Number Prefix: 12HHC

XHHC Hydrostatic-Set Retrievable Packer

Casing Size		Weight Range		Maximum OD		Minimum ID		Pressure Rating		Temperature Rating	
in.	mm	lb/ft	kg/m	in.	mm	in.	mm	psi	mbar	°F	°C
9 5/8	244.50	47 - 53.5	69.94 - 79.62	8.35	212.08	5.82	147.82	12,500	861 844	400	204.44
9 7/8	250.825	62.8	93.98	8.35	212.08	5.82	147.82	12,500	861 844	400	204.44

High-Pressure/High-Temperature Packers

This section covers high-pressure/high temperature (HP/HT) hydraulic-set and wireline-set packers. These packers are typically used in HP/HT or geothermal applications. Halliburton is the industry leader in HP/HT packer development through ongoing packer research, development, and testing.

HP/HT Hydraulic-Set Permanent Packer

The Halliburton HP/HT hydraulic-set packer, designed for extreme-service conditions, is a permanent, millable packer set in the well casing to divert casing to tubing flow. In addition to high-pressure and temperature ratings, this packer offers high tubing-to-packer load ratings. It features a tubing thread top connection and is hydraulically set by applying pressure to the plugged tubing string. A special shear sleeve attached to the bottom slip carrier allows the packer to set without the need for mandrel movement. It has a low profile for greater running clearance to help reduce problems that may occur when running in highly deviated and horizontal wells. Premium metal-to-metal (MTM) thread connections add to packer dependability. Halliburton's unique double opposing barrel slips and HP/HT element systems are unsurpassed in the industry for efficient gripping and sealing against the casing wall under the toughest of conditions.

Applications

- Hot, corrosive, deep, deviated, production, or test completions
- Single-trip production and/or injection wells
- Offshore completions that require a single-trip packer

Features

- Case-carburized opposing barrel slips
- Metal-to-metal backup shoes with 360° casing wall contact for reliable element containment
- MTM thread connections
- Low profile
- Self-contained hydraulic setting system
- No mandrel-movement operation

Options

- Element package suitable for particular temperatures, pressures, and fluids
- Metallurgy suitable for particular completions, stimulations, and produced fluids
- Various latch heads available if required



HP/HT Hydraulic-Set
Permanent Packer

Ordering Information

Specify: casing size and weight; tubing size; weight; grade; and thread; service environment (standard, %H₂S, %CO₂, amines/other chemicals, chloride content, pressures, temperatures, etc.); maximum differential pressure requirement.

Part Number Prefix: 212TNT, 212HNT, 212THT, 212HHT

HP/HT Wireline-Set Permanent Packer

Halliburton high-pressure/high temperature (HP/HT) wireline-set permanent packers are designed for the most severe completion environments. These packers are based on the rugged and reliable Perma-Series® packer and are designed specifically for high-pressure/high-temperature, hostile environments. Compatible seal assemblies and various receiving heads, including metal-to-metal (MTM) latching type, are available.

Applications

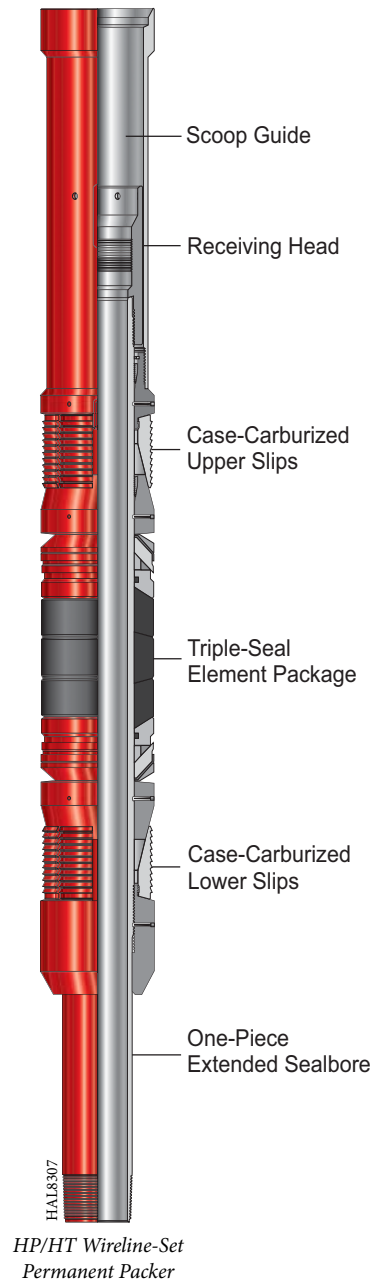
- Hot, corrosive, deep, deviated, production, or test completions
- Pressures up to 18,000 psi (124.1 MPa)
- Temperatures up to 550°F (288°C)

Features

- HP/HT triple-seal element package with metal-to-metal backup shoes with 360° casing wall contact for reliable element containment
- Contoured case-carburized external c-slips for maximum casing contact with uniform engagement to minimize casing damage or deformation
- Extended seal mandrel eliminates o-ring or thread connections for the sealbore extension
- Premium metallurgy suitable for hostile environments
- Scoophead for easy entry of seal assembly in highly deviated wellbores
- Can be set on wireline or with hydraulic setting tool

Options

- Element package suitable for particular temperatures, pressures, and fluids
- Metallurgy suitable for particular completions, stimulations, and produced fluids
- Length of sealbore
- Type of top connection
- Hydraulic-set design



HP/HT Wireline-Set
Permanent Packer

Ordering Information

Specify: casing size and weight; tubing size; weight; grade; and thread; service environment (standard, %H₂S, %CO₂, amines/other chemicals, chloride content, pressures, temperatures, etc.); maximum differential pressure requirement.

Part Number Prefix: 212NWT

High-Performance Retrievable Packers

HPH Retrievable Packer

The Halliburton hydraulic-set HPH retrievable packer combines the pressure and temperature ratings and reliability of a permanent packer with the convenience of a retrievable packer. The HPH retrievable packer incorporates unique design features that result in a retrievable packer suitable for use at differential pressures up to 15,000 psi (103.4 MPa) at temperatures up to 400°F (204°C). The barrel slip uniformly distributes the packer to casing loads. The propped element package reliably seals the high pressures and can be retrieved with the packer.

Applications

- Hot, corrosive, deep, deviated, production, or test completions
- Where permanent packer retrieval without milling is preferred
- Wells where minimum casing deformation or damage is desired
- Single-trip completions
- Multi-packer installations
- Liner top installation
- Monobore

Features

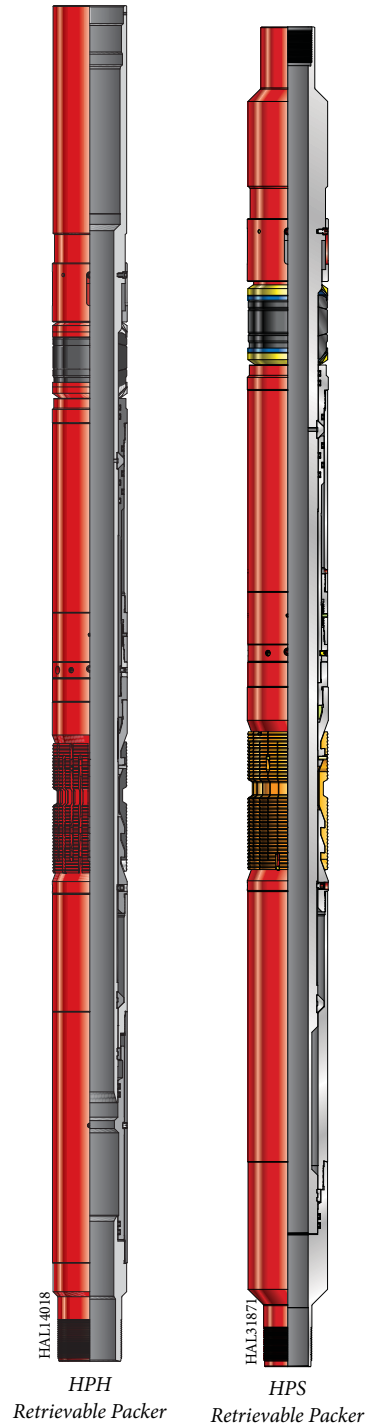
- Propped element package
- Barrel slip
- Metal-to-metal backup shoes with 360° casing wall contact for element containment and reliability
- Slips located below element package for protection from debris
- No-mandrel movement operation
- Premium metallurgy and elastomers
- Also available as cut-to-release (HPS packer)

Benefits

- No milling required to retrieve HPH packer
- Suitable for unsupported casing
- Nondamaging to casing
- Maximum sealing and retrieving reliability
- Designed to minimize rig time during completion and decompletion

Ordering Information

Specify: casing size and weight; tubing size; weight; grade; and thread; service environment (standard, %H₂S, %CO₂, amines/other chemicals, chloride content, pressures, temperatures, etc.); maximum differential pressure requirement.
Part Number Prefix: 12HPH, 12HPS



HPH Retrievable Packer

Casing Size		Casing Weight		Maximum OD		Minimum ID	
in.	mm	lb/ft	kg/m	in.	mm	in.	mm
7	177.80	26-32	38.69-47.62	5.92	150.36	3.80	96.52
		29-32	43.16-47.62	5.92	150.36	3.72	94.48
		29-32	43.16-47.62	5.92	150.36	3.88	98.55
		32-35	47.62-52.08	5.82	147.82	3.72	94.48
		35-38	52.08-56.54	5.730	145.54	2.75	69.85
7 5/8	193.68	39	58.03	6.405	162.69	2.972	75.49
		47.1	70.08	6.125	155.57	3.32	84.32
8 5/8	219.08	55.83	83.08	7.125	180.97	4.60	116.84
		57.4	85.41	7.125	180.97	3.32	84.32
		61.4	91.36	6.935	176.14	3.32	84.32
9 5/8	244.48	47-53.5	69.94-79.62	8.35	212.09	5.06	128.52
						5.80	147.32
9 7/8	250.83	62.8-66.4	93.46-98.81	8.25	209.55	4.465	113.41
10 3/4	273.05	97.1-99.5	144.50-148.07	8.64	219.46	4.65	118.11

Hydro-Set, Hydro-Release HP1 Packers

The hydro-set, hydro-release HP1 production packer is a fully retrievable, low stress, high-performance production packer able to transmit the forces generated by the tubing into the casing without overstressing the casing. It is ideal for single-string completions in hostile environments with a premium thread-up to connect directly to the tubing.

The HP1 packer sets and performs like a permanent packer, yet it can be retrieved by methods other than milling.

The packer is set in one trip on the production string by the application of tubing pressure. The packer system is released by running a dedicated punch on slickline or coiled tubing and then applying tubing pressure. Once released, the tool can be retrieved together with production tubing.

Applications

- Where retrieval of the packer without milling is preferred so as not to damage the casing or in long-reach, highly deviated, or horizontal wells where milling and subsequent cleanup operations may prove difficult and lengthy. Also, where high tensile/compressive loads and pressure differentials exist but damage to the casing must be prevented
- Where running a flanged-up and tubing-displaced completion in one trip is an advantage
- In single-trip completions incorporating TCP gun assemblies in the suspended tailpipe
- Where long-term reliability is required and well conditions dictate the use of corrosion resistant alloy materials and selected elastomers

Features

- Anti-pre-set mechanism
- Full circumference slips
- Multi-cone slips
- Release mechanism protected from produced fluids
- Hydraulic test of all seals before running in hole

Benefits

- A high-performance, retrievable packer designed to prevent pre-set during installation and to help ensure recovery during workover



*Hydro-Set, Hydro-Release
HP1 Packer*

Hydro-Set, Hydro-Release HP1 Packers

Casing/Tubing				Thread		Maximum OD		Minimum ID		Length		Pressure Rating			
Size		Weight										Above		Below	
in.	mm	lb/ft	kg/m	in.	mm	in.	mm	in.	mm	in.	mm	psi	bar	psi	bar
7	177.80	29	43.16	3 1/2	88.90	5.990	152.27	2.810	71.42	129.900	3299.46	5,000	344.50	5,000	344.50
		32	47.62	4 1/2	114.30	5.910	149.98	3.810	96.82	164.560	4179.82	7,500	516.75	7,500	516.75
		35	52.09			5.820	147.70	3.530	89.53	141.430	3592.32				
9 5/8	244.48	47	69.94	5 1/2	139.70	8.440	214.37	4.500	114.30	178.880	4543.55	5,000	344.50	5,000	344.50
				3 1/2	88.90			2.770	70.23						
				5 1/2	139.70			4.500	114.30	166.650	4232.91	6,500	447.85	6,500	447.85
								4.630	117.47	153.400	3896.36				
								4.560	115.82	172.050	4370.07				
								4.500	114.30	178.880	4543.55	5,000	344.50	5,000	344.50
		53	78.87	4.630	117.47	152.400	3870.96								
				3.810	96.82	153.400	3896.36	7,500	516.75	7,500	516.75				
				4 1/2	114.30	8.300	210.82	4.630	117.47	153.400	3896.36				
												5 1/2	139.70	7,500	516.75
9 7/8	250.82	62	92.26	5 1/2	139.70	8.390	212.98	3.810	96.82	153.400	3896.36	7,500	516.75	7,500	516.75
10 3/4	273.05	60	89.28	7	177.80	9.405	238.88	5.963	151.46	182.000	4622.8	5,000	344.50	5,000	344.50

Ordering Information

Specify: casing size and weight; tubing size; weight; grade; and thread; service environment (standard, %H₂S, %CO₂, amines/other chemicals, chloride content, pressures, temperatures, etc.); maximum differential pressure requirement.

Part Number Prefix: P.722HP1

Feed-Through Packers

Halliburton feed-through packers are based on successful, field-proven platforms highlighted in this catalog. The most suitably adaptable packer chassis is customized to allow for accommodation of feed-through lines, setting, retrieval, and performance capabilities and are available in various ISO 14310 validation grades up to V0.

Feed-through applications include, but are not limited to, permanent reservoir monitoring, chemical injection, distributed temperature sensing (DTS), SmartWell® completions, electrical submersible pump (ESP) completions, or any other feed-through application.

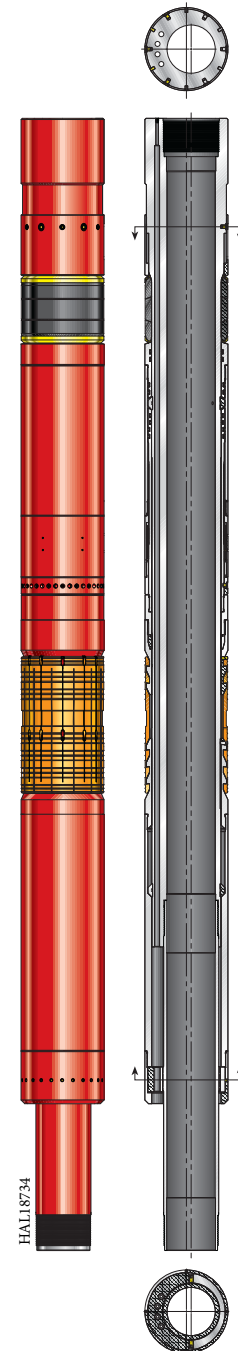
- Permanent packers can be used for ultra HP/HT applications when control-line bypass is required.
- HPS packers are used for high-performance completions when a retrievable feed-through packer is required. The MFT packer is the customized derivative.
- HF1 packers are the high-performance derivative of the HP1 packer which offers punch, shifting tool, or cut-to-release options.
- AHC packers used as hydraulic-set, retrievable feed-through packers become MFC packers.
- AHR packers used as a donor for hydraulic-set, shear release feed-through packers are MFA packers.
- ESP packers are usually based on RDH or AHC packer chassis and are hydraulic-set, shear-to-release
- Feed-through, hydrostatic-set packers are also available in certain casing and mandrel size configurations.
- DHC packers offer multiple feed-through possibilities for SmartWell completions, ESP, and dual bore with control line penetration requirements.

MFT Packer

The MFT packer is a retrievable, hydraulic-set, cut-to-release, multi-string production or injection packer set in casing to divert casing-to-tubing flow. It is based upon the proven HPH packer and features the same barrel slip and element package to provide high-performance operation. The packer can be set conventionally with tubing pressure via control line or in some sizes interventionally. Once set, an external barrel-type slip holds the packer against well pressures from above and below. The internal slips maintain the packer in the set position. After it is run and set, the packer can be released by cutting the mandrel. When properly and completely cut, the packer will release and all of the packer components along with the tailpipe can be retrieved from the well. A locator sub is installed on the lower end of the assembly for locating the cutter.

Benefits

- Compatible with feed-through equipment
- Easy installation can help make marginally economic wells attractive to produce
- Easy retrievability means less time spent on workovers
- Chemical injection and pressure monitoring ports can easily be added through the packer without significant costs
- Long-term, dependable performance; elastomeric seals have been minimized



MFT Packer

Part Number Prefix:
12MFT, 12MFC, 12MFA

Hydraulic-Set DHC Packer

The Halliburton DHC packer is a retrievable, hydraulic-set dual bore or multi-feed-through production packer that is set in casing to divert casing-to-tubing flow. It is set by pressuring up the tubing string against a plugging device below the packer or can be control-line set in certain applications.

A bi-directional, case-carburized barrel slip holds the packer against well pressures from above and below. An internal body lock ring maintains the packer in the set position until the mandrel is cut to retrieve the packer.

Applications

- Annular safety valve or SmartWell® completion
- Single-trip applications
- Multi-string completions with control-line feed-through
- Dual-bore completions

Features

- API 11D1-V3 validated
- Single-trip capability
- Barrel slip design
- Single piece mandrel construction
- Triple seal multidurometer element

Benefits

- Multiple mandrel penetrations possible
- Cut to release
- Various casing sizes available

Hydraulic-Set DHC Packer

Casing Size		Casing Weight		Packer OD	
in.	mm	lb/ft	kg/m	in.	mm
9 5/8	244.48	47	69.94	8.41	213.6
		43.5-53.5	64.73-79.62	8.30	210.8
9 5/8 / 9 7/8	244.48 / 250.82	47-53.5 / 62.8	69.94-79.62 / 93.45	8.35	212.1
10 3/4	273.05	55.5	82.59	9.455	240.1
		55.5-60.7	82.59-90.33	9.455	240.1
		60.7-65.7	90.33-97.77	9.375	238.1
13 3/8	339.72	68-72	101.2-107.14	12.13	308.1



Retrievable Sealbore Production Packers

Halliburton offers wireline- and hydraulic-set Versa-Trieve® sealbore packers.

Wireline-Set Versa-Trieve® Packers

Versa-Trieve® Packers—VTA, VBA, VCA, VCH, VBS, VGP, VGH, VSA

Versa-Trieve packers are production packers designed for intermediate and high-pressure well applications. Numerous features in these compact, millable packers incorporate the production features of permanent packers plus the added feature of retrievability. External components are easily millable if conventional release is not possible. Versa-Trieve packers can be either hydraulic- or wireline-set and are designed to be retrieved by a pulling tool with a straight pull of the tubing. When hydraulically set, Versa-Trieve packers are ideal for deviated or directional wells where it is difficult to run mechanical-set packers.

Applications

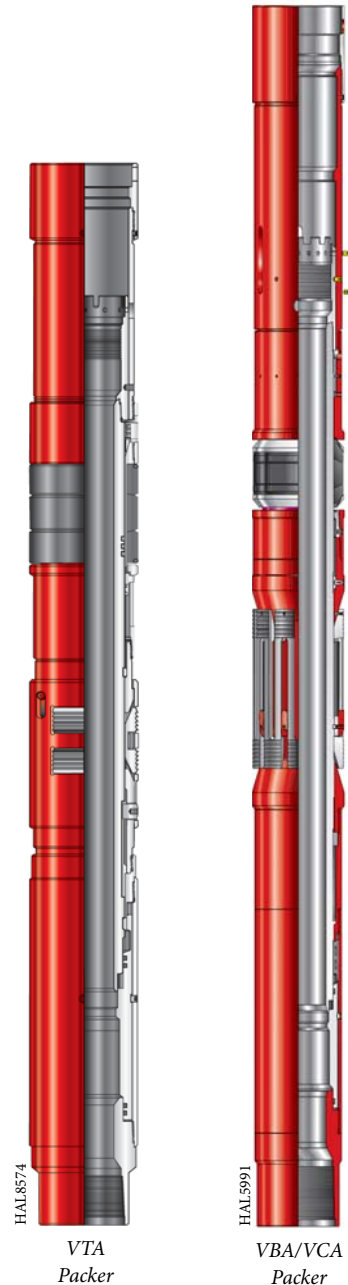
Versa-Trieve packers can be used in a variety of applications:

- Gravel pack
- High-rate water pack
- Ex-tension PacSM service
- FracPacSM service
- Horizontal completion assemblies
- Production sealbore packer
- Suspending guns for tubing-conveyed perforating
- Stimulation

Features

- Packers can be run, set, and retrieved without any tubing rotation in most cases.
- The hydraulic setting mechanism is contained in the setting tool, reducing the number of potential leak paths left down hole.
- The scoop-head design saves time by providing a guide for landing seal units and accessories.
- The spring-loaded, case-carburized internal locking slips maintain maximum element compression.
- The multi-durometer element package is designed to seal against high and low pressures and against casing irregularities. (Standard Nitrile or optional AFLAS® element packages are available. Special packer and element designs exist to suit high-pressure applications.)
- Element mandrel design on the VTA packers provides a positive means of locking the upper components to the slips. (The element mandrel allows for the full setting force to be applied to the elements and slips during setting operations. If milling operations are required, the upper components will not spin.)

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- Lower anchor lug allows torque to be transmitted through the packer during running and pulling operations.
- Bi-directional case-carburized slips or a barrel slip hold the packer securely against well pressures in all casing grades (including P110 and harder) from above and below.
- Case-carburized slips or a barrel slip hold the packer securely against well pressures in all casing grades (including P110 and harder) from above and below.
- Location of the slips beneath the elements simplifies releasing and retrieval. Debris is kept off the slips.
- A flow-isolated, C-ring release mechanism locks the packer in the set position until located and unlatched by the retrieving tool. Pushing through tight spots is safer than with traditional collet-release mechanisms.
- Packer is designed for simple release and retrieval.

Versa-Trieve® packers are available with the following top sub configurations:

- Versa-Latch® assembly
- Upper sealbore assembly (VSA, VBS)

Threaded bottom subs are provided as a separate item for running gravel pack flow subs, sealbore extensions, or tubing tailpipe of varying types and sizes. The -Z designates that the packer has no bottom sub.

VCH and VGH Packers

The VCH packer is a VCA packer specially designed for use with the HPT heavy hang weight soft release tool. The VGH packer is a VGP packer designed for the HPT tool.



VGP Packers

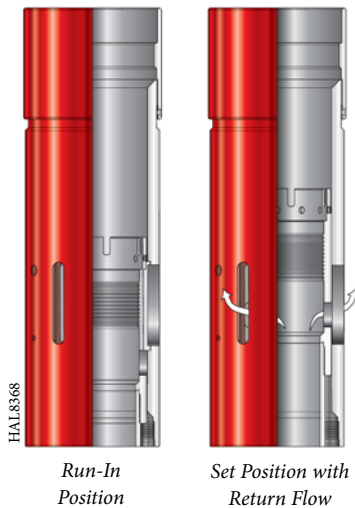
The large bore Versa-Latch® gravel pack (VGP) style packer is an integral part of the Versa-Trieve® packer family of tools. It provides a larger bore than the VTA packers (in a given casing size and weight) by eliminating the element mandrel while still retaining many of the Versa-Trieve packer line features. Because the element mandrel is eliminated, the lower element retainer is pinned to the packer mandrel, and an additional set of anti-rotation pins has been added for ease of milling. The VGP packer also includes a top snap guide for collet locations.

Upper Sealbore VSA Versa-Trieve® Packer

For applications where a larger bore is required through the seal assembly, Halliburton can provide an upper sealbore packer head design that has a larger ID than the packer mandrel and can accept a short seal assembly and Versa-Latch® locator.

Weight-Down Versa-Trieve® and PGP Packers

All the packers described previously can be provided as weight-down packer designs for weight-down squeeze and circulate gravel packing and FracPac™ systems. The weight-down packers utilize the slots in the top sub and setting sleeve. These slots align when the packer is in the set position. This provides a flow path for return fluids during a gravel packing operation or pressure monitoring during a FracPac™ system operation.



HPW Packer

The HPW packer is a retrievable single-string packer with design features of a large bore permanent packer. The packer has a bi-directional barrel slip below the elements. The packer will hold high pressure from above and below. The polished mandrel and top sub is designed to accept an indexing muleshoe. It is designed to be set on an MPW multi-position tool. The packer is pulled by a straight upward pull with the VRA retrieving tool or by a cut-to-release option.

Thermal Versa-Trieve® Packer

The thermal Versa-Trieve® packer is designed with a high-temperature elastomer package and o-rings, providing a packer suitable for extremely high temperatures. The elements are made of materials suitable for service where operations such as steam injection and “huff ‘n’ puff” exist.

Sealbore and Millout Extensions

Halliburton’s packer sealbore extensions extend the polished surface below the packer. This enables the use of longer sealing units to compensate for tubing contraction or elongation. The extensions are available in standard 8-, 20-, and 24-ft (2.44-, 6.10-, and 7.32-m) increments.

Halliburton millout extensions provide a large ID below the packer sealbore or sealbore extension, which allows a single-trip packer milling/retrieving tool to be used when tubing is run below the packer assembly.

PGP Permanent Sealbore Packer

The PGP packer is a Perma-Series® permanent packer that has been modified so it can be run on the MPT tool. The packer has an upper setting sleeve and a longer bottom sub with a thread that is compatible with standard flow subs and closing sleeves. The PGP packer has the benefit of withstanding higher pressures and temperatures than the retrievable Versa-Trieve® family of packers but must be milled out when workover operations are required.



HPW
Packer



Thermal
Versa-Trieve® Packer



PGP Type
Permanent Packer

Versa-Trieve® Packers

Casing Size		Casing Weight		Packer OD		Packer Bore		Production Seal Assembly ID	
in.	mm	lb/ft	kg/m	in.	mm	in.	mm	in.	mm
4 1/2	114.30	9.5-11.6	14.14-17.26	3.82	97.03	2.380	60.45	1.74	44.07
5	127.00	15-18	22.32-26.79	4.09	103.89	2.550	64.77	1.81	45.97
5 1/2	139.70	23-23.8	35.22-35.41	4.44	112.78	2.750	69.85	1.93	48.95
		20-23	29.76-34.22	4.50	114.30				
		17-20	20.83-25.30	4.60	116.84				
		14-17	20.83-25.30	4.67	118.62				
6 5/8	168.28	34.5	51.34	5.35	135.89	3.250	82.55	2.35	59.69
		28	41.66	5.53	140.46				
		24	35.72	5.73	145.54				
		20-24	29.76-35.71	5.73	145.54				
7	177.80	49.5-50.1	73.66-74.56	5.30	134.62	3.000	76.20	2.35	59.69
		49	72.92	5.35	135.89	3.250	82.55	2.35	59.69
		41-44	61.01-65.47	5.53	140.46				
		41-42.7	61.01-63.54	5.565	141.35				
		38	56.55	5.73	145.54				
		32-38	47.62-56.54	5.73	145.54	3.88	98.55	3.05	77.47
		38	56.55	5.725	145.42				
		32-35	47.62-52.08	5.82	147.83				
		32	47.62	5.89	149.61				
		26-32	38.69-47.62	5.92	150.37				
		26-29	38.69-43.15	6.015	152.78				
		26-29	38.69-43.15	6.00	152.78				
		23-29	34.22-43.15	6.00	152.40				
		23	34.22	6.14	155.96				
		23	34.22	6.19	155.96				
		17-23	25.30-34.22	6.19	157.23				
		17-20	25.30-29.76	6.275	159.51				
7 5/8	193.68	45.3-47.1	67.41-70.08	6.14	155.96	3.880	98.55	3.050	77.47
		42.4-47.1	35.72-70.09	6.19	157.23				
		39-42.8	58.03-63.69	6.275	159.51				
		39	58.04	6.44	163.58				
		29.7-39	44.19-58.03	6.44	163.58				
		29.7-33.7	44.19-50.15	6.56	166.62				
		33.7	50.15	6.58	167.13				
		26.4-29.7	39.28-44.19	6.62	168.15				
7 3/4	196.85	46.1	68.6	6.28	159.96	3.880	98.55	3.050	77.47
		46.1-48.6	68.6-72.32	6.19	157.23				
8 1/8	206.38	45.5	67.71	6.87	174.5	4.600	116.84	3.800	96.52
8 5/8	219.08	57.4-61.4	85.42-91.37	6.985	177.42	4.600	116.84	3.800	96.52
		60.7	90.32	7.00	177.8	4.600	116.84	3.800	96.52
		61.4	91.37	7.00	177.8	4.600	116.84	3.800	96.52
		55.83	83.08	7.125	180.95	4.600	116.84	3.800	96.52
		55.83-57.4	82.94-85.42	7.125	180.98	4.600	116.84	3.800	96.52
		52-54	77.38-80.36	7.16	181.86	4.600	116.84	3.800	96.52
		40-44	59.53-65.47	7.43	188.72	5.000	127.00	3.850	97.79
		28-36	41.67-53.57	7.63	193.80	3.880	98.55	3.05	77.47
8 3/4	222.25	32-36	47.62-53.57	7.64	194.06	3.880	98.55	3.05	77.47
		49.7	73.96	7.43	188.72	5.000	127.00	3.85	97.79

Versa-Trieve® Packers

Casing Size		Casing Weight		Packer OD		Packer Bore		Production Seal Assembly ID	
in.	mm	lb/ft	kg/m	in.	mm	in.	mm	in.	mm
9 5/8	244.48	70.3-71.8	104.61-106.84	7.89	202.69	5.000	127.00	3.850	97.79
		53.5	79.62	8.30	210.82	3.880	98.55	3.05	77.47
						5.000	127.00	3.85	97.79
						6.000	152.40	4.86	123.44
		47-53.5	64.74-79.62	8.30	210.82	3.880	98.55	3.05	77.47
						5.000	127.00	3.85	97.79
						6.000	152.40	4.86	123.44
		43.5-53.5	64.73-79.62	8.30	210.82	3.880	98.55	3.05	77.47
						5.000	127.00	3.85	97.79
		43.5-47	64.74-69.94	8.45	214.63	3.880	98.55	3.05	77.47
						5.000	127.00	3.85	97.79
						6.000	152.40	4.86	123.44
		40-47	59.53-69.94	8.46	214.88	5.000	127.00	3.85	97.79
		36-43.5	53.57-64.73	8.52	216.41	3.880	98.55	3.05	77.47
		40-43.5	59.53-64.73	8.52	216.41	5.000	127.00	3.85	97.79
						6.000	152.40	4.86	123.44
						3.880	98.55	3.05	77.47
		36-40	53.57-59.53	8.60	218.44	5.000	127.00	3.85	97.79
						3.880	98.55	3.05	77.47
9 3/4	247.65	59.2	88.10	8.30	210.82	5.000	127.00	3.85	97.79
9 7/8	250.83	62.8	93.46	8.30	210.82	5.000	127.00	3.85	97.79
		66.9	99.56	8.30	210.82	6.000	152.40	4.86	123.44
10 1/8	257.18	79.22	117.89	8.30	210.82	5.000	127.00	3.85	97.79
10 3/4	273.05	51-65.7	75.90-97.77	9.31	236.47	5.000	127.00	3.85	97.79
		60.7-65.7	90.33-97.77	9.34	237.24	6.000	152.40	4.86	123.44
11 3/4	298.45	65-71	96.72-105.65	10.30	261.62	6.000	152.40	4.86	123.44
11 7/8	301.63	71.8	106.85	10.34	273.05	6.000	152.40	4.86	123.44
13 3/8	339.73	68-72	101.18-107.14	12.13	308.10	6.000	152.40	4.86	123.44

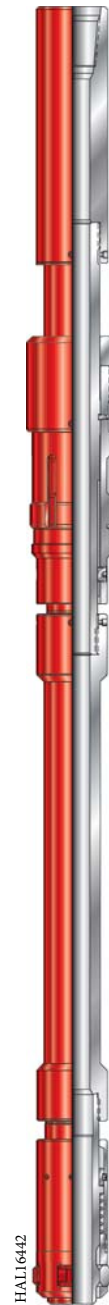
Versa-Trieve® Packer Retrieving Tool

Versa-Trieve® retrieving tools are used to retrieve Versa-Trieve packers. To retrieve the packer, the retrieving tool is run on a workstring, stung into the packer bore, and latched into the Versa-Trieve packer. Once latched, the tool locates the packer's release sleeve and moves it upward, allowing the packer to be released. Additional upward movement of the workstring releases the sleeve and moves the packer mandrel up, which permits the packer slips to retract and the packer to be retrieved.

No rotation is necessary in the pulling operation. To release the pulling tool from the packer, only a 1/4 turn to the right is needed on the J-latch-type locators. Additional rotation (six to eight turns) is required to release from the Versa-Latch® locator from the packer top sub.

Benefits

- No rotation required for pulling operation
- Straight-through ID allows washing down to the packer
- Easy to release with 1/4 turn on lug-type tools (VRT type) and six to eight turns on a Versa-Latch tool (VRA type)
- Rugged lug-type design engages the release sleeve



*Versa-Trieve® Packer
Retrieving Tool*

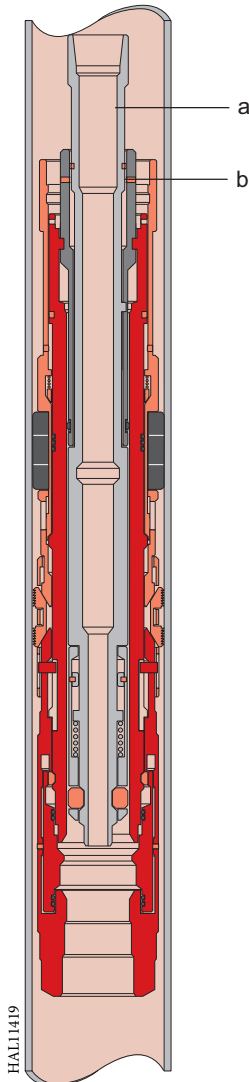
Ordering Information

Specify: casing size and weight; tubing size; weight; grade; and thread; service environment (standard, %H₂S, %CO₂, amines/other chemicals, chloride content, pressures, temperatures, etc.); maximum differential pressure requirement.

Part Number Prefix: 12VRA, 12VRR, 12VRT

Schematic 1—Engaging

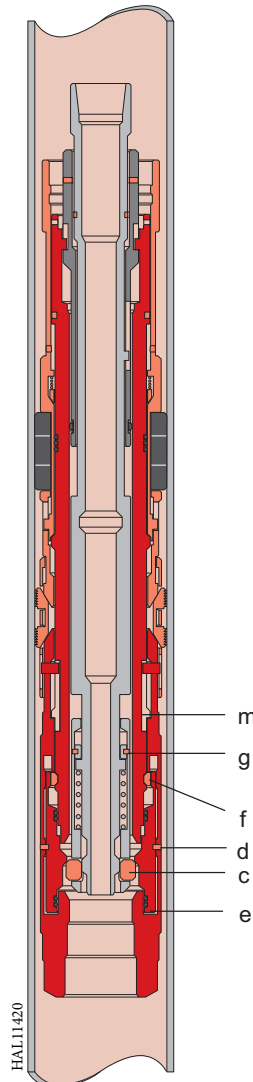
After the tubing and seal assembly have been removed, a Versa-Trieve® retrieving tool is lowered into the packer until the locator no-goes on the packer top sub. An upstrain on the workstring (a) verifies that the retrieving tool is latched into the packer and shears the locator to the mandrel shear pin (b).



Schematic 1

Schematic 2—Releasing

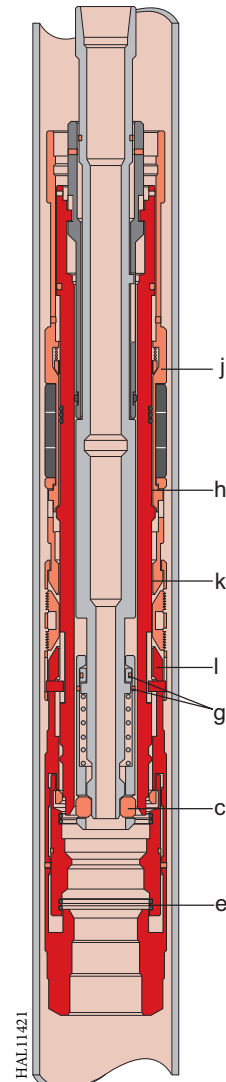
With the retrieving tool properly latched to the packer, the workstring is slacked off until the retrieving tool begins to take weight. An upstrain applied to the retrieving tool allows the lugs of the retrieving tool (c) to engage the packer release sleeve. Continued upstrain will shear the shear pins (d) in the release sleeve, allowing the release sleeve (e) to move upward and toward the release ring (f).



Schematic 2

Schematic 3—Retrieving

Additional upstrain shears the pins (g), causing the lugs of the retrieving tool (c) to retract. Upward movement of the packer's mandrel causes the shoulder (h) to raise the element mandrel (j) and top wedge (k). The top wedge is pulled out from under the slips and pulls on the slip carrier, which pulls the slips off of the lower wedge (l). The shoulder (m) catches the lower edge and supports the tailpipe below the packer.



Schematic 3

Retrievable Mechanical-Set Packers

This section describes Halliburton Versa-Set®, Perma-Lach®, and Guiberson® G-6 packers.

Versa-Set® Packers

The Versa-Set packer uses a modular design concept to combine all the best high-performance features of a mechanical-set retrievable packer with the ability to set on wireline. Three setting modules are available for a wide variety of applications.

The Versa-Set packer can be set on wireline using the economical WLS module. A standard XL on-off tool with integral landing nipple profile provides for a temporary plug at the top of the packer while the production tubing is being run. Setting on wireline allows lubricating into the wellbore without having to kill the well, which minimizes damage to sensitive formations. It also provides for precise setting depth control. Once the packer is set on wireline and the tubing string is engaged, the packer can be released and retrieved in a half-trip—no retrieving tool is required. The packer can also be set on wireline when either of the mechanical-set modules (with or without clutch) are used. These modules allow the packer to be moved and reset multiple times before retrieving.

The Versa-Set packer can easily be converted at the wellsite to run mechanically on tubing. Operation is simple 1/4 turn for right-hand set and release. The element package can be packed off in either compression or tension. Heavy-duty jay-lugs are capable of handling most treating conditions and also provide emergency straight-pull release.

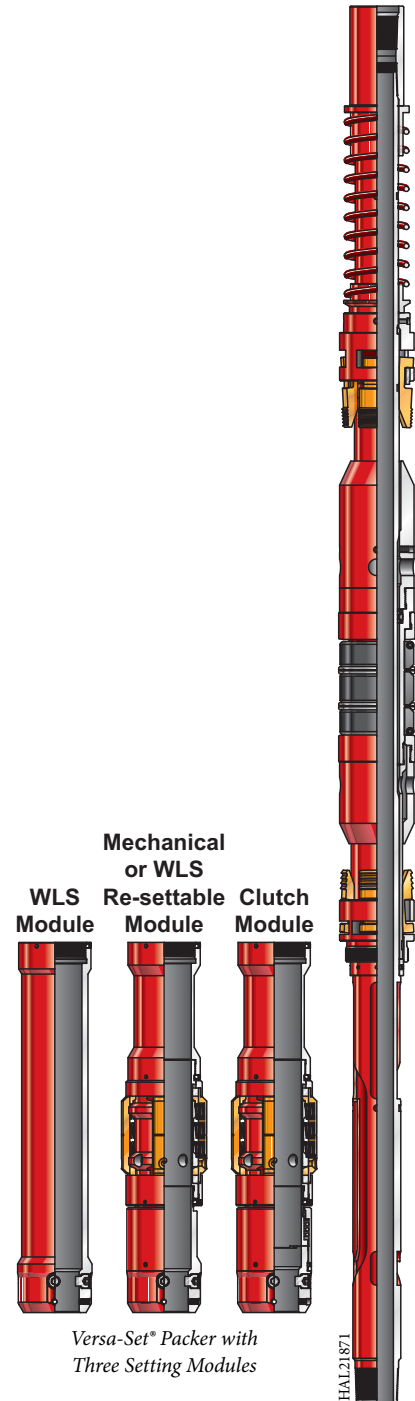
The Versa-Set packer provides a full opening for most tubing-casing combinations and is available in pressure ratings up to 10,000 psi to meet most well conditions. The packer can be trimmed for sour service applications.

Applications

- Production, testing, injection, stimulation, tubing-conveyed perforating, and temporary plugging
- Any application requiring that the packer be lubricated into the wellbore with a plug in place
- TCP or other completions requiring tailpipe

Features

- Wireline-set or mechanical-set
- Temporary plug may be installed at top of packer
- Holds in both directions with opposing slip design
- Pressure ratings to 10,000 psi (68.9 MPa)
- Single-trip retrievable
- Mechanically re-settable on tubing with simple, 1/4 turn J-slot operation when using either mechanical-set module
- High-performance triple seal multidurometer element package
- Internal bypass
- Emergency shear release
- Large ID
- Accepts a wide variety of completion accessories
- Sour service trim available



Benefits

- Three setting modules available for maximum versatility
- Easily converts to wireline-set or mechanical-set in the field
- Can be lubricated into well under pressure thereby minimizing damage to sensitive formations
- Provides both the accuracy of wireline setting and the capability of a mechanical-set retrievable packer
- Provides the versatility to run the packer either on wireline, slickline (with Halliburton DPU® downhole power unit), or tubing
- Profile nipples and/or pump-out plugs can also be run below the packer

Wireline-Set Operation

- When trimmed with any of the three setting modules, packer may be set with a standard wireline setting tool or DPU® downhole power unit with wireline adapter kit
- Uses standard XL on-off tool
- May be run and set with plug installed in on-off tool gudgeon (top of packer)
- Operates as a standard mechanical double-grip packer with on-off tool once set on wireline
- One-quarter turn right-hand release
- Re-settable on tubing when either of the mechanical-set modules are used
- Emergency straight-pull release

Mechanical-Set Operation

- May be run and set on tubing with either of the mechanical-set modules
- One-quarter turn right-hand set
- Can be packed off in either compression or tension
- Can be left in compression, tension, or neutral position
- One-quarter turn right-hand release
- Emergency straight-pull release
- Clutch (mechanical-set with clutch module) helps assure reliable setting operations by removing excess tubing torque



Versa-Set® Retrievable Packer
with Clutch
(Wireline Operation)



Versa-Set® Retrievable Packer
with Clutch
(Mechanical Operation)

Versa-Set® Packer

Casing Size		Casing Weight		Maximum OD		Minimum ID		Tubing Size	
in.	mm	lb/ft	kg/m	in.	mm	in.	mm	in.	mm
4 1/2	114.30	9.5-11.6	14.14-17.20	3.82	97.03	1.923	48.84	2 3/8	60.33
		11.6-13.5	17.26-20.09	3.76	95.50				
		15.1	22.47	3.66	92.96				
5	127.00	11.5-15	17.11-22.32	4.22	107.19	1.923	48.84	2 3/8	60.33
		15-18	22.32-26.79	4.10	104.14				
5 1/2	139.70	13-15.5	19.35-23.07	4.77	121.16	1.985	50.42	2 3/8	60.33
						2.36	59.94	2 7/8	73.03
		15.5-20	23.07-29.76	4.61	117.09	1.985	50.42	2 3/8	60.33
						2.36	59.94	2 7/8	73.03
		20-23	29.76-34.23	4.50	114.30	1.985	50.42	2 3/8	60.33
						2.36	59.94	2 7/8	73.03
7	177.80	17-23	25.30-34.23	6.18	156.97	1.975	50.17	2 3/8	60.33
						2.423	61.54	2 7/8	73.03
						2.853	72.47	3 1/2	88.90
		23-29	34.23-43.16	6.00	152.40	1.975	50.17	2 3/8	60.33
						2.423	61.54	2 7/8	73.03
						2.853	72.47	3 1/2	88.90
		29-32	43.16-47.62	5.91	150.11	1.975	50.17	2 3/8	60.33
						2.423	61.54	2 7/8	73.03
						2.853	72.47	3 1/2	88.90
		32-35	47.62-52.09	5.80	147.32	1.975	50.17	2 3/8	60.33
						2.423	61.54	2 7/8	73.03
						2.853	72.47	3 1/2	88.90
7 5/8	193.68	24-29.7	35.72-44.20	6.70	170.18	2.423	61.54	2 7/8	73.03
						2.853	72.47	3 1/2	88.90
		33.7-39	50.15-58.04	6.45	163.83	2.423	61.54	2 7/8	73.03
						2.853	72.47	3 1/2	88.90

Ordering Information

Specify: casing size and weight; tubing size; weight; grade; and thread; service environment (standard, %H₂S, %CO₂, amines/other chemicals, chloride content, pressures, temperatures, etc.); maximum differential pressure requirement.

Part Number Prefix: 12VSW, 12VSM, 12VSC

Perma-Lach® PLS Packers

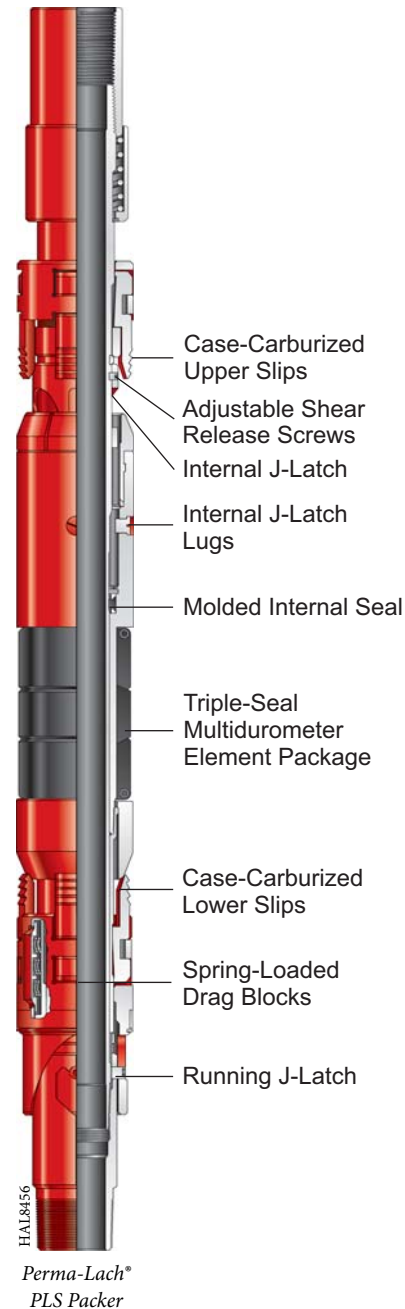
Perma-Lach® PLS packers are multiple-use, retrievable packers designed to be run and set on tubing, released and moved, and set again without tripping the tubing. They can be retrieved, redressed if necessary, and used repeatedly. The 10K Perma-Lach PLS packers operate the same way as standard Perma-Lach PLS packers, but they have been tested and rated to 10,000 psi (68.9 MPa).

Applications

- Testing and zone stimulation
- As a tubing anchor (When the well goes on pump, the packer can be used to place the tubing in tension for more efficient pumping.)
- For temporary bridge plug applications (An overshot tubing seal divider may be made up immediately above the Perma-Lach packer, making it possible to plug off the zone below the packer and come out of the hole with tubing.)

Features

- Internal J-latch holds the packer set
- Generous fluid bypass area through the packer to speed run-in and equalization times
- No o-rings throughout the packer
- Opposed spring-loaded, case-carburized slips manufactured with controlled depth penetration and consistent maximum hardness across the tooth section to provide a reliable gripping force even after multiple setting operations
- Molded seals resist flow cutting or washout during bypass
- Triple-seal multidurometer element package contains a nested dual garter spring end element design and mandrel anti-extrusion ring rated for up to 10,000 psi (68.9 MPa) differential
- Standardized components allow interchangeability
- Automatic relatch after release with the running J-latch
- Secondary shear release provides a method of retrieving packer if tubing cannot be rotated



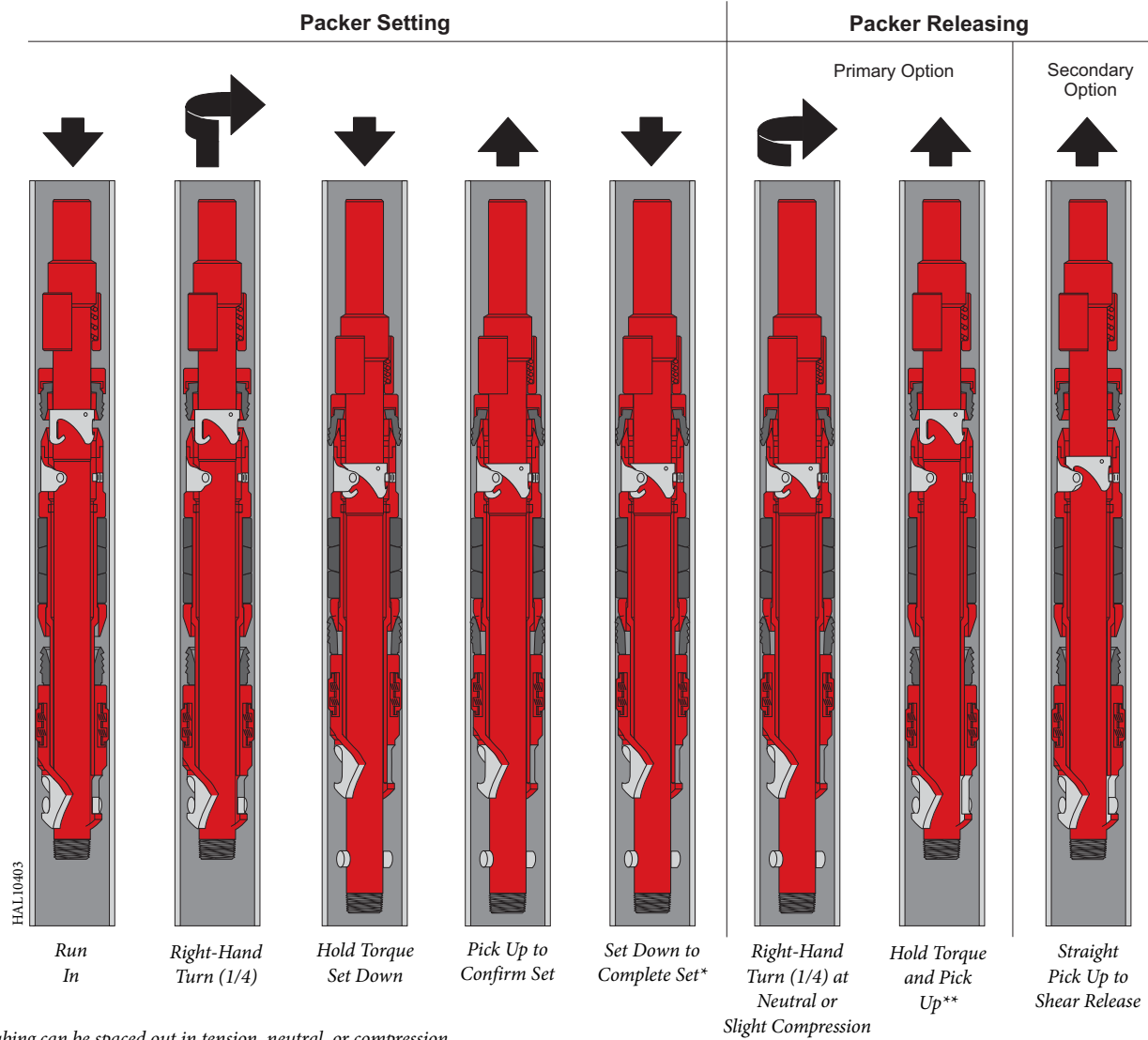
Ordering Information

Specify: casing size and weight, tubing size; weight; grade; and thread, service environment (standard, %H₂S, %CO₂, amines/other chemicals, chloride content, pressures, temperatures, etc.), maximum differential pressure requirement.

Part Number Prefix: 12PLS

Benefits

- Tubing can be spaced out in tension, neutral, or compression
- Requires only 1/4 turn of tubing at packer to set or release
- Allows adjustment of shear value in the field
- Can be broken down, redressed, and reassembled in the field and run back into the well
- Available with 10,000 psi (68.9 MPa) pressure rating in most sizes



*Tubing can be spaced out in tension, neutral, or compression.

**Packer now ready to retrieve or reset

Perma-Lach® PLS Packers

Casing Size*		Casing Weight		Packer OD		Packer ID*		Tubing Size**	
in.	mm	lb/ft	kg/m	in.	mm	in.	mm	in.	mm
2 7/8	73.03	6.4	9.52	2.28	57.91	0.92	23.37	1.66	42.16
3 1/2	88.90	9.2-10.2	13.69-15.18	2.81	71.37	1.33	33.78	2.06	52.32
4	101.60	9.5-11.6	14.14-17.26	3.28	83.31	1.56	39.62	1.90	48.26
4 1/2	114.30	9.5-11.6	14.14-17.26	3.80	96.52	1.92	48.77	2 3/8	60.33
		11.6-13.5	17.26-20.09	3.76	95.50	1.92	48.77	2 3/8	60.33
5	127.00	15-18	22.32-26.79	4.09	103.89	1.92	48.77	2 3/8	60.33
5 1/2	139.70	13-17	19.35-25.30	4.70	19.35	1.97	50.04	2 3/8	60.33
						2.36	59.94	2 7/8	73.03
		20-23	29.76-34.23	4.50	114.3	1.97	50.04	2 3/8	60.33
						2.36	59.94	2 7/8	73.03
6 5/8	168.28	20-24	29.76-35.72	5.70	144.78	2.36	59.94	2 7/8	73.03
		28-32	41.67-47.62	5.50	139.70	2.36	59.94	2 7/8	73.03
7	177.80	20-29	29.76-43.16	6.02	152.91	2.36	59.94	2 7/8	73.03
						2.97	75.44	3 1/2	88.90
		26-32	38.69-47.62	5.90	149.86	2.36	59.94	2 7/8	73.03
						2.97	75.44	3 1/2	88.90
						2.36	59.94	2 7/8	73.03
7 5/8	193.68	26.4-29.7	39.29-44.20	6.66	169.16	2.80	71.12	3 1/2	88.90
		29.7-39	44.20-58.04	6.44	163.58	2.36	59.94	2 7/8	73.03
						2.97	75.44	3 1/2	88.90
8 5/8	219.08	28-36	41.67-53.57	7.62	193.55	2.98	75.69	3 1/2	88.90
		38-49	56.55-72.92	7.33	186.18	2.98	75.69	3 1/2	88.90
9 5/8***	244.48	36-43.5	53.57-64.74	8.54	216.92	3.86	98.04	4 1/2	114.30
		43.5-53.5	64.74-79.62	8.31	211.07	3.00	76.20	3 1/2	88.90

*Minimum ID is dependent on tubing size and thread type selected.

**Other casing and tubing sizes available on request.

***9 5/8 available in PL version only.

Guiberson® G-6 Packer

Halliburton's Guiberson® G-6 packer is a double-grip, retrievable, single-string packer designed for service in water-flooding and enhanced oil recovery. Because many of these wells are shallow, the packer was designed to allow the packing elements to be packed off with either tubing tension or compression at the packer. The double-grip feature allows the option of leaving tubing in compression or tension and makes the packer suitable for use with accessories such as on-off tools, downhole shut-off valves, and other related completion tools.

Applications

- Ideally suited for enhanced oil recovery projects
- CO₂ and steam injection service with specific elastomer systems
- Up to 5,000 psi (34.45 MPa) differential pressure typical

Features

- Short compact length
- Compatible with plastic coating, sour service trim, or special flow wet metallurgy
- Internal bypass valve provides reliable running, releasing, and retrieving
- Innovative packing element system for positive pressure-enhanced pack-off
- Opposing slip design holds forces from either direction while loaded slip remains on the low pressure side of the tool
- Emergency shear release
- Take-up feature allows pressure differential to increase and lock in added rubber compression

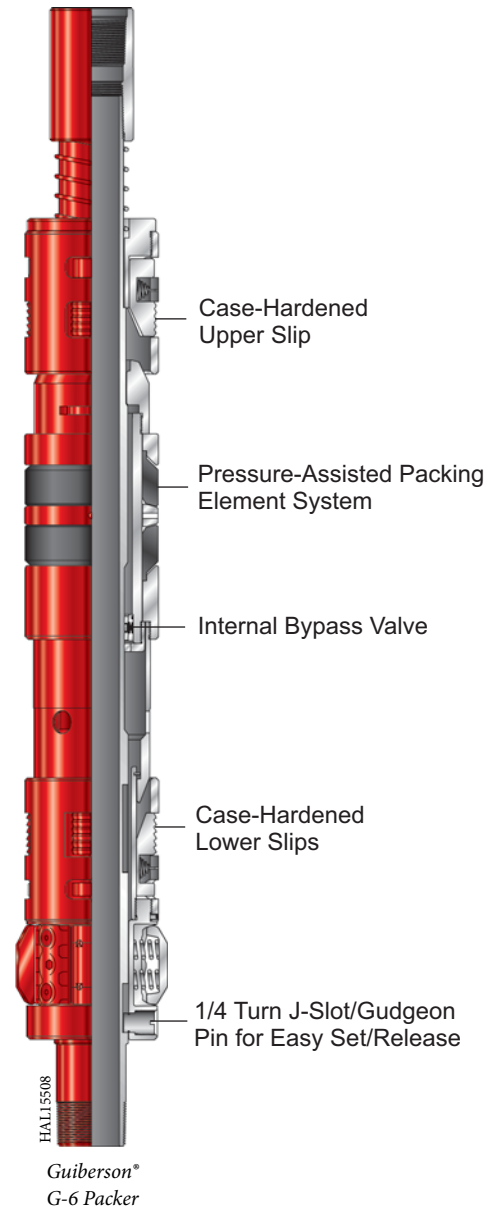
Benefits

- Requires only 1/4 turn of the tubing at the packer to set or release
- Tension or compression packs off the rubber system
- Tubing can be spaced out in tension, neutral, or compression
- Field-redressable
- May be run with a variety of completion accessories

Operation

The G-6 packer is set by lowering the tubing to the setting depth. The tubing is picked up to allow for the setting stroke (8 in.) plus the desired tubing load. One-quarter turn right-hand rotation of the tubing at the packer with set down motion sets both slip systems. Packing elements can then be packed off with either compression or tension at the packer.

The packer is retrieved by applying 1/4 turn right-hand rotation of the tubing at the packer and picking up. Opening of the internal bypass valve allows pressure to equalize across the packer. Continued upward movement will release the packer and automatically re-J the tool into the running position. In the event the tool cannot be unset conventionally, a safety shear system is provided for emergency release.



Guiberson® G-6 Packer

Casing Size		Casing Weight		Packer OD		Minimum ID		Tubing Size	
in.	mm	lb/ft	kg/m	in.	mm	in.	mm	in.	mm
4	101.60	9.25-11.6	13.77-17.26	3.295	83.69	1.545	39.24	1.90	48.26
4 1/2	114.30	9.5-13.5	14.14-20.09	3.75	98.17	1.94	49.22	2 3/8	60.33
5	127.00	11.5-15	17.11-22.32	4.25	107.95	1.94	49.22	2 3/8	60.33
		15-18	22.32-26.79	4.13	104.78				
5 1/2	139.70	13-15.5	19.35-23.07	4.75	120.65	1.94 or 2.44	49.22 or 61.93	2 3/8 or 2 7/8	60.33 or 73.03
		17-20	25.30-29.76	4.63	117.48				
		20-23	29.76-34.22	4.50	114.30				
5 3/4	146.10	17-20	25.30-29.76	4.85	123.19	1.94	49.22	2 3/8	60.33
6 5/8	168.30	17-24	25.30-35.72	5.69	144.45	2.50	63.50	2 7/8	73.03
		26-32	38.67-47.62	5.50	139.70				
7	177.80	17-20	25.30-29.76	6.13	155.58	1.94 or 2.50 or 3.00	49.22 or 63.50 or 76.20	2 3/8 or 2 7/8 or 3 1/2	60.33 or 73.03 or 88.90
		22-26	32.74-38.69	6.00	152.40				
		28-32	41.67-47.62	5.88	149.23				
7 5/8	193.70	20-26.4	29.76-39.29	6.69	169.88	2.50	63.50	2 7/8	73.03
		29.7-39	44.20-58.04	6.44	163.53				
8 5/8	209.10	28-36	41.67-53.57	7.56	192.10	3.00	76.20	3 1/2	88.90
		40-49	59.53-72.92	7.38	190.50				
9 5/8	244.50	32.3-40	48.07-59.53	8.50	215.90	4.00	101.60	4 1/2	114.30
		43.5-47	64.74-69.94	8.38	212.72				
		53.5	79.62	8.25	209.55				

Ordering Information

Specify: casing size and weight; tubing size; weight; grade; and thread; service environment (standard, %H₂S, %CO₂, amines/other chemicals, chloride content, pressures, temperatures, etc.); maximum differential pressure requirement.

Part Number Prefix: 12GF, D

Retrievable Hydraulic-Set Packers

This section describes PHL, RH, G-77, and RDH packers.

Hydraulic-Set Perma-Lach® PHL Packers

Using many features from the mechanical-set Perma-Lach® packer design, this hydraulic-set retrievable packer offers high performance at low cost. Because the packer has no holddown buttons, potential leakage from the o-rings is eliminated. A simple change of packer elastomers equips the tool for H₂S or higher temperature service.

The tool's small diameter speeds running in. Setting is achieved by running the packer in on tubing, dropping a ball, or setting a slickline plug and applying pressure. No tubing manipulation is required. To release, pull straight up on the production tubing. Setting pressures can be field-adjusted, allowing multiple packers to be run and set during a single trip. Tubing can also be spaced out before the packer is set.

Applications

- Single or multi-packer production completions
- Can be used as the lower packer in dual completions
- Can be set at predetermined pressures or selectively set in multizone completions
- Up to 7,500 psi (51.7 MPa) differential pressure typical

Features

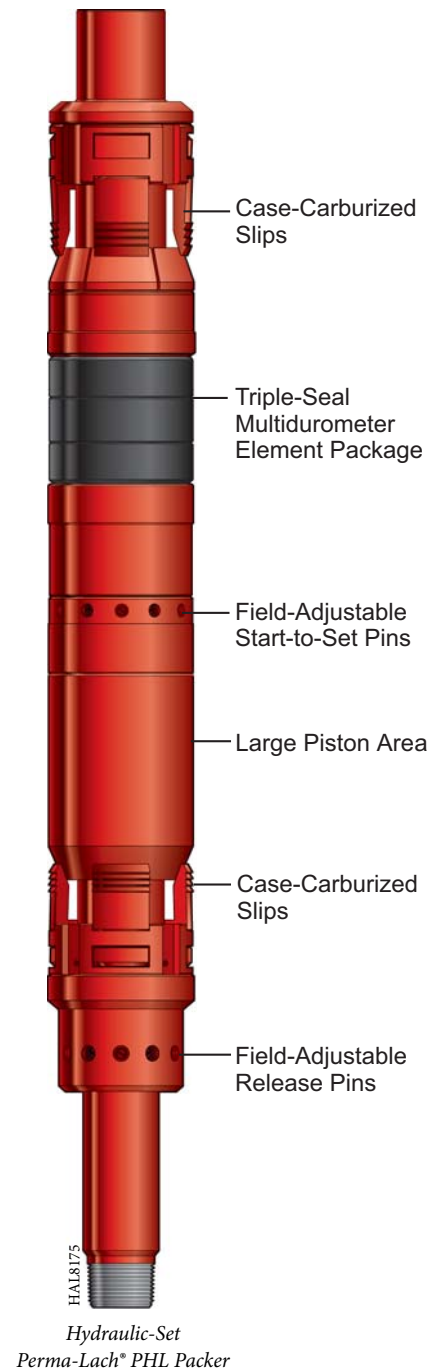
- Small OD
- Triple-seal multidurometer elements
- Shear screws allow operator to vary the setting pressure and release force
- Release mechanism is protected from premature shearing
- No mandrel movement when setting

Benefits

- High performance at low cost
- Setting pressures can be adjusted in the field
- No tubing manipulation is required for setting
- Requires only a straight pull to release
- Multiple packers can be run and set in one trip
- Tubing can be spaced out before the packer is set

Options

- Premium metallurgy and elastomers
- Selective set



Hydraulic-Set Perma-Lach® PHL Packers

Casing Size		Casing Weight		Tubing Size		Packer OD	
in.	mm	lb/ft	kg/m	in.	mm	in.	mm
4 1/2	114.30	9.5-13.5	14.14-20.09	2.38	60.33	3.76	96.16
5 1/2	139.70	13-17	19.35-25.30	2.38	60.33	4.70	119.38
		13-17	19.35-25.30	2.88	73.03	4.70	119.38
		17-23	25.30-34.22	2.38	60.33	4.52	114.30
		17-23	25.30-34.22	2.88	73.03	4.52	114.30
6 5/8	168.28	24-28	35.72-41.67	3.50	88.90	5.61	142.49
7	177.80	23-29	34.22-43.16	2.88	73.03	5.98	151.89
				3.50	88.90	5.98	151.89
		29-35	43.16-52.09	2.88	73.03	5.83	147.83
				3.50	88.90	5.83	147.83
7 5/8	193.68	29.7-39	44.20-58.04	2.88	73.03	6.47	164.29
				3.50	88.90	6.47	164.29
9 5/8	244.48	40-47	59.53-69.94	3.50	88.90	8.45	214.68
				4.50	114.30	8.45	214.68
		43.5-53.5	64.74-79.62	3.50	88.90	8.23	209.12
				4.50	114.30	8.23	209.12

Ordering Information

Specify: casing size and weight; tubing size; weight; grade; and thread; service environment (standard, %H₂S, %CO₂, amines/other chemicals, chloride content, pressures, temperatures, etc.); maximum differential pressure requirement.

Part Number Prefix: 12PHL

Hydraulic-Set RH Packers

The hydraulic-set RH packers are designed to be set by pressuring up the tubing string against a plugging device below the packer.

RH packers lock-set mechanically using spring-loaded internal slips. A lower set of mechanical slips holds the packer set against downward forces and an upper set of pressure-actuated button slips holds the packer set against differential pressure from below. The packers are released and retrieved by a straight upward pull on the tubing string.

Applications

- In deep or deviated wells

Features

- Tungsten-carbide disk inserts in the holddowns are protected from the casing wall as the packer is lowered into the well.
- Triple-seal multidurometer elements have two hard end elements with molded self-retracting springs and a softer middle element for casing irregularities and low pressures.
- Contoured case-carburized slips conform without belling the casing.
- Dovetail slip design helps release the packer when retrieving.
- Optional heads allow use of various latching arrangements.
- Shear screws allow operator to vary the setting pressure and release force.
- Release mechanism is protected from premature setting.

Benefits

- Has a low-pressure setting requirement so rig pumps usually provide enough pressure
- Allows varied setting pressures with varied numbers of shear screws
- Requires only a straight pull to release
- Allows adjustment of pull-to-release in the field
- Works well in all types of casing



Hydraulic-Set
RH Packer

Ordering Information

Specify: casing size and weight; tubing size; weight; grade; and thread; service environment (standard, %H₂S, %CO₂, amines/other chemicals, chloride content, pressures, temperatures, etc.); maximum differential pressure requirement

Part Number Prefix: 12RH

Hydraulic-Set RH Single Production Packers

Casing Size		Casing Weight		Packer OD		Packer ID*	
in.	mm	lb/ft	kg/m	in.	mm	in.	mm
4 1/2	114.30	9.5-13.5	14.14-20.09	3.75	95.25	1.62	41.15
		9.5-11.6	14.14-17.26	3.83	97.28	1.92	48.77
		11.6-15.1	17.26-22.47	3.65	92.71	1.53	38.86
5	127.00	11.5-13	17.11-19.35	4.31	109.47	1.94	49.28
		15-18	22.32-26.79	4.12	104.65	1.93	48.95
5 1/2	139.70	13-17	19.35-25.30	4.62	117.35	1.94	49.28
		17-23	25.30-34.22	4.50	114.30	1.94	49.28
6	152.40	20-26	29.76-38.69	5.00	127.00	1.92	48.77
6 5/8	168.28	17-24	25.30-35.72	5.73	145.54	2.89	73.41
		24-32	35.72-47.62	5.50	139.70	2.38	60.45
7	177.80	17-23	25.30-34.22	6.19	157.23	2.38	60.45
		20-26	29.76-38.69	6.12	155.45	2.37	60.20
		23-29	34.22-43.16	5.88	149.35	2.38	60.45
				6.00	152.40	2.89	73.41
		26-32	38.69-47.62	5.94	150.88	2.36	59.94
		29-32	43.16-47.62	5.88	149.35	2.89	73.41
						2.34	59.44
				5.77	146.56	2.89	73.41
		29-35	43.16-52.09	5.83	148.08	2.38	60.45
7 5/8	193.68	24-29.7	35.72-44.20	5.50	139.70	2.38	60.45
		24-29.7	35.72-44.20	6.64	168.66	2.91	73.91
9 5/8	244.48	29.7-39	44.20-58.04	6.40	162.56	2.89	73.41
		36-40	53.57-59.53	8.62	218.95	3.00	76.20
		40-47	59.53-69.94	8.38	212.85	2.34	59.44
				8.38	212.85	3.00	76.20
				8.38	212.85	4.73	120.14
		47-53.5	69.94-79.62	8.31	211.07	2.34	59.44
				8.31	211.07	3.00	76.20
				8.31	211.07	4.73	120.14

*Other IDs available depending on tubing size selected.

Hydraulic-Set Guiberson® G-77 Packer

Halliburton's Guiberson® G-77 packer is a hydraulically set, mechanically held pressure-balanced packer that is ideal for deep, high-pressure oil and gas production. The hydraulic-setting feature combined with the pressure balance system makes this packer ideal for multizone and deviated-hole applications. The G-77 packer is hydraulically actuated and pressure-balanced to offset pressure fluctuations and reversals when the packer is in the set position.

Applications

- Multizone production or injection
- Single-trip completions
- Deviated wellbores
- Up to 5,000 psi (34.5 MPa) differential pressure typical

Features

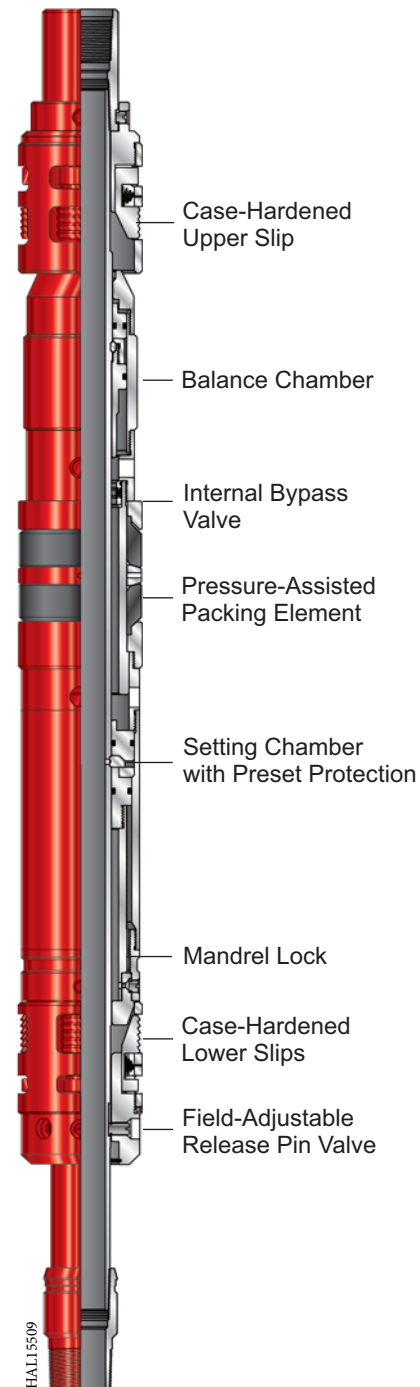
- Opposing slip design holds forces from either direction while loaded slip remains on low pressure side of the tool
- Compatible with packer accessories
- Innovative packing element system for positive pressure-enhanced pack-off
- Solid pre-set prevention system
- Internal bypass valve

Benefits

- Packer does not require tubing manipulation to set
- Multiple packers can be set close together
- Handles differential pressure from below without excessive releasing shear values
- Field-redressable
- Releasing shear value is field-adjustable without disassembly

Operation

The G-77 packer is set by establishing a temporary plug in the tubing below and applying setting pressure to the tubing. The packer is released by straight upstrain to the tubing string in excess of the previously selected releasing shear value.



Hydraulic-Set
Guiberson® G-77 Packer

Hydraulic-Set Guiberson® G-77 Packer

Casing Size		Casing Weight		Packer OD		Minimum ID		Tubing Size	
in.	mm	lb/ft	kg/m	in.	mm	in.	mm	in.	mm
4 1/2	114.30	9.5-13.5	14.14-20.09	3.75	95.25	1.94	49.23	2 3/8	60.33
5	127.00	11.5-15	17.11-22.32	4.25	107.95	1.94	49.23	2 3/8	60.33
		15-18	22.32-26.79	4.13	104.78				
5 1/2	139.70	13-15.5	19.35-23.07	4.75	120.65	1.94	49.23	2 3/8	60.33
		17-20	25.30-29.76	4.63	117.48				
		20-23	29.76-34.22	4.50	114.30				
6 5/8	168.30	17-24	25.30-35.72	5.69	144.47	2.44	61.93	2 7/8	73.03
		26-32	38.67-47.62	5.50	139.70				
7	177.80	17-20	25.30-29.76	6.13	155.58	2.44 or 3.00	61.93 or 76.20	2 7/8 or 3 1/2	73.03 or 88.90
		22-26	32.74-38.69	6.00	152.40				
		28-32	41.67-47.62	5.88	149.23				
7 5/8	193.70	20-26.4	29.76-39.29	6.69	169.88	3.00	76.20	3 1/2	88.90
		29.7-39	44.20-58.04	6.44	163.53				
8 5/8	209.10	28-36	41.67-53.57	7.56	192.10	3.00	76.20	3 1/2	88.90
		40-49	59.53-72.92	7.38	190.50				
9 5/8	244.50	32.3-40	48.07-59.53	8.50	215.90	4.00	101.60	4 1/2	114.30
		43.5-47	64.74-69.94	8.38	212.72				
		53.5	79.62	8.25	209.55				

Ordering Information

Specify: casing size and weight; tubing size; weight; grade; and thread; service environment (standard, %H₂S, %CO₂, amines/ other chemicals, chloride content, pressures, temperatures, etc.); maximum differential pressure requirement.

Part Number Prefix: 12GGG, D

Hydraulic-Set RDH Dual Production Packers

RDH hydraulic-set packers are designed to be set by pressuring up the tubing string against a plugging device below the packer.

RDH packers lock-set mechanically using spring-loaded internal slips. A lower set of mechanical slips holds the packer set against downward forces and an upper set of pressure-actuated button slips holds the packer set against differential pressure from below. The packers are released and retrieved by a straight upward pull on the tubing string.

Applications

- In deep or deviated wells requiring dual completion strings

Features

- Tungsten-carbide disk inserts in the holddowns are protected from the casing wall as the packer is lowered into the well.

- Triple-seal multidurometer elements have two hard end elements with molded self-retracting springs and a softer middle element for casing irregularities and low pressures.
- Contoured case-carburized slips conform without deforming casing.
- Dovetail slip design helps release the packer when retrieving.
- Optional heads allow use of various latching arrangements.
- Shear screws allow operator to vary the setting pressure and release force.

Benefits

- Has a low-pressure setting requirement so rig pumps usually provide enough pressure
- Allows varied setting pressures with varied numbers of shear screws
- Requires only a straight pull to release
- Allows adjustment of pull-to-release in the field
- Works well in all types of casing

Hydraulic-Set RDH Dual Production Packers

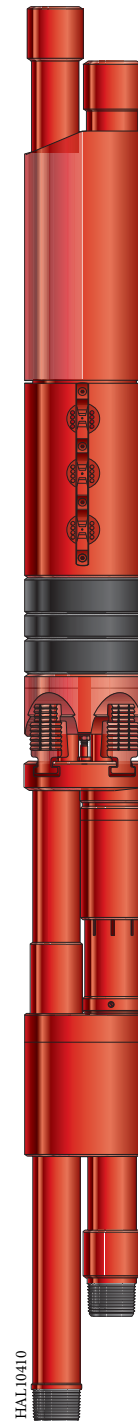
Casing Size		Casing Weight		Packer OD		Primary ID*		Secondary ID*	
in.	mm	lb/ft	kg/m	in.	mm	in.	mm	in.	mm
6 5/8	168.29	20-28	29.76-41.67	5.63	143.00	1.94	49.28	1.56	39.62
7	177.80	20-26	29.76-38.69	5.94	150.88	1.94	49.28	1.94	49.28
		23-29	34.22-43.16	6.03	153.16	2.38	60.45	1.53	38.86
		26-32	38.69-47.62	5.94	150.88	1.94	49.28	1.94	49.28
		35	52.09	5.95	151.13	2.38	60.45	1.53	38.86
7 5/8	193.68	26.4-29.7	39.29-44.20	5.84	148.34	1.92	48.77	1.54	39.12
				6.64	168.66	1.94	49.28	1.94	49.28
		29.7-39	44.20-58.04	6.64	168.66	2.37	60.20	1.91	48.51
				6.48	164.59	1.91	48.51	1.91	48.51
9 5/8	244.48	36-43.5	53.57-64.74	6.44	163.58	2.38	60.45	1.93	49.02
				8.62	218.95	2.90	73.66	2.90	73.66
		40-47	59.53-69.94	8.44	214.38	2.90	73.66	2.44	61.98
				8.44	214.38	2.90	73.66	2.90	73.66
				8.50	215.90	2.89	73.41	2.90	73.66
				8.50	215.90	3.81	96.77	1.92	48.77
		43.5-53.5	64.74-79.62	8.34	211.84	2.90	73.66	2.44	61.98

*Other IDs available depending on tubing size selected.

Ordering Information

Specify: casing size and weight; tubing size; weight; grade; and thread; service environment (standard, %H₂S, %CO₂, amines/other chemicals, chloride content, pressures, temperatures, etc.); maximum differential pressure requirement, primary or secondary string set, secondary latch type.

Part Number Prefix: 12RDH



HAL10410

Hydraulic-Set
RDH Packer

Seal Assemblies for Hydraulic-Set RDH Dual Packers

Head Configurations for RDH Packers

The following head configurations for RDH packers are available for both short-string and long-string set packers:

- J-slot latch that releases with only a 1/4 right-hand turn of the tubing at the packer
- Collet latch that requires no rotation for latching or unlatching; it indicates this action at the surface
- Collet/sleeve latch that operates the same as the collet latch except the sleeve has to be shifted to lock and unlock the collet

Options

These head configurations are available with molded or premium seal elements.

Dual Hydraulic-Set Packer Latch Selection Chart

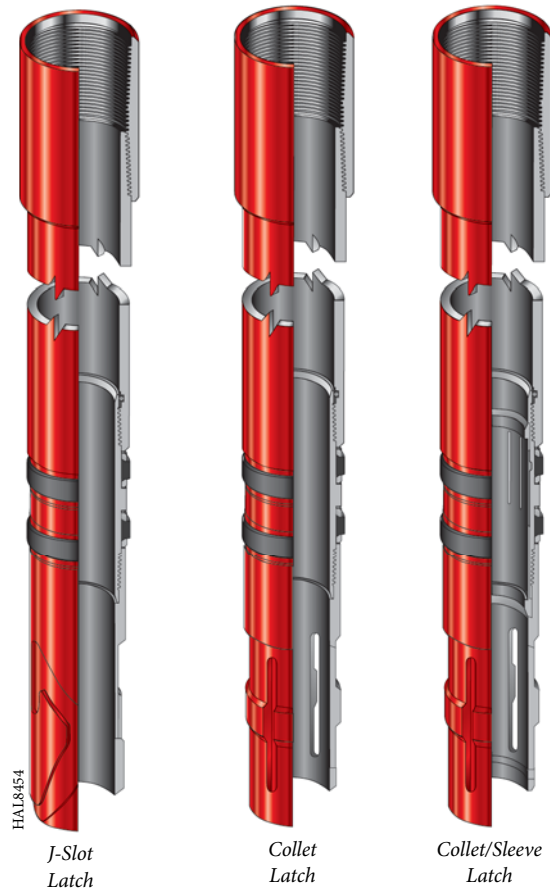
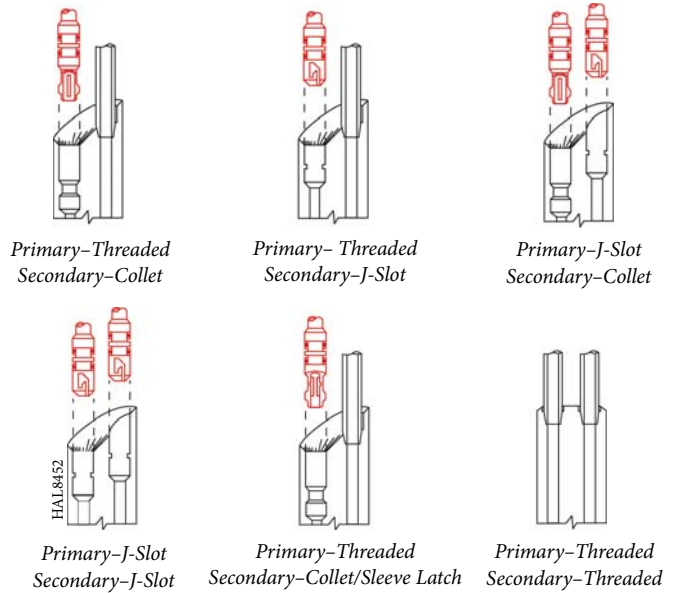
Type Latch	Latch Arrangement*	
	Primary	Secondary
LT	Threaded	Threaded
LB	Threaded	Collet
LA	Threaded	J-Latch
LSC	Threaded	Steamline Collet
LJ	J-Latch	J-Latch
LC	J-Latch	Collet
SSC	Collet (sliding sleeve)	Collet (sliding sleeve)
SC	Collet (sliding sleeve)	Collet

*Other combinations available

Ordering Information

Specify: casing size and weight; sizes; weights; grades; and threads for all tubing strings; latch type for each tubing string (refer to dual hydraulic-set packer latch selection chart); service environment (standard, %H₂S, %CO₂, amines/other chemicals, chloride content, pressures, temperatures, etc.); approximate weight to be suspended below packer; selective set (Y/N); maximum differential pressure requirements; setting string and desired setting pressure; desired shear release value.

Part Number Prefix: 12RDH



Special Application Packers

This section describes ZoneGuard®, electric submersible pump (ESP), utility, and R4™ treating packers.

ZoneGuard® FP Packer

The ZoneGuard FP isolation packer is designed for openhole and cased hole applications that require multizone isolation for stimulation, fracturing, or general production operations. The packer uses a uniquely designed element package with backup and deployment system to deliver consistent sealing performance in a defined range of openhole and cased hole conditions. It features a compact design ideal for short radius well conditions, which is also a benefit in wells where a large number need to be run.

The ZoneGuard FP packer is run as an integral part of the production casing or tubing string and is set with hydraulic pressure by setting a plug beneath the tool.

Applications

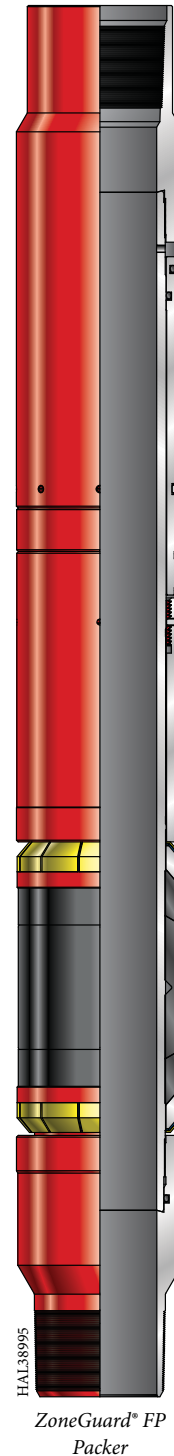
- Horizontal or vertical completions
- Openhole isolation
- Stimulation
- Fracturing
- Stage cementing
- Water shutoff
- Gas shutoff
- Production testing
- Selective production

Features

- Long-term isolation reliability
- Small running OD
- Compact length for short radius well conditions
- Multi-piece/multi-durometer element package
- Adjustable setting shear value
- Internal locking system
- Hydraulic set

Benefits

- Compact and economic design suited for multizone fracturing treatments
- Selective production management in horizontal wellbores
- Ability to reliably control inflow or injection within selected sections of the wellbore



ZoneGuard® FP Packer

SAP Number (Legacy)	Tool Connection	Maximum Metal OD in.	Minimum ID in.	Minimum Borehole in.	Maximum Borehole in.	Length in.	Temperature Rating °F	Pressure Rating	
								Minimum Bore ID psi	Maximum Bore ID psi
102311255 (12OHL4552392-D)	4 1/2 Blank	5.75	3.900	5.875	6.5	52.06	325	10,000	8,000
102284144 (12OHL4552390-D)	4 1/2 13.5 lb API-LC	5.75	3.900	5.875	6.5	47.91	325	10,000	8,000
102312189 (12OHL4552391-D)	4 1/2 13.5 lb API-BC	5.75	3.900	5.875	6.5	47.91	325	10,000	8,000
102292931 (12OHL5572460-D)	5 1/2 23.0 lb API-LC	7.25	4.660	7.75	8	50.78	325	10,000	10,000
102292939 (12OHL5572461-D)	5 1/2 23.0 lb API-BC	7.25	4.660	7.75	8	52.94	325	10,000	10,000
102292942 (12OHL5572462-D)	5 1/2 Blank	7.25	4.660	7.75	8	50.78	325	10,000	10,000
102295852 (12OHL5582460-D)	5 1/2 23.0 lb API-LC	8.00	4.670	8.50	8.75	47.6	325	10,000	8,000
102295921 (12OHL5582461-D)	5 1/2 23.0 lb API-BC	8.00	4.670	8.50	8.75	50.26	325	10,000	8,000
102295861 (12OHL5582462-D)	5 1/2 Blank	8.00	4.670	8.50	8.75	51.70	325	10,000	8,000
102327173 (12OHL6682570-D)	24.0 lb 6 5/8 HC VAM®* TOP	8.20	5.800	8.50	8.75	80.77	325	6,000	6,000

*VAM® is registered trademark of Vallourec Oil and Gas France.

ZoneGuard® HE Packer

The ZoneGuard® HE packer is designed for situations where a high-expansion sealing element is required and when large variations in hole gauge diameter are anticipated. The packer uses a multi-durometer element package with a unique backup and deployment system to deliver unsurpassed sealing performance in a wide range of openhole conditions. The packer can be used for zonal isolation in formation control, selective stimulation and fracturing applications.

The ZoneGuard HE packer is run as part of the completion string and can be set either hydraulically with plug set below or with an isolation straddle tool when plugging is not possible. It contains a hydrostatic assist feature, which helps maintain positive pressure on the packer at all times.

Applications

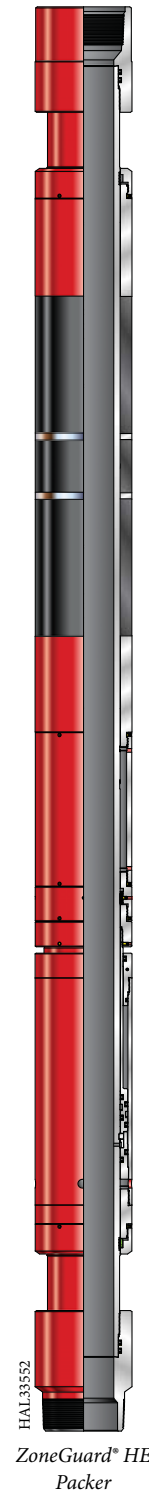
- Openhole applications (horizontal or vertical)
- Water shutoff
- Gas shutoff
- Stimulation
- Production testing
- Isolation
- Selective production
- Stage cementing

Features

- Small running OD
- Large element expansion capabilities
- Multi-piece/multi-durometer element package
- Adjustable setting shear value
- Internal locking system
- Long-term isolation reliability
- Hydraulically set plugged tubing and isolation straddle
- Hydrostatic assist allows positive pressure on element

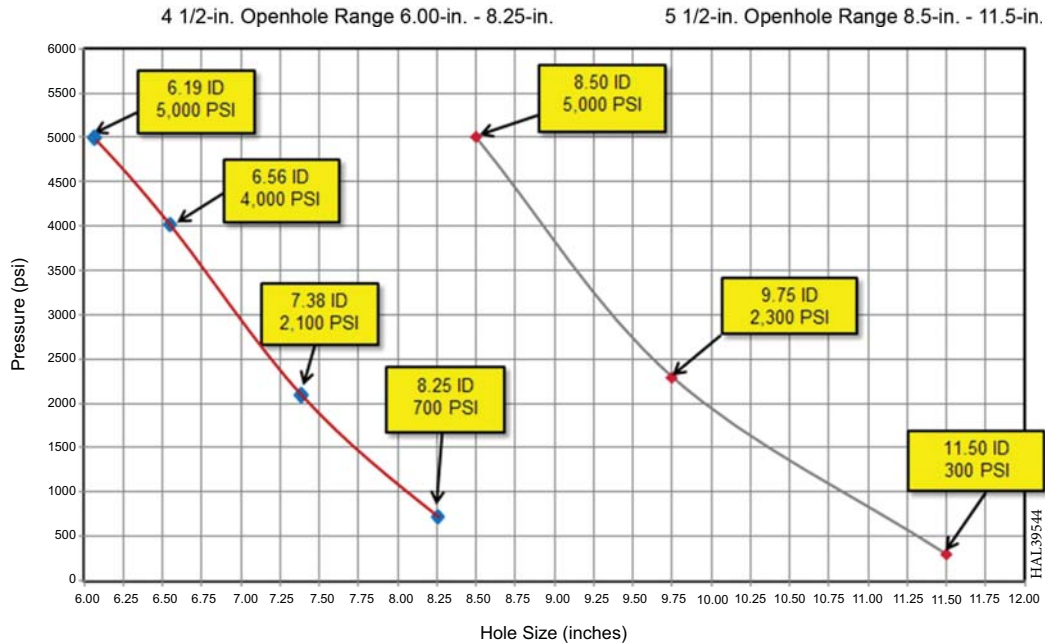
Benefits

- Selective production management in horizontal wellbores
- Ability to reliably control inflow or injection within selected sections of the wellbore
- Wide range of openhole isolation capabilities with one packer design



ZoneGuard® HE Openhole Packer Test Data

Differential Rating vs. Hole Size at 250°F



ZoneGuard® HE Packer

SAP Number (Legacy)	Tool Connection	Maximum Metal OD in.	Minimum ID in.	Minimum Borehole in.	Maximum Borehole in.	Length in.	Temperature Rating °F	Pressure Rating	
								Minimum Bore ID psi	Maximum Bore ID psi
102227674 (12OHN4560354-A)	4 1/2 12.60 lb VAM®** TOP	5.630	3.490	6	8 1/4	130.70	325	5,000	700
102143148 (12OHN4560353-A)	4 1/2 11.6 lb NEW VAM	5.630	3.490	6	8 1/4	130.70	325	5,000	700
102143147 (12OHN4560352-A)	4 1/2 12.75 lb API-EU	5.630	3.490	6	8 1/4	129.40	325	5,000	700
102144046 (12OHN5580493-A)	5 1/2 17.0 lb NEW VAM	8.000	4.670	8 1/2	11 1/2	142.33	325	5,000	300
102144021 (12OHN5580492-A)	5 1/2 17.0 lb API-LC	8.000	4.670	8 1/2	11 1/2	136.83	325	5,000	300
102338610 (12OHN6680490-F)	6 5/8 32 lb VAM TOP	8.000	4.890	8 1/2	11 1/2	140.88	325	5,000	300

*VAM® is registered trademark of Vallourec Oil and Gas France.

ZoneGuard® SR Packer

The ZoneGuard® SR (Short Radius) packer is designed for situations where tight radius well conditions are anticipated. The packer uses a multi-durometer element package with a unique backup and deployment system to deliver unsurpassed sealing performance in a wide range of openhole conditions. The design features an anti-preset feature to help protect against premature setting of the packer during running operations. The packer can be used for zonal isolation in formation control, selective stimulation, and fracturing applications.

The ZoneGuard SR packer is run as an integral part of the production casing or tubing string. The packer is set with hydraulic pressure either by setting a plug beneath the tool, or when plugging the completion string is not possible, by running an isolation straddle tool across the packer.

Applications

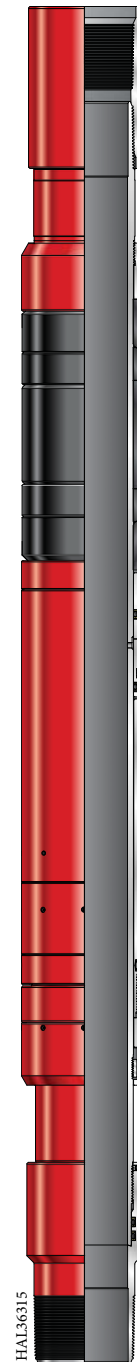
- Horizontal or vertical completions
- Openhole isolation
- Stimulation
- Stage cementing
- Water shutoff
- Gas shutoff
- Production testing
- Selective production

Features

- Long-term isolation reliability
- Small running OD
- Compact length for short radius well conditions
- Anti-preset feature
- Multi-piece/multi-durometer element package
- Adjustable setting shear value
- Internal locking system
- Multiple hydraulic setting methods

Benefits

- Ideally suited for short radius and higher differentials
- Selective production management in horizontal wellbores
- Ability to reliably control inflow or injection within selected sections of the wellbore



ZoneGuard® SR
Packer

ZoneGuard® SR Packer

SAP Number	Tool Connection in.	Maximum Metal OD in.	Minimum ID in.	Minimum Borehole in.	Maximum Borehole in.	Length in.	Temperature Rating °F	Pressure Rating		Legacy Number
								Minimum Bore ID psi	Maximum Bore ID psi	
102053962	4 1/2 Blank	5.65	3.85	5.87	6.25	70.48	325	10,000	7,000	12OHS4552382-B
102053960	4 1/2 13.5 lb API-LC	5.65	3.85	5.87	6.25	60.04	325	10,000	7,000	12OHS4552380-B
102309734	4 1/2 15.1 lb API-LC	5.81	3.724	5.87	6.25	60.61	325	10,000	7,000	12OHS4552372-E
102159273	4 1/2 13.5 lb API-LC	5.9	3.85	6.25	6.50	60.61	325	10,000	10,000	12OHS4562380-B
102123418	5 1/2 23.0 lb API-LC	7.25	4.67	7.50	7.87	80.24	325	10,000	8,000	12OHS5572460-D
102123419	5 1/2 23.0 lb API-BC	7.25	4.67	7.50	7.87	80.24	325	10,000	8,000	12OHS5572461-D
102123420	5 1/2 Blank	7.25	4.67	7.50	7.87	80.24	325	10,000	8,000	12OHS5572462-D
102107206	5 1/2 23.0 lb API-LC	8.00	4.67	8.50	8.75	80.36	325	10,000	7,000	12OHS5582460-D
102148005	5 1/2 23.0 lb API-BC	8.00	4.67	8.50	8.75	80.36	325	10,000	7,000	12OHS5582461-D
102148006	5 1/2 Blank	8.00	4.67	8.50	8.75	80.36	325	10,000	7,000	12OHS5582462-D
102267222	5 1/2 17.0 lb VAM® TOP	8.00	4.67	8.50	8.75	82.24	325	10,000	7,000	12OHS5582463-A
102260641	6 5/8 32.0 lb VAM TOP	8.25	4.89	8.50	8.75	82.25	325	10,000	7,000	12OHS6682490-D

*VAM® is registered trademark of Vallourec Oil and Gas France.

Electric Submersible Pump Packer

The electric submersible pump (ESP) packer is a single-trip, single-bore, hydraulic-set packer with a dual head designed to reduce the overall costs of submersible pump completions. It is made up on the tubing string allowing a full bore production tubing ID and passage of an electric cable and control line.

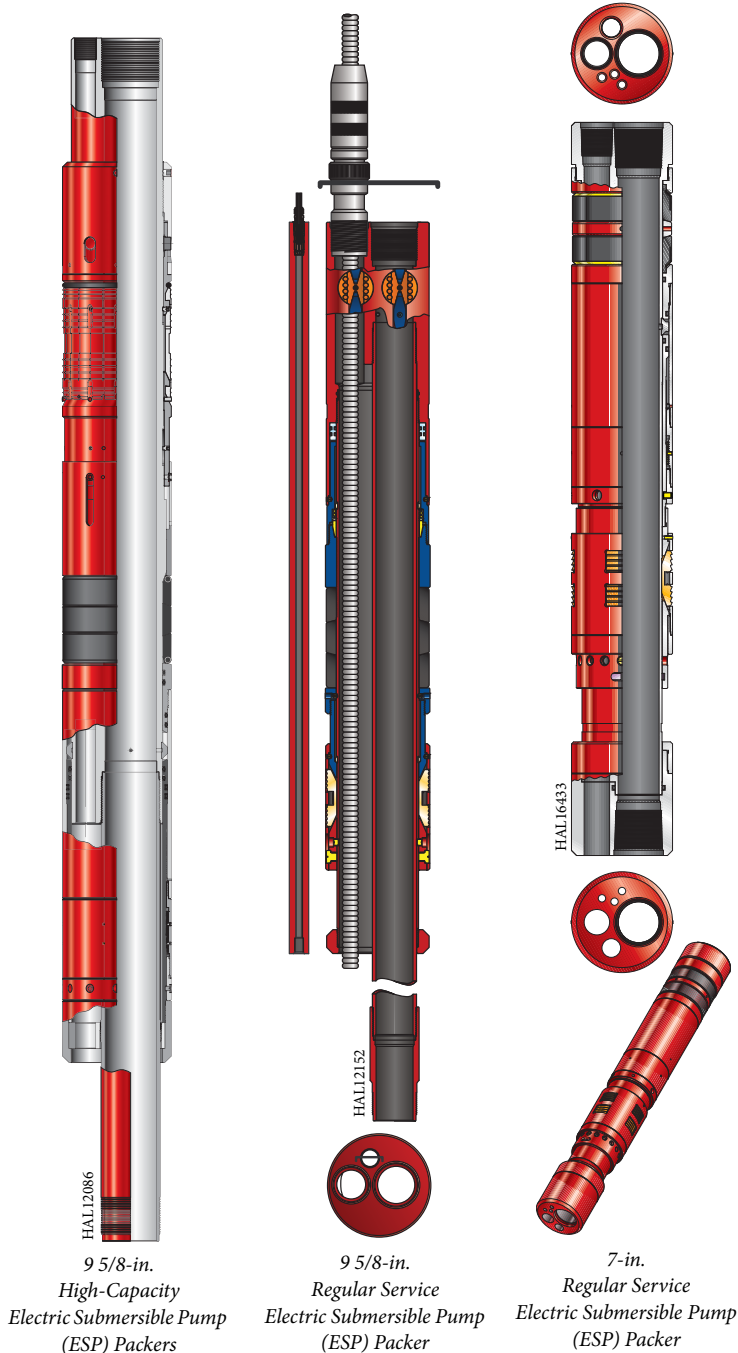
The ESP packer is designed to set with low hydraulic pressure. This allows the packer to be set shallow or in deviated wellbores. Concentric, symmetric elements require less force to compress to the set position, and the elements distribute the setting load more evenly for more dependable operation.

The triple-seal multidurometer element package is designed to maintain a seal against high or low pressures and to compensate for casing irregularities. Holddown buttons have been eliminated, reducing the number of potential leak paths.

Case carburized bi-directional slips anchor the packer against well pressure from above or below. Slip design offers exceptional hang weight capability. Straight pull shear release allows simple retrieval without a retrieving tool. Shear value is field adjustable. No tubing rotation is required to set or release.

Features and Benefits

- Compatible with feed-through equipment.
- Easy installation can help make marginally economic wells attractive to produce.
- Easy retrievability means less time spent on workovers. (ESP workovers often occur more frequently than conventional completion workovers.)
- Chemical injection and pressure monitoring ports can easily be added through the packer without significant costs.
- Field redress is quick, easy, and cost effective. Also, only two flow-wetted parts means lower costs when conditions require corrosion-resistant alloys.
- Performance is long-term and dependable. Elastomeric seals have been minimized. Only six o-rings in the assembly increases reliability.



B-Series Wireline-Retrievable Packers

B-series packers are large bore pack-off devices which can be set at a predetermined point anywhere in the tubing or casing.

A slip system on the tool's outer body anchors the packer to the wellbore while a packing element provides a pressure seal. The tools are typically run and retrieved using conventional well intervention methods along with Halliburton RO and PK service tools.

Applications

B-series packers are 'utility' packers designed for a wide range of applications.

The base model is the BB0 packer, which is primarily used as an anchoring device for the suspension of various well intervention tools, such as water injection valves, downhole gauges, and safety valves (Storm Choke® safety valves). It may also be used as a plugging device by attaching a remote equalizing sub (i.e., pump-out plug) or prong equalizing sub.

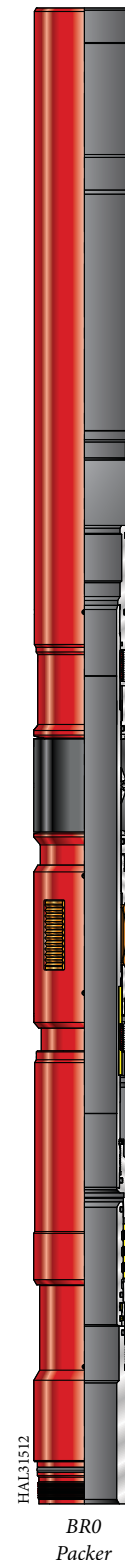
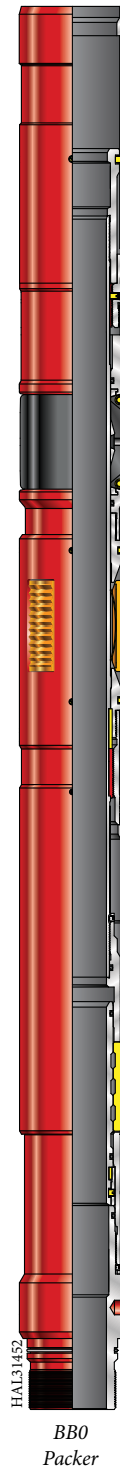
The BB0 model forms the base for both the BR0 sealbore packer and the BV0 velocity string packer.

The BR0 packer incorporates an upper seal receptacle, making it ideal for use in modular straddle systems in addition to the standard BB0 packer applications.

The BV0 packer utilizes a Ratch-Latch™ type running head which is designed for use with the Halliburton CT hydraulic running tool. This configuration allows large weights to be suspended from the packer while running in hole and makes the BV0 packer ideal for velocity string, long isolation straddle, and sand control applications.

Features

- One-piece dual modulus packing element
- Compact, modular design
- High running and retrieval speeds
- Large footprint segmented slip mechanism
- Slip mechanism located below packing element
- Slips mechanically retained on retrieval
- Controlled setting action
- OD components rotationally locked
- Field redressable



Benefits

- Modular design allows economical conversion of BB0 and BR0 variants for use in different operational conditions
- Compact design allows deployment in restricted work areas where limited height is available above the well and access is difficult
- Slip design and controlled setting action helps ensure the stresses exerted on the casing or tubing are evenly distributed thus preventing damage
- Mechanically retained slips reduce premature setting risk while running in hole and hanging upon retrieval
- Packing element design enhances its ability to return to its original shape upon release, reducing hanging up risk
- Slip mechanism position offers protection from casing debris thus improving reliability
- BB0 and BR0 packers can be run and retrieved on slickline, electric line, coiled tubing, or workstring; the BV0 packer deployment and retrieval method is dependent on the application

B-Series Wireline-Retrieval Packers

Casing/Tubing				Maximum OD		Minimum ID		Pressure Rating			
Size		Weight						Above		Below	
in.	mm	lb/ft	kg/m	in.	mm	in.	mm	psi	bar	psi	bar
2 7/8	73.03	6.4	9.52	2.280	57.91	1.000	25.40	5,000	344.50	5,000	344.50
		8.7	12.94	2.140	54.36			5,000	344.50	5,000	344.50
3 1/2	88.90	9.2	13.69	2.720	69.09	1.375	34.93	3,000	206.70	3,000	206.70
4 1/2	114.30	10.2	17.26	3.650	92.71	2.340	59.44	4,000	275.76	3,000	206.70
		12.7	18.75	3.650	92.71	2.340	59.44	4,000	275.76	4,000	275.76
		13.5	20.09	3.650	92.71	2.340	59.44	5,000	344.50	5,000	344.50
		15.1	22.47	3.650	92.71	2.340	59.44	4,000	275.76	4,000	275.76
5	127.00	15.0	22.31	4.280	108.71	2.500	63.50	4,000	275.76	4,000	275.76
		18.0	26.78	4.090	103.89	2.400	60.96	4,000	275.76	4,000	275.76
				3.970	100.84	2.400	60.96	2,000	137.88	2,000	137.88
		20.3	30.20	4.090	103.89	2.400	60.96	5,200	358.48	5,200	358.48
5 1/2	139.70	17.0	25.29	4.530	115.06	2.625	66.68	5,000	344.50	4,500	310.26
				4.470	113.54	2.625	66.68	3,000	206.70	3,000	206.70
		20.0	29.75	4.530	115.06	2.625	66.68	5,000	344.50	5,000	344.50
				4.470	113.54	2.625	66.68	3,000	206.70	3,000	206.70
				4.460	113.28	2.625	66.68	4,000	275.76	3,500	241.15
		23.0	34.21	4.460	113.28	2.625	66.68	5,000	344.50	4,500	310.26
		26.0	38.68	4.280	108.71	2.340	59.44	5,000	344.50	5,000	344.50
6 5/8	168.30	32	47.60	5.500	139.70	3.400	86.36	5,000	344.50	5,000	344.50
7	177.80	26.0	38.68	5.935	150.75	3.625	92.08	3,000	206.70	3,000	206.70
		29.0	43.15	5.720	145.29	3.625	92.08	5,000	344.50	4,000	275.76
		32.0	47.60	5.720	145.29	3.625	92.08	5,000	344.50	5,000	344.50
		35.0	52.06	5.720	145.29	3.625	92.08	5,000	344.50	5,000	344.50
		38.0	56.54	5.635	143.13	3.625	92.08	4,000	275.76	4,000	275.76

Part Number Prefix: P.803BB0, P.803BR0, P.803BV0

Evo-Trieve® Intervention Packer

The Evo-Trieve® intervention packer is a retrievable large bore packoff device for superior technical and operational performance. This high-performance packer can be set at a predetermined point anywhere in the tubing or casing.

Evolved from the industry-leading B Series monobore intervention product family, the Evo-Trieve packer system features a robust cage-type slip system that anchors the packer to the wellbore while a packing element provides a bi-directional pressure seal.

The Evo-Trieve intervention packer may be run and retrieved with conventional well intervention systems in conjunction with field-proven Halliburton service tools. The packer is V0 qualified per ISO 14310 to 5,000 psi and up to 275°F.

Applications

The Evo-Trieve intervention packer is a modular utility packer designed to be adaptable for a wide range of monobore intervention applications.

- Primarily used as an anchoring device for the suspension of well intervention tools including water injection valves, downhole gauge, and Storm Choke® safety valves
- May be used as an ISO-qualified plugging device by attaching a remote equalizing sub (pump-out plug) or prong equalizing sub
- Serves as base conversion unit for Evo-Trieve upper sealbore packer and used in conjunction with RS2 anchor latch assemblies
- Used as part of a system to provide isolation straddle solutions of any length

Features

- Supplied with H₂S packing element system as standard for critical well deployment and reduced inventory management
- Packing element positioned above slips for improved debris protection
- Robust slip system permits no over-expansion in set position while mechanically locked in retracted position when released
- Body lock ring system retains element packoff force during pressure reversals
- Field-proven sleeve-type release mechanism
- Anti-reset ring helps provide improved retrieval capabilities by locking out external components
- Easily redressed in field or at rig site



*Evo-Trieve®
Intervention Packer*

Part Number Prefix: P.803EB0; P.803ER0;

Benefits

- Design allows operator to quickly and economically adapt for use in different operational requirements
- Compact design allows deployment in restricted work areas where limited rig-up height availability makes for difficult access
- Controlled setting action and slip design helps ensure stress exerted on the casing or tubing is evenly distributed thus preventing damage
- Mechanically retained slips reduce premature setting risks while running in hole and hanging up on retrieval
- Packing element design enables it to return to original shape upon release, reducing hanging up risk
- Slip mechanism position offers protection from casing debris for improved reliability
- May be run and retrieved on slickline, electric line, coiled tubing, or workstring

Evo-Trieve® Intervention Packer

Casing/Tubing				Maximum OD		Minimum ID		To Pass Restriction		Pressure Rating - V0 Tested				Temperature	
Size		Weight								Above		Below			
in.	mm	lb/ft	kg/m	in.	mm	in.	mm	in.	mm	psi	bar	psi	bar	°F	°C
5 1/2	139.70	17	25.30	4.53	115.06	2.63	66.68	4.56	115.87	5,000	344.50	5,000	344.50	275	135
		20	29.76												
7	177.8	29	43.16	5.720	145.29	3.625	92.08	5.750	146.1	5,000	344.50	5,000	344.50	275	135

GO™ Packer

Halliburton's GO™ packer is designed for use in macaroni tubing. Its single-seal element works in a wide range of temperatures. Its automatic J-latch arrangement is useful in running, setting, and releasing the packer. Slips help anchor this packer in the well.

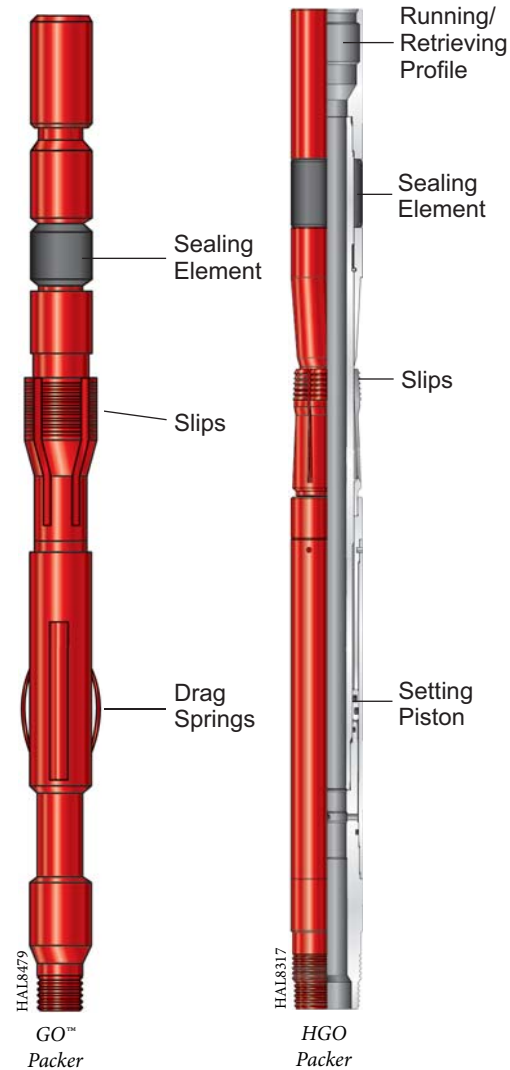
HGO Packer

The HGO packer is a hydraulic-set version of the GO packer designed for use in coiled tubing completions. The HGO packer is run on coiled tubing and set hydraulically. The running tool releases from the packer, leaving the packer and coiled tubing suspended in the well. The packer is retrieved by straight pull on the packer with a coiled tubing-deployed retrieving tool.

Ordering Information

Specify: casing size and weight; type (GO, CP, HGO); thread type; size; and weight for top and bottom connections; service environment (standard, %H₂O, %CO₂, amines/other chemicals, chloride content, pressures, temperature, etc.).

Part Number Prefix: 12GO—GO packer, 12CP—cup packer, 12HGO—hydraulic-set GO packer



GO™ Packers

Casing Size		Casing Weight		Maximum OD		Minimum ID*		Tubing Size**	
in.	mm	lb/ft	kg/m	in.	mm	in.	mm	in.	mm
2 3/8	60.33	4.6	6.85	1.84	46.74	1.00	25.40	1.34	33.40
		4.7	6.99	1.84	46.74	1.00	25.40	1.32	34.14
2 7/8	73.03	6.4-6.5	9.52-9.67	2.28	57.91	1.38	35.05	1.66	42.16
3 1/2	88.90	9.3	13.84	2.81	71.37	1.73	43.94	2.06	52.40
				2.81	71.37	1.90	48.26	2.38	60.33
4 1/2	114.30	11.6-13.5	17.26-20.09	3.64	92.46	1.525	38.74	1.900	48.26
5	127.00	15-18	22.32-26.79	4.06	103.12	1.98	50.29	2.38	60.33
7	177.80	29-32	43.16-47.62	5.77	146.56	2.43	61.72	2.88	73.03

*Minimum ID dependent on tubing size and thread type selected.

**Other tubing sizes available on request.

R4™ Treating Packer

The Halliburton R4™ treating packer is a full opening, retrievable packer designed to be run in casing as a tension-type hookwall packer. Drag springs are attached to the slip retracting sleeve to furnish a friction drag so that the J-slot can be operated. A left-hand thread safety joint is provided to allow the tubing to be backed off and removed from the well if the packer cannot be retrieved. The R4 treating packer, although designed to be run as a tension packer, may be used as a conventional compression hookwall type packer without any modifications by running the packer upside down. It is designed for a 5,000 psi (34.5 MPa) differential working pressure.

Applications

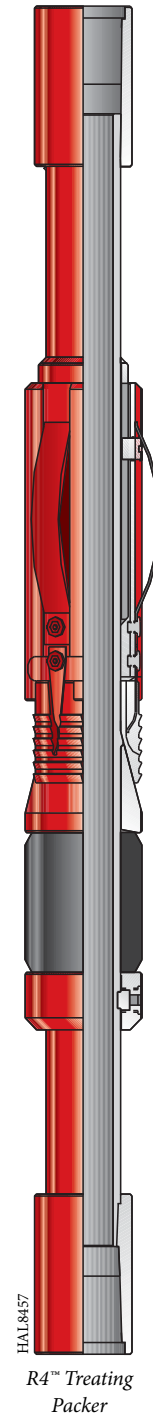
- Water injection
- Treating packer

Features

- Tension-set for shallow wells
- Emergency shear release

Benefits

- Left-hand thread safety joint



Ordering Information

Specify: casing size and weight; tubing size; weight; grade; and thread; service environment (standard, %H₂S, %CO₂, amines/other chemicals, chloride content, pressures, temperatures, etc.); maximum differential pressure requirement.

R4™ Treating Packer

Casing				Top and Bottom EUE Tubing Thread*		Minimum Tool ID		Maximum Tool OD	
Size		Weight Range							
in.	mm	lb/ft	kg/m	in.	mm	in.	mm	in.	mm
2 7/8	73.0	6.5	9.67	1.32	33.4	0.88	22.4	2.22	56.4
3 1/2	88.9	9.3-10.3	13.84-15.32	1.90	48.3	1.34	34.0	2.75	69.8
4 1/2	114.3	9.5-13.5	14.14-20.09	2 3/8	60.3	1.94	49.3	3.75	95.3
5	127.0	11.5-18	17.11-26.79	2 3/8	60.3	1.94	49.3	4.06	103.1
5 1/2	139.7	13-23	19.35-34.23	2 7/8	73.0	2.36	59.9	4.50	114.3
6 5/8	168.3	24-32	35.71-47.62	2 7/8	73.0	2.36	59.9	5.37	136.4
7	177.8	17-35	25.30-52.09	2 3/8	60.3	2.36	59.9	5.75	146.1
8 5/8	209.1	24-49	35.71-72.91	3 1/2	88.9	2.88	73.2	7.25	184.2
9 5/8	244.5	29.3-53.5	43.60-79.60	3 1/2	88.9	2.88	73.2	8.25	209.6

*All are 8 Rd EUE tubing thread except the 2 7/8-in. (73.0 mm) packer which has a 10 Rd EUE tubing thread and the 3 1/2-in. (88.9 mm) packer which has a 10 Rd NU tubing thread.

Packer Completion Accessories

This section describes the Halliburton line of packer completion accessories, including:

- Sealbore and millout extensions
- Seal assemblies
- Seal units
- Polished bore receptacle and seal units
- Travel joints
- Adjustable unions
- Twin-flow assemblies
- On-off tools
- Sealing plugs and backpressure valves
- Catcher subs
- Anvil® plugging system
- Mirage® disappearing plug
- Tubing safety joints
- Tubing tester flapper valve
- Setting adapter kits

Sealbore and Millout Extensions

Halliburton's packer sealbore extensions extend the polished surface below the packer. This enables the use of longer sealing units to compensate for tubing contraction or elongation. The extensions are available in standard 8-, 20-, and 24-ft (2.44-, 6.10-, and 7.32-m) increments.

Halliburton millout extensions provide a large ID below the packer sealbore or sealbore extension, which allows a single-trip packer milling/retrieving tool to be used when tubing is run below the packer assembly.

Applications

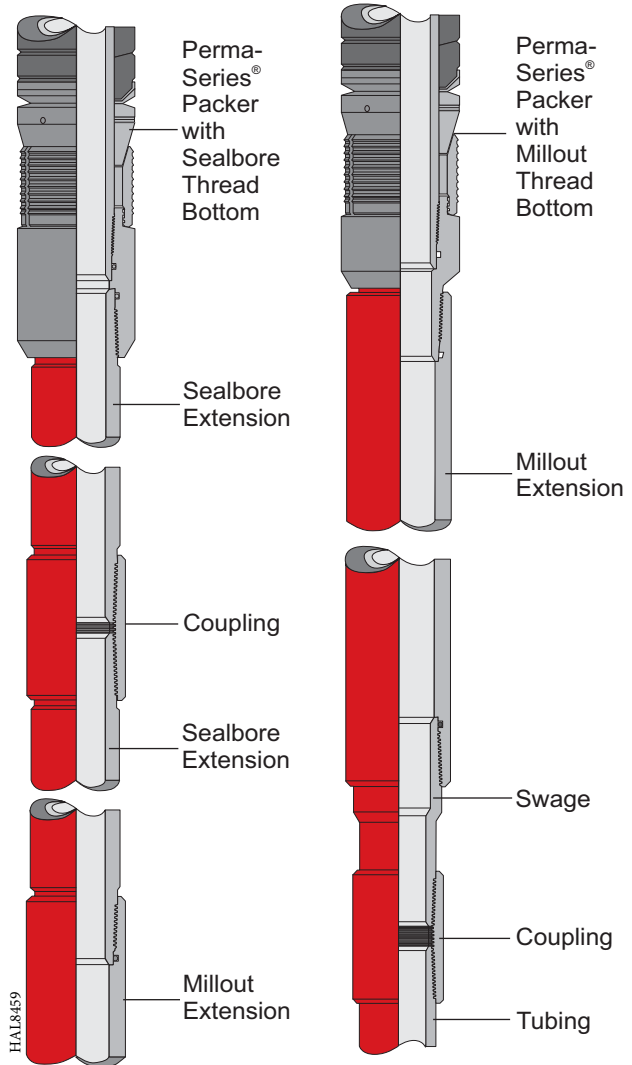
- For using longer sealing units to compensate for tubing contraction or elongation (sealbore extensions)
- To allow a single-trip packer milling retrieving tool to be used when tubing is run below the packer assembly (millout extensions)

Benefits

- Enables the use of longer sealing units to compensate for tubing contraction or elongation (sealbore extensions)

Options

- Sealbore extensions available in 8-, 20-, and 24-ft (2.44-, 6.11-, and 7.32-m) increments



Sealbore and Millout Extension

Ordering Information

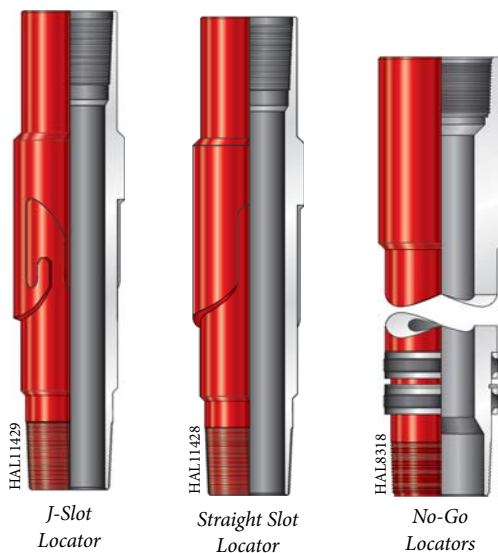
Specify: casing size and weight, packer with which assembly will be used, packer bore, length of sealbore extension (standard lengths = 8, 20, and 24 ft), service environment (standard, %H₂S, %CO₂, amines/other chemicals, chloride content, pressures, temperatures, etc.), tubing material, bottom thread, special material or assembly thread requirements, if applicable.

Part Number Prefix: 212

Seal Assemblies

Halliburton seal assemblies for permanent and retrievable sealbore-type packers act as a seal between the tubing string and the packer. If components will be exposed to corrosive fluids, the seal's standard alloys may be plated for extra protection or manufactured from corrosive-resistant alloys. A Halliburton seal assembly consists of three major components:

- Locator
- Molded or premium seal units
- Muleshoe guide, collet, or production tubing



J-Slot, Straight-Slot, and No-Go Locators

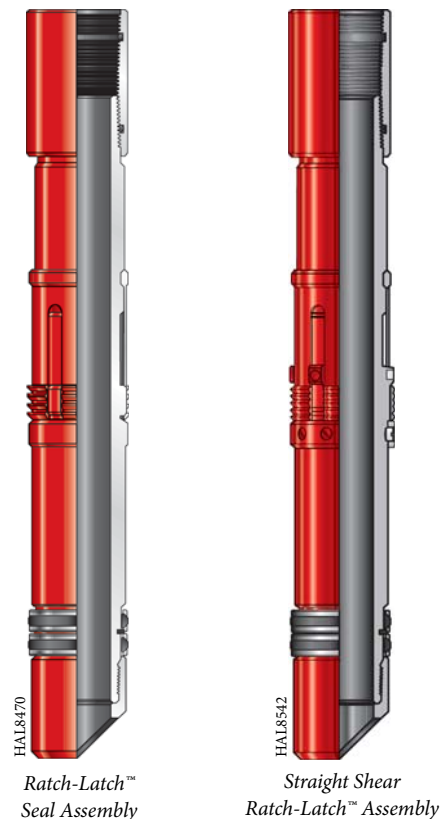
The J-slot locator anchors the tubing string to the packer. The J-slot allows automatic latching by slacking off tubing to engage the J-slot lugs in the packer. The latch releases when the tubing is lowered, rotated approximately 1/4 turn to the right at the packer, and then pulled upward.

The straight-slot locator allows the locator slots to engage the lugs in the packer. The seal assembly may be spaced across the packer and sealbore with the locator above the packer to allow for tubing elongation. Removal from the packer requires a straight upward pull of the tubing.

No-go locators are used with permanent packers with the short Ratch-Latch™ type receiving head. These locators provide a positive locating stop for the tubing at the packer. These locators can be provided with the correct spacing between the no-go and seals because the lower end is machined to accept either molded seals or premium seal units.

Ratch-Latch™ Seal Assembly

No rotation is needed to install the Ratch-Latch seal assembly. This positive latch system minimizes seal movement in the well. Right-hand tubing rotation is used to disengage the Ratch-Latch assembly. It may be relatched or pulled with the tubing for redress operations. The straight shear Ratch-Latch assembly can be disengaged by either rotation or straight pull.



Ordering Information

Specify: packer bore; latch type (J-slot, straight slot, Ratch-Latch™, no-go); special makeup thread requirements if applicable (standard or premium); nominal makeup length required (2-, 4-, 6-, 8- or 10-ft standard lengths available); service environment (standard, %H₂S, %CO₂, amines/other chemicals, chloride content, pressures, temperatures, etc.); material (L-80 alloy, P-125 alloy, 9Cr-1Mo, or Incoloy® 925) molded seal units are designed with a high-modulus rubber compound (Nitrile or Fluorel) that is bonded to two metal backup shoes. The shoes are designed to pressure-energize the seals, increasing differential pressure so the element seals more tightly. These seal units have proved effective in wells with highly abrasive fluids and in wells when the unit must be set and released many times.

Seal Units

Halliburton offers molded, crimp, and premium seal units. Molded and crimp seals are recommended for applications in which the seal will be going in and out of the sealbore. Premium seals are ideal for applications requiring the seal units to remain in the sealbore.

All seal units are designed for easy field redress. The geometry of the seals enables them to provide a more positive seal as they are subjected to higher pressures and temperatures. Many types of seal units are available, and others can be designed to fit specific needs.

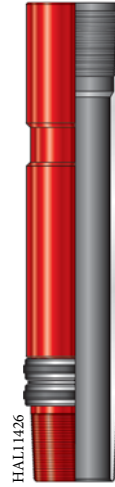
Premium seal units are v-packing seal systems of either elastomeric or plastic materials with a plastic and metal backup system. The lips of the v-ring are designed to have an initial interference fit with the sealbore. The lips respond to pressure increases by flexing outward. Various backup material combinations can be used with each v-packing seal to make it suitable for different temperatures and pressures.

Crimp seals, in combination with centering rings and debris barrier rings, can be fitted on very long seal mandrels, which eliminates threaded connections typically used on successive chevron stacks. The crimp seal consists of a rubber seal ring that is fitted to a groove on the seal mandrel. The seal is retained to the groove by a metal ring crimped over the seal. It is available in various elastomers and can be fitted with anti-extrusion backup rings.

While it has long been considered a best practice in HP/HT sealbore applications to limit seal movement by using a Ratch-Latch™ or seal anchor, treating jobs or production cycles under these conditions can reach the mechanical limit of the completion. ESET premium seal units were developed specifically for these applications which require dynamic sealing capabilities.



Crimp Seal



Molded Seal Unit



Cross-section of Molded Seal



Premium Seal Unit



Cross-section of Premium Seal



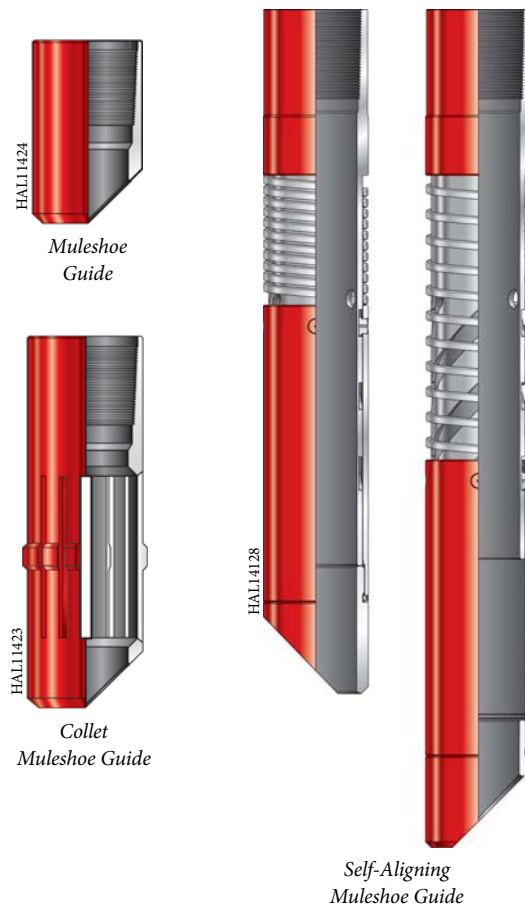
ESET Premium Seal Unit

Muleshoe Guides

Muleshoe guides provide a means to guide the end of the tubing away from the casing wall, then enter liner tops or the packer bores. The length of the muleshoe varies with the application, from centralization, to seal guide and protection, to flow isolation sleeve.

Collet muleshoe guides combine the features of the muleshoe with an indicator collet to provide a surface indication of the packer seals entering or leaving a packer bore. Push-through and no-go type collets are available for indication on the packer or a special ID sub below the packer.

Self-aligning muleshoe guides allow the end of the guide to rotate and orient with the liner top without rotation of the tubing. This tool is especially useful in dual wellbore or horizontal completions where control of tubing rotation downhole is difficult.



Ordering Information

Seal Units and Muleshoe Guides

Specify: packer bore; special makeup thread requirements, if applicable (standard or premium); service environment (Standard, %H₂S, %CO₂, amines/other chemicals, chloride content, pressures, temperatures, etc.); material (L-80 alloy, 9Cr-1Mo, or Incoloy® 925).

Part Number Prefixes: 212G, 212SDG, 212COL

Polished Bore Receptacle and Seal Units

An optional upper polished bore receptacle (PBR) above the packer and seal system has been designed for a variety of hydraulic-set retrievable or Perma-Series® packers to provide a stroke of up to 30 ft (9.14 m). This system allows for a floating seal to compensate for tubing elongation or contraction, and it maintains the large bore through the packer, PBR, and seal assembly. The system also retains the single-trip feature.

Applications

- Used with retrievable or Perma-Series® packers in applications which require large bores, dynamic sealing, and the ability to retrieve the upper completion

Features

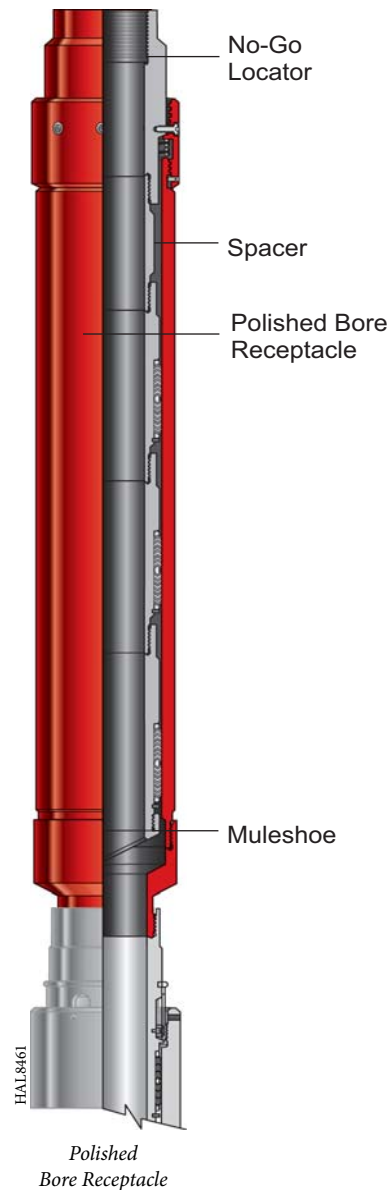
- Metal-to-metal premium thread connections
- A standard option wiper ring/debris barrier
- Minimum thread connections
- Single-piece lengths of 10 and 30 ft (3.05 and 9.14 m)
- Variable spacing shear ring
- Right-hand rotation releases the PBR from the packer

Benefits

- Seal units and the PBR are retrievable without disturbing the production packer
- Wide variety of corrosion-resistant alloys and elastomers available
- Predetermined space out available through a variable spacing shear ring
- Maintains maximum available IDs through the packer assembly
- Allows a single trip
- Allows easy retrieval of the seal units and PBR without disturbing the production packer

Options

- With a special packer head arrangement, an optional straight shear release Ratch-Latch™ assembly is also available for completions in which rotation release is not possible
- 10-, 20-, 30-ft stroke options available
- Corrosion-resistant alloys and elastomers available
- Metal-to-metal sealing capability in closed position



Ordering Information

Specify: casing size and weight; thread type; size; and weight for top of seal assembly; type of connection to packer (Ratch-Latch™ or seal anchor); seal space out (full closed or stroked out position); service environment (standard, %H₂S, %CO₂, amines/other chemicals, chloride content, pressures, temperatures, etc.); tubing material, seal assembly stroke; if applicable; seal type on seal units and anchor/latch assembly.

Part Number Prefix: 212PBA

Travel Joints

Halliburton travel joints provide alternate methods of compensating for tubing contraction and elongation in producing, injection, and disposal wells. They are run as an integral part of the tubing string and have a full tubing drift ID that is compatible with other downhole control equipment. After packers and associated equipment are in place, travel joints may be activated. Depending on the application, travel joints may be activated by shear pins or standard wireline tools.

Options

- Allow tubing torque below by engagement of clutch, spline, or lug and slot
- Available in versions that may be locked in open, mid-stroke, or closed positions
- Available in standard tubing sizes
- May be trimmed with standard or sour service seal packages
- Available in versions equipped with centralizer rings to guard against abnormal seal wear and to clean accumulated solids from the inner mandrel during the operation of the well

Swivel Travel Joints

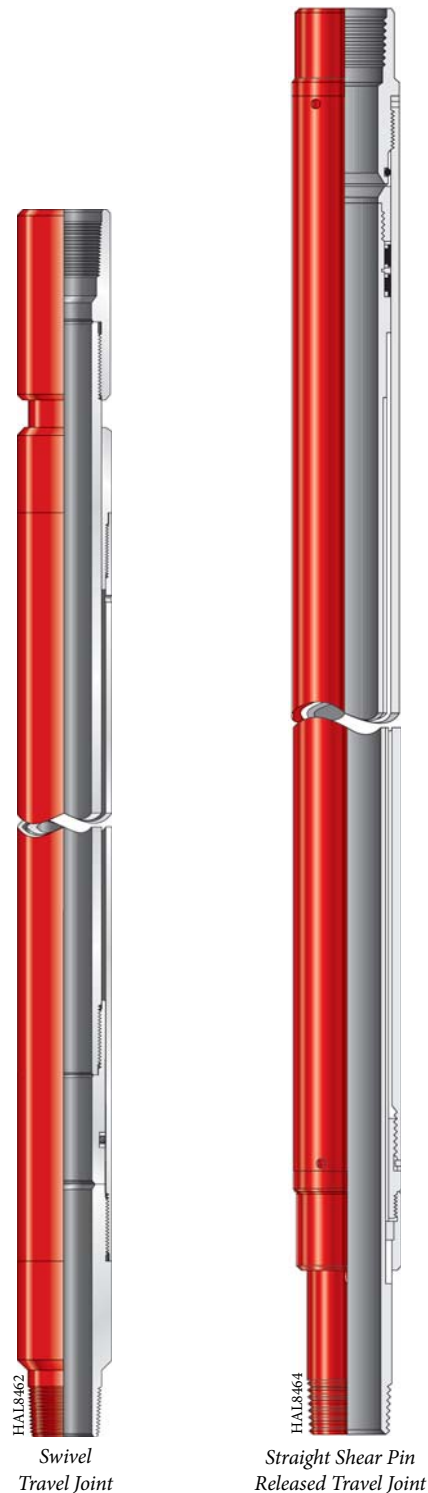
Swivel travel joints help with spacing out between packers or near the surface. They offer continuous 360° rotation while isolating the tubing from the annulus.

Straight Shear Pin Released Travel Joints

These travel joints may be sheared by tension or compression. They also may be sheared during initial completion operations or left in the pinned condition to be sheared at a predetermined value later. These travel joints can be pinned at any point in the stroke. The release valve can be varied by the number of shear screws. Splined designs are available when torque must be transmitted through the travel joint.

Ordering Information

Specify: casing size and weight; tubing size; weight; grade; and thread; type (wireline-release, H-slot and shear-pin release, straight shear-pin release, swivel, swivel with clutch, ability to lock or pin in opened, mid-stroke, or closed positions); stroke; service environment (standard, %H₂S, %CO₂, amines/other chemicals, chloride content, pressures, temperatures, etc.); tensile/hand weight requirements; special material or elastomer requirements, if applicable



Keyed Travel Joints

Keyed travel joints are similar to the swivel type but are keyed the length of the travel joint. The keys allow torque to be transmitted through the full stroke of the travel joint.

Long Space Out Travel Joints

The Halliburton long space out travel joint (LSOTJ) was designed to ease the space out process required to install a production tubing string in a subsea completion. The joint is designed to stroke closed under a compressive load after a production seal assembly has been landed in a sealbore packer. After the joint strokes, the production tubing can be lowered until the subsea tubing hanger lands in the subsea tubing head spool.

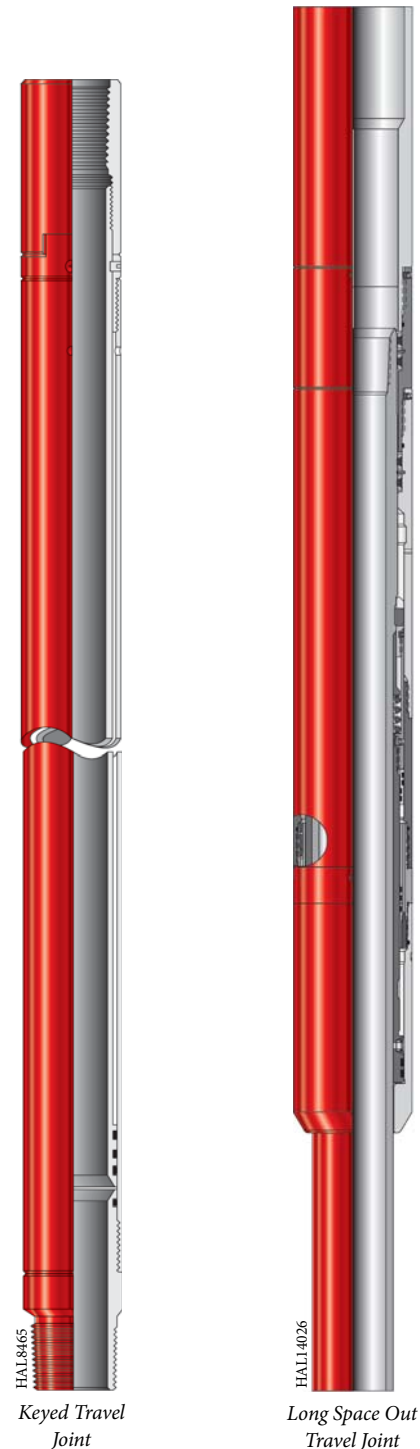
Travel Joints

Size		OD		ID		Stroke	
in.	mm	in.	mm	in.	mm	ft	m
2 3/8	60.33	3.65	92.71	1.92	48.77	10	3.05
						20	6.10
2 7/8	73.03	4.28	108.71	2.37	60.20	10	3.05
						20	6.10
3 1/2	88.90	5.03	127.76	2.97	75.44	10	3.05
						20	6.10

Other sizes available upon request.

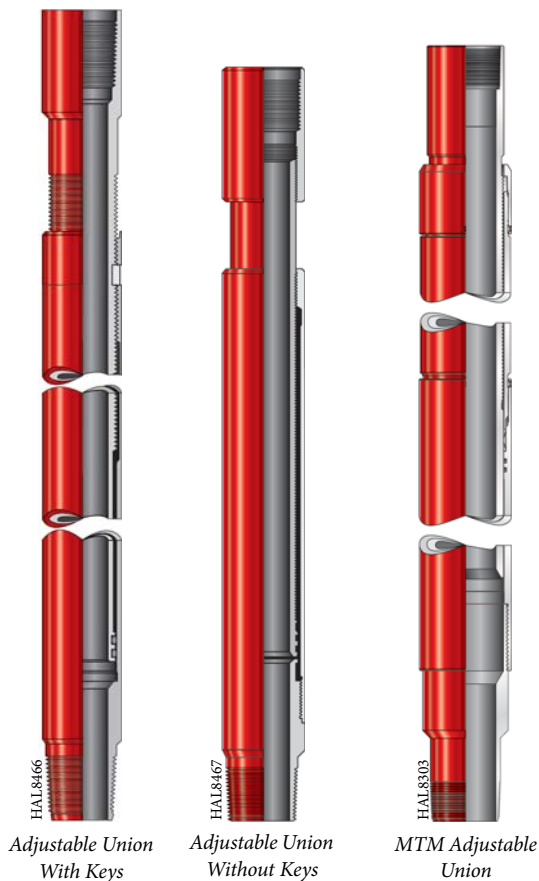
Specific ID dimension may change with selection of premium or API thread connection.

Part Number Prefix: 31TO



Adjustable Unions

Halliburton adjustable unions are installed in the tubing string to facilitate critical spacing out at the surface and between subsurface components. They may also be keyed to prevent future rotation. Threading on the adjustable unions stops travel. Adjustable unions with keys allow tubing torque to packers and other equipment below. Adjustable unions without keys are used when tubing torque below the union is not required. The metal-to-metal adjustable union does not have elastomeric seals exposed to well pressure.

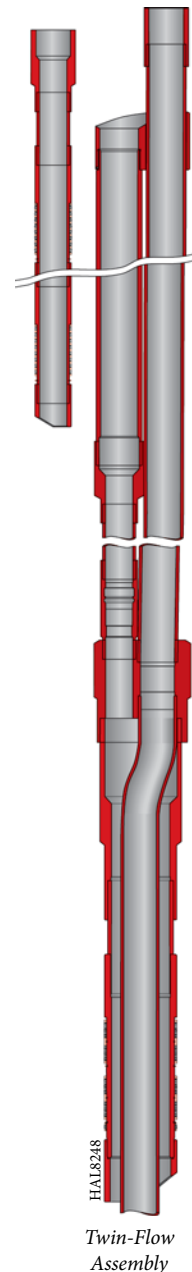


Adjustable Unions With and Without Keys

Size		OD		ID		Stroke	
in.	mm	in.	mm	in.	mm	in.	mm
2 3/8	60.33	2.83	71.88	1.93	49.02	12.0	304.80
						24.0	609.60
2 7/8	73.03	3.50	88.90	2.44	61.98	12.0	304.80
						24.0	609.60
3 1/2	88.90	4.54	115.32	2.90	76.66	12.0	304.80
						24.0	609.60
4 1/2	114.30	5.58	141.73	3.92	99.57	24.0	609.60

Twin-Flow Assemblies

Halliburton twin-flow assemblies are used in production or injection wells where dual or isolated flow above the packer is desired. The twin-flow assembly has seals on the lower end that seal in the polished bore of the upper packer. The upper section of the assembly has a primary string that is threaded above and below the twin-flow head. The secondary tubing string connection can be designed with either floating seals or a latch-type connection.



On-Off Tools

The XL on-off tool is a short, compact, overshot-type receptacle with a J that engages automatically and releases with 1/4 turn left-hand rotation.

When mounted above a packer, this tool provides a means of disconnecting the tubing string from the packer so that remedial work can be performed without unseating the packer. An integral nipple profile in the mandrel provides a means of blanking off the zone below the packer with a wireline-retrievable plug. After the remedial stage has been completed, the tubing is re-engaged at the on-off tool. The tubing plug may then be pulled by wireline to place the well on production.

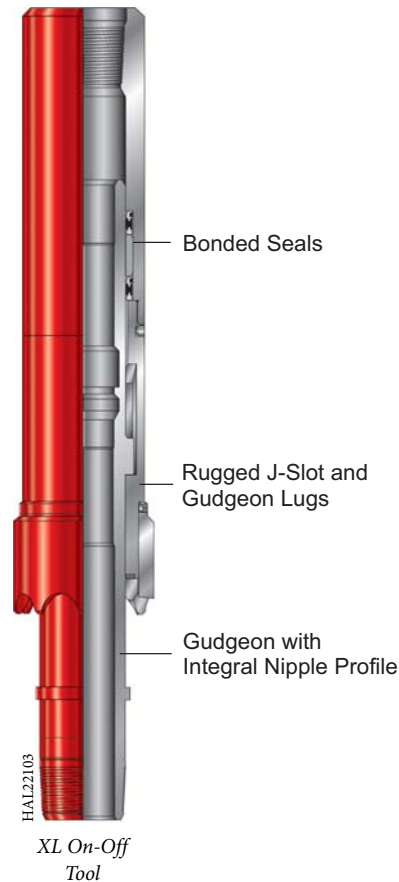
A replaceable seal system in the overshot section provides a positive seal on the polished mandrel. Standard seals are HNBR with optional Nitrile, Fluorel®, and AFLAS® fluoro rubber. A heavy one-piece seal body eliminates o-ring connections. The overshot washover shoe provides for rotation through sand or other debris for maximum washing action and easy re-engagement. Washover shoes are available for all casing sizes without changing out the J-slot.

The XL on-off tool is built for strength and durability. The integral cast J-slot contains no welds. Heavy gudgeon lugs on the polished mandrel provide high tensile strength, while materials and heat treatments are consistent with sour service requirements. Packer setting and releasing forces are taken in the lower section of the seal mandrel rather than the profile area above.

*AFLAS is a registered trademark of Asahi Glass Co.
Fluorel is a registered trademark of Dyneon, LLC.*

XXL On-Off Tool

The type XXL on-off tool retains all the benefits of the XL tool but allows 2 ft of movement before disconnecting.



XL On-Off Tool

Casing Size		Casing Weight		Maximum OD		Mandrel Seal Surface OD		Tubing Size Box Thread Up Pin Thread Down
in.	mm	lb/ft	kg/m	in.	mm	in.	mm	in.
4 1/2	114.30	9.5-13.5	14.14-20.09	3.75	95.25	2.62	66.42	2 3/8 8 Rd EUE
5 1/2	139.70	13-23	19.35-34.23	4.38	111.13	2.62	66.42	2 3/8 8 Rd EUE
				4.38	111.13	3.00	76.20	2 7/8 8 Rd EUE
7	177.80	17-38	25.30-56.54	5.50	139.70	2.62	66.42	2 3/8 8 Rd EUE
				5.50	139.70	3.00	76.20	2 7/8 8 Rd EUE
				5.75	146.05	4.00	101.60	3 1/2 8 Rd EUE
7 5/8	193.68	20-39	29.76-58.04	6.25	158.75	4.00	101.60	3 1/2 8 Rd EUE
9 5/8	244.48	36-71.8	53.57-106.84	8.02	203.71	2.62	66.42	2 3/8 8 Rd EUE
						3.00	76.20	2 7/8 8 Rd EUE
						4.00	101.60	3 1/2 8 Rd EUE

Ordering Information

Specify: casing size and weight; tubing size; weight; grade; and thread; service environment (standard, %H₂S, %CO₂, amines/other chemicals, chloride content, pressures, temperatures, etc.); maximum differential pressure requirement.

Part Number Prefix: 212XLH

Sealing Plugs and Backpressure Valves

A sealing plug can be used to convert the packers to bridge plugs when testing, squeezing, or performing well stimulation work above a sealbore packer such as Halliburton Perma-Series® or Versa-Trieve® packers. Halliburton offers many plugs that can help reduce the possibility of high pressure damage in the zone below the packer. When properly seated, Halliburton sealing plugs provide a fluid- or gas-tight seal and withstand pressures within the limits of the packer.

Retrievable Sealing Plugs

Retrievable sealing plugs are available in two types:

- J-slot plug latches into lugs in the packer and seals against pressure from above or below
- Locator plug provides a seal against pressure from above the packer only

Sealing plugs can be dropped into the casing if no possibility of a low fluid level exists, or they can be run in on tubing. A bypass incorporated into the J-slot sealing plug aids in the plug's retrieval since it equalizes pressure above and below the packer when the plug is unlatched and an upward pull is taken on the tubing. A J-slot overshoot is used to retrieve the J-slot plug. A ratchet-type overshoot is used to retrieve the locator plug.

Ordering Information

Specify: mating packer part number or packer type and bore size, latch type (J-slot, straight slot, Ratch-Latch™, No-Go), top thread and tubing material/grade, service environment (standard %H₂S, %CO₂, amines/ other chemicals, chloride content, pressures, temperatures, etc.), mandrel material (L-80 alloy, 9 Cr-1Mo, Incoloy® 925), seal type for Ratch-Latch locators (Nitrile, Fluorel®, Rytan®, Viton®, PEEK™), special bottom thread requirements if applicable (standard or premium).

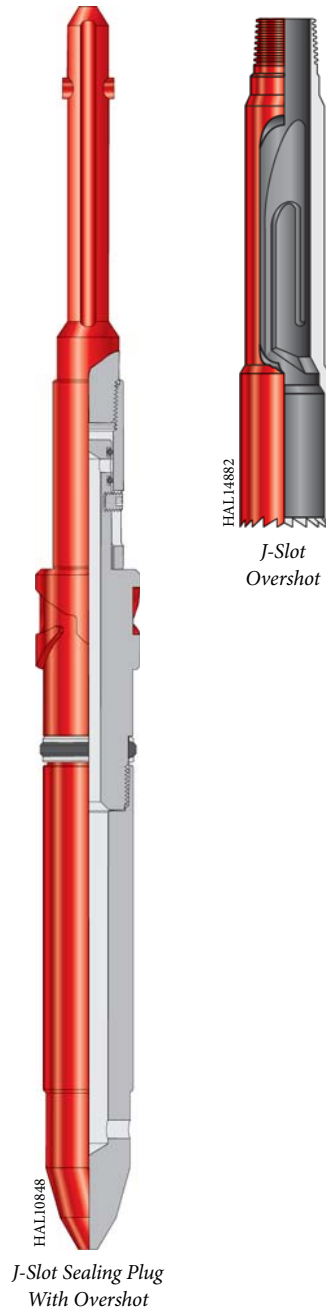
Fluorel is a registered trademark of Dyneon, LLC.

Rytan is a trademark of Phillips Petroleum Co. - Polyphenylene Sulfide

*Viton is a registered trademark of DuPont Dow Elastomers, LLC.-
Fluorocarbon*

PEEK is a trademark of ICI Americas, Inc. - Poly-Ether-Ether-Ketone

Incoloy is a registered trademark of Huntington Alloys.



Straight Shear Plug

Several straight shear plug designs are available, depending on the packer type with which they are used. The straight shear plug is designed to either latch into the lugs or thread in the top of the packer or into a recess at the lower end of the packer. The plug is retrieved by a straight pull.



Straight Shear Plug

Expendable Sealing Plugs

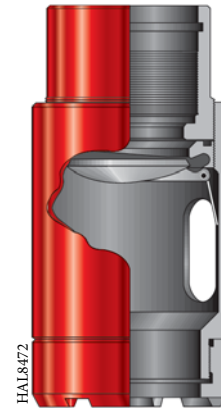
Expendable sealing plugs may be assembled in the bottom of a sealbore-type packer. Expendable sealing plugs are millable and may be expended out the bottom of the packer by applying tubing weight as the stinger or sealing unit enters the packer. The plug's releasing mechanism is pressure-balanced, and the plug is designed to hold any pressure within the limits of the packer.



Expendable Sealing Plug

Backpressure Valves

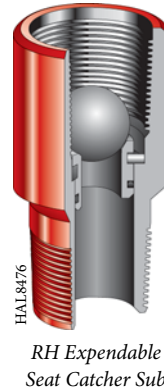
Backpressure valves are attached to the bottom of a packer to shut off flow from below when the sealing unit and tailpipe are removed. These flapper-type valves seal against a resilient seal and metal seat.



Backpressure Valve

Catcher Subs

Catcher subs for hydraulic-set packers are installed on the tubing to catch the seal ball that sets the packer. They have a full-open ID when the ball is expended. The seat-type catcher sub, called the RH, has the seat expended with the ball. The collet catcher sub (CCS) is used when landing nipples or other restrictions are installed below.



RH Expendable
Seat Catcher Sub



Collet
Catcher Sub

RH Expendable Seat Catcher Subs

Size		OD		Ball OD		Seat ID		ID After Seat Extended*	
in.	mm	in.	mm	in.	mm	in.	mm	in.	mm
1.660	42.16	2.20	55.88	1.00	25.40	0.86	21.84	1.37	34.80
1.900	48.26	2.50	63.50	1.31	33.27	0.98	24.89	1.49	37.85
2 3/8	60.33	2.91	73.91	1.00	25.40	0.86	21.84	1.91	48.51
				1.50	38.10	1.29	32.77		
2 7/8	73.03	3.50	88.90	1.00	25.40	0.86	21.84	2.36	59.94
				1.87	47.50	1.73	43.92		
3 1/2	88.90	4.06	103.12	2.25	57.15	1.73	43.92	2.88	73.15
4 1/2	114.30	5.52	140.21	1.75	44.45	1.48	37.59	3.92	99.57

*Also considered OD of ball seat. Care must be taken to avoid running this tool above smaller IDs, such as landing nipples, in the tubing string.

Ordering Information

Specify: casing size and weight; tubing size; weight; grade; and thread; service environment (standard, %H₂S, %CO₂, amines/other chemicals, chloride content, pressures, temperatures, etc.); maximum differential pressure requirement.

Part Number Prefix: 12RH

Collet Catcher Subs

Top Thread Size		OD		Collet ID Before Expended		Ball OD		Minimum ID After Expended	
in.	mm	in.	mm	in.	mm	in.	mm	in.	mm
1.90	48.26	2.50	63.50	1.25	31.75	1.44	36.58	1.52	38.61
2 3/8	60.33	3.06	77.72	1.56	39.62	1.75	44.45	1.92	48.77
2 7/8	73.03	3.67	93.22	2.00	50.80	2.12	53.85	2.38	60.45
3 1/2	88.90	4.50	114.30	2.50	63.50	2.75	69.85	2.80	71.12
5 1/2	139.70	6.52	165.61	4.32	109.73	4.50	114.3	4.67	118.62

Ordering Information

Specify: type (collet or expendable ball and seat); thread type; size; and weight; service environment (standard, %H₂S, %CO₂, amines/other chemicals, chloride content, pressures, temperatures, etc.); tubing material; desired shear-out value; tubing ID restrictions above and below the catcher sub.

Part Number Prefix: 12CCS

DP1 Anvil® Plugging System

The DP1 Anvil® plugging system is a temporary tubing plugging device that allows the operator to perform multiple pressure tests of the production tubing prior to packer setting and on command provides full bore, through-tubing access without well intervention. The plug incorporates a solid metal mechanical barrier removed by applying tubing pressure a predetermined number of times. No special surface equipment is required for operation and no debris results after actuation.

The Anvil plug comes complete with a range of accessories to accommodate pre-fill, circulation, and secondary override facility.

Applications

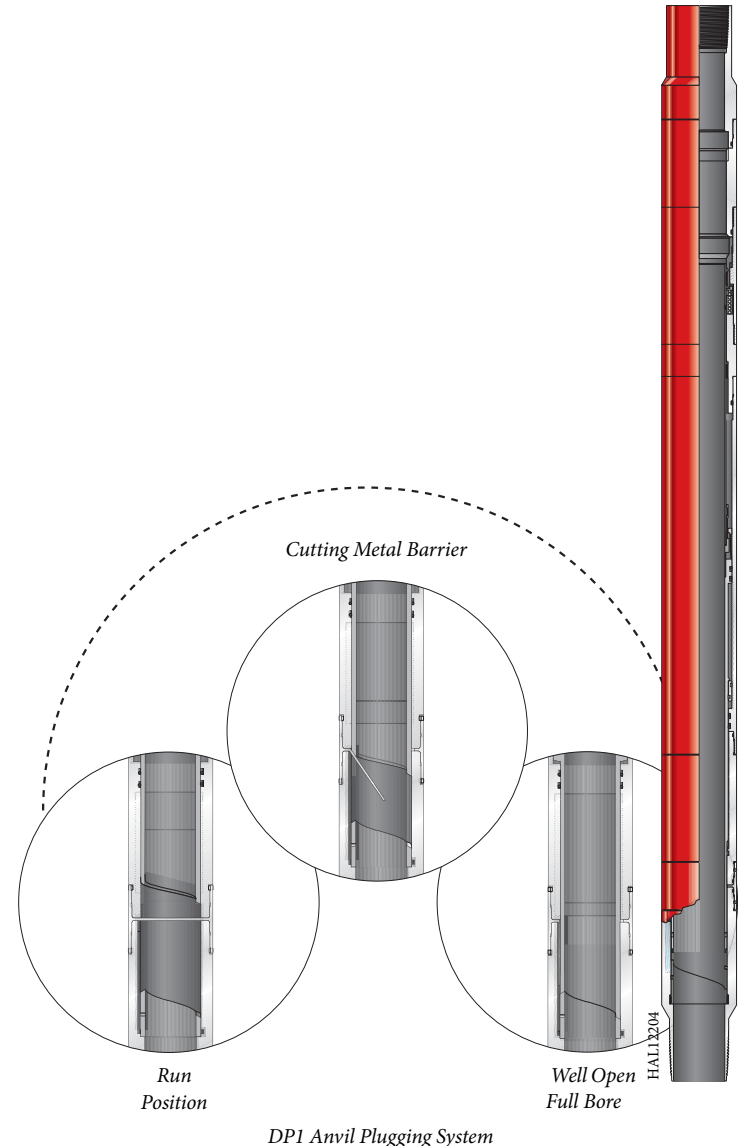
This tool is ideally suited for deep, highly deviated, or horizontal wells that would normally require coiled tubing to run plugs to set hydraulically activated equipment.

Features

- Solid metal barrier
- Depth limited only by ratings of hydrostatic chamber housings
- Low pressure differentials to activate
- Feedback at surface of tool activation
- No waiting on plug to open
- No debris
- No non-standard surface equipment required
- Mechanical override
- Easily millable

Benefits

- Totally interventionless completion installation
- Can use as a deep, downhole barrier for wireline valve removal
- After activation, Anvil plug produces no residual debris
- No special surface equipment to occupy premium space on the rig floor



Part Number Prefix: P.208DP1

DP1 Anvil® Plugging System

Casing Size		Thread				Maximum OD		Minimum ID		Pressure Rating			
		Size		Weight						Above		Below	
in.	mm	in.	mm	lb/ft	kg/m	in.	mm	in.	mm	psi	bar	psi	bar
7	177.80	3 1/2	88.90	9.2	13.69	5.890	149.61	2.890	73.41	6,000	413.40	2,000	137.80
						6.000	152.40						
		4	101.60	10	14.88	5.875	149.23						
						4 1/2	114.30	12	17.86				
		13	19.35										
9 5/8	244.48	5 1/2	139.70	17	25.28	7.650	194.31	4.625	117.48	6,000	413.40	2,000	137.80

LA0 Liquid Spring-Actuated Anvil® Plugging System

The LA0 liquid spring-actuated Anvil® plugging system is a pressure cycle-operated device that allows the operator to perform tubing tests or set hydraulically activated tools, then have full tubing bore ID without wireline or coiled tubing intervention. Operation of the liquid spring device is carried out by a ratchet mechanism that travels each time pressure is applied for the predetermined number of cycles. When the predetermined number of pressure cycles has been applied, communication from the tubing to the control line is opened, allowing pressure to activate the Anvil plugging system.

Applications

This tool is ideally suited for deep, highly deviated, or horizontal wells, which would normally require coiled tubing to run plugs for setting hydraulically activated equipment and where it is necessary to apply a number of tubing pressure tests before activation of hydraulically operated downhole tools.

Features

- Control fluid sealed within actuation mechanism
- Solid metal barrier
- Surface indication of successful tool activation
- No waiting on plug to open
- No debris
- Circulating sub
- Mechanical override
- Easily millable

Benefits

- Control fluid in the actuation mechanism is sealed and therefore protected from contamination.
- Unlike nitrogen pre-charged systems, the liquid spring mechanism eliminates well-specific setup and enhances long term suspension capability.

- Fluid used to communicate with the hydraulically activated device is carried in with the liquid spring module and therefore remains separate from the tubing fluid at all times, reducing the chance of control line blockages.
- Prior to activating the release mechanism, a solid metal plate is in place to form a downhole barrier that satisfies most operators' plugging policy.
- Sudden pressure drop at the device at activation will be detectable at surface.
- The Anvil plug completely opens at the time of actuation. Controlled metallurgical and dimensional properties help ensure consistent and reliable full opening.
- After activation, the plug produces no residual debris to potentially cause problems during subsequent operations.
- To allow tubing to self-fill while running, a circulating sub or fill up sub can be run in conjunction with the Anvil plug.
- In the event the plug does not activate normally, it can be mechanically overridden using a dedicated tool run on wireline or coiled tubing.
- The Anvil plug can also be milled with no rotating parts and only the metal plate to be removed.



LA0 Liquid Spring-Actuated Anvil® Plugging System

Part Number Prefix: P.208LA0

LS0 Liquid Spring-Actuation Device

The LS0 liquid spring-actuation device is a pressure cycle-operated, self-contained tool used to initiate any hydraulically activated downhole device. The tool allows setting of equipment after the application of a predetermined number of pressure cycles. The operation of the tool is carried out by a ratchet mechanism that travels each time pressure is applied. When the predetermined number of pressure cycles have been applied, communication from the tubing to the control line is opened, allowing pressure to reach the target device.

Applications

The liquid spring actuation device is used when it is necessary to apply a number of tubing pressure tests before activation of hydraulically operated downhole tools such as the Twin Flow Absolute Isolation system.

Features

- No atmospheric chamber or nitrogen-charged chamber to limit time to actuation
- Control fluid in the actuation mechanism is sealed and therefore protected from contamination

Benefits

- Total interventionless actuation of hydraulically operated tools.
- Unlike similar actuation devices, the liquid spring is not affected by changes in pressure in the lower annulus.
- Unlike nitrogen pre-charged systems, the liquid spring mechanism eliminates well-specific setup and enhances long term suspension capability.
- Fluid used to communicate with the hydraulically activated device is carried within the liquid spring module and therefore remains separate from the tubing fluid at all times, reducing the chance of control-line blockages.



LS0 Liquid
Spring-Actuation Device

Part Number Prefix: P.224LS0

Mirage® Disappearing Plug

The Halliburton Mirage® disappearing plug enables setting the production packer without intervention, which reduces costs associated with running and pulling slickline plugs. The plug is run in as a part of the tubing interior string and disintegrates after a predetermined number of pressure cycles, leaving the tubing full bore. See the “Subsurface Flow Controls” section for additional information on the Mirage disappearing plug.



*Mirage®
Disappearing Plug*

Tubing Safety Joints

Halliburton tubing safety joints are straight shear-type safety joints run on the tubing usually below a packer. They allow for separation of the tubing in case the lower packer or sealing unit becomes stuck and cannot be pulled with existing tubing. Their shear strength can be adjusted by adding or removing shear screws.



*Tubing
Safety Joint*

SO Straight Shear Tubing Safety Joint

Size		OD		ID		Maximum Shear Valve	
in.	mm	in.	mm	in.	mm	lb	kg
2.063	52.40	2.38	60.45	1.50	38.10	24,000	10 886
2 3/8	60.33	3.02	76.71	1.92	48.77	40,000	18 144
2 7/8	73.03	3.64	92.46	2.36	59.94	40,000	18 144
3 1/2	88.90	4.27	108.46	2.90	73.66	50,000	22 680
4	101.60	4.73	120.14	3.52	89.41	40,000	18 144
4 1/2	114.30	5.19	131.83	3.85	97.79	80,000	36 288
5 1/2	139.70	6.22	157.99	4.98	126.49	120,000	54 432

Ordering Information

Specify: casing size and weight; tubing size; weight; grade; and thread; service environment (standard, %H₂S, %CO₂, amines/other chemicals, chloride content, pressures, temperatures, etc.); maximum differential pressure requirement.

Part Number Prefix: 12SO

Seal Selection for Packers

To select the appropriate seal—static, dynamic, nonactive, or active—consider the following factors:

- Maximum pressure differential
- Maximum and minimum temperature
- Well fluids
- Seal application

The static seals remain stationary in the sealbore while dynamic seals move. Temperature, pressure, or movement

cycling within the sealbore calls for active seals; conversely, nonactive seals cannot undergo these changes. These facts, as well as application, availability, price, and well environment must be considered in seal selection.

Other Seal Technology

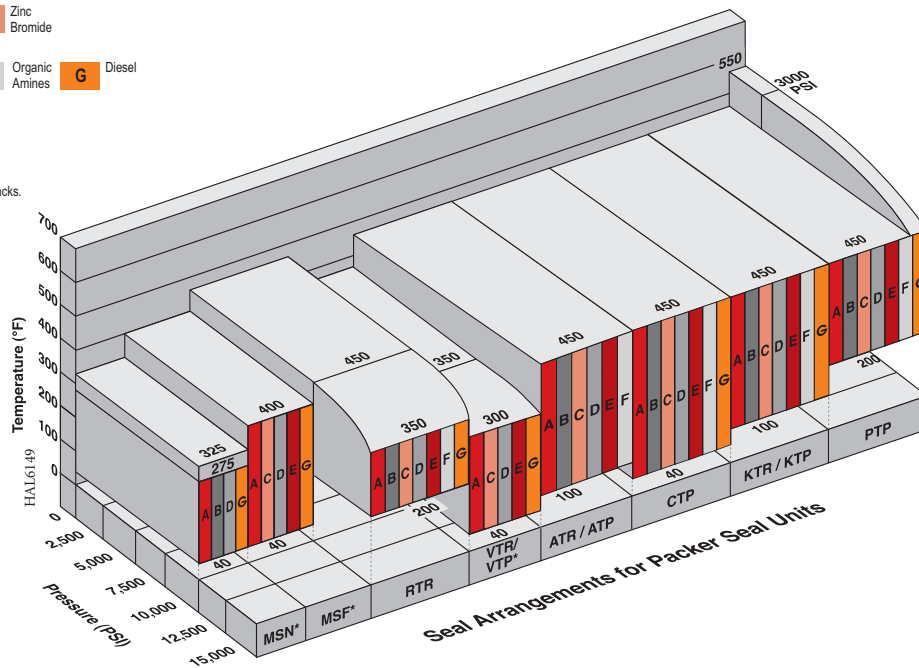
Other seal technologies, including metal-to-metal seals, are available from Halliburton representatives.

Halliburton Packer Seal Unit Selection Guide

A Sweet Crude	B Hydrogen Sulfide	C Zinc Bromide
D Steam	E Carbon Dioxide	F Organic Amines
G Diesel		

V=Viton
 R=Ryton
 K=Kalrez
 T=Teflon
 P=Peek
 All other seal units are premium v-packing stacks.

A=Aflas
 C=Chemraz
 N=Nitrile
 F=Fluorel
 MS=Molded Seal



*Good to 325°F if seal static. Good in water soluble amines—no aromatic hydrocarbon carrier and in CO₂ if seal static.

Note: Higher pressure ratings available. Contact your local Halliburton representative.

	KTR	KTP	RTR	VTR	VTP	ATR	ATP	PTP	CTP
	RYTON	PEEK	RYTON	RYTON	PEEK	RYTON	PEEK	PEEK	PEEK
	TEFLON	TEFLON	TEFLON	TEFLON	TEFLON	TEFLON	TEFLON	TEFLON	TEFLON
	KALREZ	KALREZ	RYTON	VITON	VITON	AFLAS	AFLAS	PEEK	CHEMRAZ

AFLAS® is a registered trademark of Asahi Glass Co. - Propylene - Tetrafluoroethylene Copolymer.

Chemraz® is a registered trademark of Greene, Tweed & Co., Inc. - Perfluoro Elastomer.

Kalrez® is a registered trademark of DuPont Dow Elastomers, LLC - Perfluoro Elastomer.

PEEK™ is a trademark of ICI Americas, Inc. - Poly-Ether-Ether-Ketone.

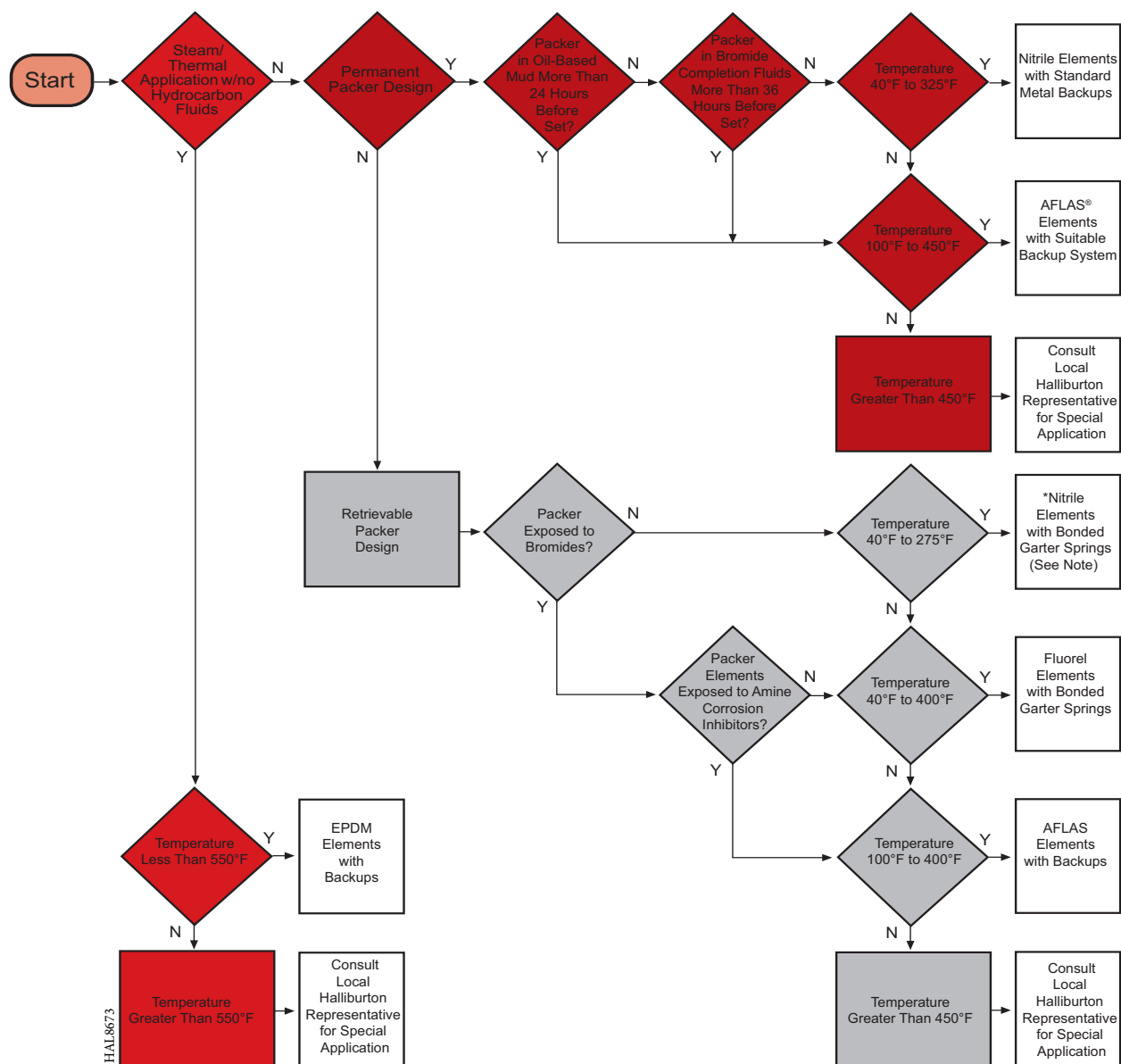
Ryton® is a registered trademark of Phillips Petroleum Company - Polyphenylene Sulfide.

Teflon® is a registered trademark of DuPont - PTFE Polytetrafluoroethylene.

Viton® is a registered trademark of DuPont Dow Elastomers, LLC - Fluorocarbon.

KTR® is a registered trademark of Halliburton.

Packer Element Selection Chart



*Note: Nitrile elements exposed to H₂S will experience varying levels of hardening.

Please consult your Halliburton Representative and supply the following information:
%H₂S, temperature, length of exposure, completion fluids, and produced fluids data.

AFLAS is a registered trademark of Asahi Glass Co., Ltd.
Teflon is a registered trademark of DuPont.
Fluorel is a registered trademark of Dyneon, LLC.

